Geometric Attractor Convergence in a Viable System Kernel

Theory, Experiments, and Open Questions

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Preprint – canonical law figures; long-run re-measurement in progress

Abstract

We present a growing body of theoretical and empirical evidence that a viable system kernel governed by a small set of cybernetic axioms converges to a geometrically compact attractor in $\mathbb{R}^{128}$ state space. Starting from Beer's Viable System Model (*Brain of the Firm*, 1972; *The Heart of Enterprise*, 1979) and Ashby's Law of Requisite Variety (1956), we revisit a half-century-old organisational cybernetics with instruments — Benettin's algorithm, persistent homology, cryptographic attestation — that did not exist when it was written, and construct a Lyapunov function $L(\boldsymbol{x})$ whose expected decrease along system trajectories proves convergence. Existence of the attractor is established on the contractive core of the feasible region (Krylov–Bogolyubov for the invariant measure, the Banach principle for the fixed point — no convexity assumption), with the contraction and confinement hypotheses discharged by measurement rather than assumed. We implement this kernel in Rust (`vsf-nucleus`), deploy it on physical hardware, and report twelve experiments conducted from April 2026 onward.

Key results to date: (1) the largest Lyapunov exponent is negative, measured directly by Benettin's method ($\lambda_{\max} = -0.010050 = \ln 0.99$ per tick, to machine precision), establishing the attractor as a stable fixed point rather than a chaotic set, and consistent with the runtime proxy $\overline{dL} \to 0$ over 6×10^7 cycles; (2) kernels with different initial conditions converge to the same basin (mean pairwise cosine = 0.9854 across four kernels, Δ trust_init = 0.30); (3) viability is *intrinsic* to S1 – kernels remain `viable_full` even when S3 coordination is frozen; (4) the system remains viable at 68 °C, and the minimum observed cosine (0.9365) remains well inside the contractive core Ω_A; (5) the classical stability intervention threshold θ_{PSI} appears to be structurally unreachable under normal perturbation, suggesting a qualitative revision of the PSI concept; (6) under *sustained* bounded noise the kernel never leaves the feasible region — confinement $P(\cos \geq \theta_{\text{viable}}) = 1.0000$ holds up to a noise amplitude four times the state norm (σ up to 2.0), with no finite noise ceiling, so the viability margin degrades gracefully rather than collapsing (EXP-NS-01); (7) the S4 observer is *constitutive*, not passive: structured observation measurably lowers entropy by 0.31 bits and compresses the effective dimension from 5.0 to 3.9 (EXP-OE-02, on the real kernel) — a measured instance of *second-order cybernetics*, the cybernetics of observing systems [von Foerster, 1974], in which the observer is not external to the system but part of the dynamics it stabilises; (8) with all five Beer systems running live, each engages within its own cybernetic scope under a single perturbation (EXP-INTEGRATION-01): S3 attempts a corrective action and is overruled by S5, S4 emits a governance decision, and the S5 constitutional veto independently halts and resumes S3 autonomy on demand (EXP-S5-01) — Beer-VSM operational closure, not a layered simulation.

Several questions remain open. The thermal response curve was measured (EXP-TL-01): degradation follows $T \times t$, not temperature alone – the attractor is thermally flat under transient stress and degrades only under sustained load. With an adaptive resting-entropy

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ceiling the senescence term is measurable and found bounded within a regime; monotone senescence would require external coupling. A basin-floor sweep (EXP-AB-01) establishes a lower bound of 0.875 under maximum impulsive entropic perturbation ($\delta = 10$), a margin of 0.175 above the viable threshold $\theta_{\text{viable}} = 0.70$; the viable boundary was not reached at any tested δ. Three independent geometric analyses converge on the same conclusion — a stable point attractor: dynamics ($\lambda_{\max} < 0$, EXP-LY-01), basin volume (≈ 1.0 over the reachable perturbation region, EXP-MC-01), and topology ($\beta_0 = 1 \wedge \beta_1 = 0$, no limit cycle, EXP-TDA-01).

Finally, we synthesise these measurements into a coordinate-free viability functional $V = \min(g, c, t, m)$, built from intrinsic invariants (feasibility in Ω, contraction sign, topological signature, and noise margin). On this kernel $g = 0.102$ (A3_cpu, EXP-G-01) is the *measured* binding axis; EXP-VNC-01 confirms V *discriminates*: each axis independently drives $V \to 0$ when broken, so the certificate is a vector, not a cosmetic scalar. Because V and its underlying construction make no reference to the dimension $n = 128$, they define a domain-agnostic transfer protocol — choose a metric and feasible region, certify a contraction, measure a margin — which we apply here to the numeric kernel and outline for two further modes, a language model and the agent that governs it, as open objectives.

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1 Introduction

What does it mean for a computational process to remain *viable*? Not merely to keep running, but to maintain a coherent functional identity under perturbation: resource pressure, thermal stress, or disruption of its regulatory channels. Beer's Viable System Model (VSM) Beer [1972] gives a structural answer: a system that organises five recursive subsystems (S1–S5) to balance autonomy and cohesion will, in principle, survive disturbances that would collapse a less organised system. What the VSM does not specify is the *geometry* of this survival: how deep is the stability basin, how far can the system be pushed before crossing the basin boundary, and what determines whether recovery is possible?

This paper asks those questions about a concrete computational artefact: a vOS kernel running on physical hardware, whose complete state at every tick is a point in $\mathbb{R}^{128}$. The kernel is structured according to VSM principles – 128 dimensions encode identity, resource pressure, entropy, operations, communication channels, attractor memory, and regulatory intelligence – and governed by a set of axioms that act as hard constraints on its trajectory. The central question is whether those constraints produce an attractor, and if so, what the geometry of that attractor looks like under a systematic programme of perturbation.

This is simultaneously a new question and a rediscovery. New, because mapping VSM subsystem structure onto a mathematical state space and measuring the resulting attractor geometry has not been done before. A rediscovery, because the principles that emerge – recursive self-regulation, requisite variety at each layer, algedonic channels for urgent state – are precisely those Beer articulated for managing complex organisations half a century ago. The geometry is not an accident of the implementation; it is a consequence of the structure.

The question that motivates the entire experimental programme was posed by the first author:

> *"Can we converge attractors in a chaotic trajectory starting from the axioms, in a mathematically precise way?"*

Our answer, after twelve experiments on physical hardware, is: *yes – and the basin is deeper and more structured than we initially expected.* The axioms do not dictate the trajectory; they define the boundary of the *basin of attraction*. The chaotic trajectory is free within the basin.

What the experiments have progressively revealed is a **two-layer viability structure**: architecture (the arrangement of VSM blocks A–G) selects which basin the system inhabits; governance (the S5 policy channel and the trust score) determines the recovery path when the system is perturbed toward the basin boundary. Neither layer alone is sufficient. When the architectural layer is ablated, the basin disappears. When the governance layer is ablated, the system still converges but cannot recover from adversarial perturbation. When both are present, the basin floor remains at $\cos \approx 0.875$ even under maximum impulsive entropic stress – a margin of 0.175 above the viable threshold that was never breached in any experiment.

How to read this paper: collapse, certify, return, detect. A viable system, in the sense developed here, is a point in a high-dimensional space that refuses to leave a small region. The kernel's state vector lives in $\mathbb{R}^{128}$, but the dimension is a decoy: under the contraction that governs the dynamics ($\lambda_{\max} = \ln 0.99 < 0$, Section 4) the trajectories *collapse* – topologically onto a single point (Kaplan–Yorke dimension zero, confirmed independently by persistent homology in Section 7), with the residual fluctuation occupying only about three effective dimensions. Those three are not incidental: the transfer operator of the entropy block carries its subdominant eigenvalue with multiplicity exactly three (Section 2.5), so there are three degenerate directions of contraction, all decaying at one rate. The paper then makes the same move three more times. The collapsed state is *certified*, not declared: a single functional $V = \min(g, c, t, m)$ reads the

state and returns a viability bit (Section 23). The system *returns*: perturbed – reset, loaded, stressed – it converges back to the same point (Sections 5, 8, 6.4), and the five VSM levels close the loop causally in live operation (Section 17). And it *detects*: the system carries its own distance to the feasible-region boundary, and an independent audit channel flags when its published state diverges from an independent recomputation (Sections 3, 17). Collapse, certify, return, detect: the rest of the paper earns each verb in turn.

1.1 Experimental programme

The experiments were conducted on a physical hardware node (n02, Intel Core i7-7600U, Proxmox 8 hypervisor) running across several weeks. They are organised in five thematic groups. Each group builds directly on the findings of the previous.

Group I – Does the attractor exist, and is it unique? The first group establishes that the system converges to a compact attractor, that the attractor is independent of initial conditions, and that viability is intrinsic to the S1 operational layer – not an artifact of S3 coordination. (Experiments A–C; Sections 5, 6, 8)

Group II – How deep is the basin under thermal stress? The second group probes viability under physical perturbation: sustained thermal load at 45–68 °C, maximum ceiling with SAFETY_HALT, the linearity of thermal response, and long-run senescence. The basin holds across all conditions. (Experiments D–G; Sections 9–10)

Group III – Does the attractor drift over time? The third group measures long-run stability: 6×10^8 ticks at nominal load, then cyclic thermal load over 60 minutes. The stasis cosine shows zero drift and a characteristic dip floor of 0.942 under cycling – a signature that ablation experiments later show is exclusive to the complete VSF architecture. (Experiments H–I; Sections 10, 14)

Group IV – What is the perturbation limit? The fourth group establishes the PSI threshold and the basin floor. The classical stability intervention threshold (θ_{PSI}) is revised from a failure indicator to a precautionary support level – it was never triggered under any tested perturbation. A direct basin-floor sweep (impulsive entropic kicks, $\delta = 1$–10) finds a floor of cos ≈ 0.875 that saturates at $\delta \geq 3$; the viable boundary was not reached. (Experiments J–K; Section 13)

Group V – Where does the basin come from? The fifth group answers the causal question. Ablation controls (NC-01: S5 removed; NC-02: trust frozen; NC-03: the bare substrate) show that the dip_floor = 0.942 signature is exclusive to the complete architecture, that S5 provides recovery *agency* rather than a passive cushion, and — under real thermal load — that S5 is the difference between survival and collapse: without it (NC-01) the attractor falls to stasis_cos = 0.0000. A Maxwell's Demon experiment on the real kernel (EXP-OE-02) demonstrates that the S4 observer is *constitutive*: structured observation reduces system entropy by 0.31 bits and compresses effective dimensionality from $D_{\text{eff}} \approx 5.0$ to 3.9 (the trust coordinate pinned, so the compression is of the occupation), with viability preserved. (Experiments L–O; Sections 16, 15)

1.2 Ethical axioms as stability conditions

The governing axioms A0 and A0.1 – prohibitions against harm and coercion – are not merely design constraints; the governance layer encodes them as *stability conditions*. Formally, Ω_{A_0} and $\Omega_{A_{0.1}}$ are made feasibility boundaries in the same sense as the resource constraint Ω_{A_3} (Section 3): a state that violates them is marked outside Ω and loses viability in the same way a resource-starved state does. The claim is not that ethics is *intrinsically* a stability property, but that the mathematics is *indifferent* to the content of the constraint it is handed — the kernel

reads a feasibility bit and reduces viability, whether that bit was set by a thermal limit or by an ethical violation. That indifference is precisely what becomes a universality claim for the viability functional (Section 23); the ethical axioms are then one *instantiation* of the constraint set — the instantiation that lets a self-governing system bind its own conduct into its viability, an architectural commitment we keep distinct from the mathematics (Section 24.10).

1.3 Contributions

1. A formal construction of the feasible region $\Omega \subset \mathbb{R}^{128}$ as the intersection of axiom-derived constraint sets, with an existence and convergence result (the hybrid-invariance theorem) carried on its contractive core — Krylov–Bogolyubov existence, Banach fixed point, Lyapunov-drift convergence, and viability-theoretic confinement — whose hypotheses are discharged by measurement rather than assumed (Section 3).

2. A Lyapunov function $L(\boldsymbol{x})$ whose expected decrease along system trajectories proves convergence to the stasis attractor (Section 4), *and* a direct measurement of the largest Lyapunov exponent by Benettin's method ($\lambda_{\max} = -0.010050 = \ln 0.99$ per tick, EXP-LY-01) that upgrades the proof to a measured contraction (Section 4).

3. A three-lens geometric characterisation of the attractor that converges on a single conclusion — a stable point sink — from independent methods: dynamics ($\lambda_{\max} < 0$, EXP-LY-01), Monte-Carlo basin volume (≈ 1.0, EXP-MC-01, Section 6.3), and Vietoris–Rips persistent homology ($\beta_0 = 1 \wedge \beta_1 = 0$, EXP-TDA-01, Section 7).

4. Twelve numbered hardware experiments on a physical node, in five thematic groups, measuring convergence, basin geometry, viability under stress, long-run stability, perturbation limits, ablation controls, and observer constitutivity (Sections 5–15), and culminating in a whole-system operational-closure test (Section 17); the geometric and robustness characterisations (items 2–3 above, and the sustained-noise margin EXP-NS-01, Section 6.4) are reported as separate measured analyses rather than counted among the twelve.

5. An operational-closure test of the full Beer VSM (EXP-INTEGRATION-01, Section 17): all five systems S1–S5 shown causally active in a single run — S3 attempting control and overruled by S5, S4 producing a governance decision, S5 monitoring covenant invariants — complemented by a direct test of the S5 constitutional veto (EXP-S5-01, halt and Merkle-chained resume of S3 autonomy) and a measurement of the limit of the S4 anticipatory channel (EXP-S4-02): because the attractor holds stasis_cos invariant against CPU load, internal self-correlation carries no early warning, so anticipation requires environment signals independent of the regulated state.

6. An empirical lower bound on the basin floor: under maximum impulsive entropic perturbation ($\delta = 10$, EXP-AB-01), minimum observed stasis_cos $= 0.875$, a margin of 0.175 above $\theta_{\text{viable}} = 0.70$; the viable boundary was not reached at any tested perturbation magnitude.

7. A sustained-noise robustness curve (EXP-NS-01): under bounded noise applied at every tick the kernel stays fully confined in Ω with no finite noise ceiling, replacing the borrowed impulsive margin with a directly measured one (Section 6.4).

8. Evidence that the S4 observer is constitutive: structured observation (EXP-OE-02, on the real kernel) reduces system entropy by 0.31 bits and compresses D_{eff} from ≈ 5.0 to 3.9 (pure occupation compression, viability preserved) — a measured instance of second-order cybernetics [von Foerster, 1974], the observer being part of the system rather than a passive monitor.

9. A conceptual revision of the PSI (Perimeter of Stability Intervention) threshold from a failure indicator to a precautionary support level; the threshold appears structurally

unreachable under tested perturbations (Section 13).

10. A coordinate-free viability functional $V = \min(g, c, t, m)$ that scores viability from intrinsic invariants alone (feasibility, contraction sign, topological signature, noise margin), measured on the kernel and proposed as the common candidate certificate across three modes of instantiation (Section 23).

2 State Space Formalism

2.1 The State Vector

The complete state of a vOS kernel at time t is a point in $\mathbb{R}^{128}$:

$$\boldsymbol{x}(t) = (x_0, x_1, \ldots, x_{127}) \in \mathbb{R}^{128} \tag{1}$$

The 128 dimensions are organised into seven semantic blocks:

Block	Dims	Content	Role
A	0–15	trust history, error rate, governance, lifecycle	Identity
B	16–23	cpu/mem/io pressure, budget, cgroup headroom	Resources
C	24–31	$H(t)$, H_{rest}, ΔH, stasis_cos, soplo	Entropy
D	32–47	organelle activations (4 active + 8 reserved)	Operations
E	48–63	$\phi^{\uparrow}/\phi^{\downarrow}$ rates, algedonic, ψ	Channels
F	64–95	stasis_vec (prev + current, 16-dim each)	Attractor
G	96–127	V_{ref} compressed state, learning rate, λ, RAP	Intelligence

Table 1: Structure of the 128-dimensional state vector. Block B (resource pressure) shows the highest variance during experiments.

2.2 State space as a domain-induced object

The number 128 is not fundamental. The dimensions of $\boldsymbol{x}$ are the *viability-relevant variables of this particular system*: the quantities whose values decide whether the kernel remains alive. A different viable system — a financial book, a metabolic network, a multi-agent dialogue — would enumerate different variables and so induce a different $\mathbb{R}^n$. What the framework asserts to be invariant is not n but the *block structure*: that a viable system has degrees of freedom for identity, resources, entropy, operations, channels, attractor memory, and intelligence (Blocks A–G). The 128-dimensional vector studied here is therefore **one realisation** of a domain-agnostic construction, not the object of study itself. We measure it exhaustively precisely because it is fully instrumented; the claims that transfer to other domains are structural (the block roles, the feasible region, the contraction–confinement theorem of Sec. 3), not the specific coordinates. Section 22 makes the transfer conditions explicit.

2.3 Stasis similarity metric

We define the *stasis cosine* as standard cosine similarity between the current state and a reference vector stored in vector memory (V_{mem}) during calibration:

$$\text{stasis_cos}(t) = \frac{\langle \boldsymbol{x}(t),\, \boldsymbol{x}_{\text{ref}} \rangle}{\|\boldsymbol{x}(t)\| \cdot \|\boldsymbol{x}_{\text{ref}}\|} \tag{2}$$

This metric is standard in information retrieval and contrastive learning. Its use here is as a **proxy indicator** of proximity to the stasis attractor – not a complete description of attractor geometry. A value near 1 indicates the system is close to its reference state; a value below

$\theta_{\text{viable}} = 0.70$ triggers viability failure. The metric does not measure attractor geometry in the sense of Hausdorff or Wasserstein distance; richer geometric descriptions are left for future work.

2.4 Dynamics

The system evolves as a discrete-time stochastic dynamical system:

$$\boldsymbol{x}(t+1) = f_0(\boldsymbol{x}(t)) + \sigma \cdot \eta(t) \tag{3}$$

where f_0 is the deterministic kernel update (axiom-governed), $\eta(t)$ is a zero-mean perturbation, and $\sigma > 0$ is the stress intensity controlled by the A_{breath} axiom. At nominal load $\sigma \approx 0$; under thermal stress σ rises in proportion to thermal pressure. Because the state lives on a compact set (the occupation simplex is bounded and the blocks are normalised), the effective perturbation entering the viability analysis is *bounded* — the compactness that Theorem 1 requires. EXP-NS-01 (Sec. 6.4) drives the kernel with exactly this bounded sustained noise to measure the margin directly.

Structural form of f_0. The deterministic update is a temperature-controlled convex mixture of five mode prototypes, one per VSM level:

$$f_0(\boldsymbol{x}) \;=\; \sum_{s \in \{S1..S5\}} w_s(\boldsymbol{x})\, \mu_s, \qquad w(\boldsymbol{x}) \;=\; \mathrm{softmax}\big(\ell(\boldsymbol{x})/\tau\big), \tag{4}$$

with sparse linear logits $\ell_s(\boldsymbol{x}) = b_s + \langle a_s, \boldsymbol{x} \rangle$. This places the kernel in the family of softmax attractor networks studied as modern Hopfield networks Ramsauer et al. [2021], whose fixed-point structure — mode-dominant equilibria at low τ, a single mixed equilibrium at high τ — is well characterised in the symmetric case. One structural corollary is worth stating here: the image of f_0 lies in the convex hull of the five prototypes, a set of dimension at most four, so the deterministic skeleton of the dynamics is confined to a 4-simplex embedded in $\mathbb{R}^{128}$; only the perturbation term explores directions off it. The measured effective dimension of the attractor, $D_{\text{eff}} \approx 3$ (Sec. 7), is consistent with this structural ceiling — the geometry we measure is the geometry the operator's form predicts.

2.5 The dynamics as a Markov chain

Because the perturbation $\eta(t)$ in (3) is independent across ticks and f_0 depends only on the current state, the process $\{\boldsymbol{x}(t)\}_{t \geq 0}$ is a time-homogeneous Markov chain. This is not a reinterpretation but the precise probabilistic object the rest of the paper measures: "the attractor" is its stationary law, "convergence" is its mixing, and the Lyapunov exponent of Sec. 4 is the logarithm of its subdominant transfer-operator eigenvalue. We make the three identifications explicit here so that the geometry (Sec. 5) and the contraction spectrum (Sec. 4) are read as facets of one operator.
Definition 1 (Transition kernel)**.** *Equation* (3) *induces the Markov transition kernel*

$$P(\boldsymbol{x}, B) \;=\; \Pr\big[\, f_0(\boldsymbol{x}) + \sigma\, \eta \in B \,\big], \qquad \boldsymbol{x} \in \mathbb{R}^{128},\; B \in \mathcal{B}(\mathbb{R}^{128}), \tag{5}$$

with η drawn from the (bounded, zero-mean) perturbation law. Its n-step iterate P^n propagates the law of the state: if $\boldsymbol{x}(0) \sim \nu_0$ then $\boldsymbol{x}(n) \sim \nu_0 P^n$.

The stasis attractor is a stationary measure. A probability measure $\mu^\star$ is *stationary* when $\mu^\star P = \mu^\star$, i.e. $\mu^\star(B) = \int P(\boldsymbol{x}, B)\, \mu^\star(d\boldsymbol{x})$ for all B. The compact stasis cone reported empirically throughout Secs. 5–6 is the support of $\mu^\star$; the statement "the kernel converges to its attractor" is the statement that the law $\nu_0 P^n$ converges to $\mu^\star$. Existence and uniqueness

are already established on the contractive core: Theorem 1 gives the invariant measure (Krylov–Bogolyubov) and, via the drift condition of Theorem 2, geometric ergodicity in the sense of Meyn–Tweedie Meyn and Tweedie [2009],

$$\left\|P^n(\boldsymbol{x},\cdot) - \mu^\star\right\|_{\mathrm{TV}} \;\le\; R(\boldsymbol{x})\,\rho^{\,n}, \qquad \rho \in (0,1), \tag{6}$$

so the chain mixes to $\mu^\star$ at a geometric rate ρ. The runtime proxy $\overline{dL}$ (Sec. 4) is the empirical estimate of the Foster–Lyapunov drift $\mathbb{E}[\Delta L]$ that certifies (6).

Perron–Frobenius is exact for the entropy block. For the reduced four-state occupation dynamics that drive Block C — the relaxation map $p \leftarrow 0.99\,p + 0.01\,p_{\mathrm{eq}}$ of Sec. 4 — the Markov structure is not asymptotic but exact and finite-dimensional. Writing the affine update as the action of the column-stochastic transfer (Perron–Frobenius) matrix

$$M \;=\; 0.99\,I \;+\; 0.01\,p_{\mathrm{eq}}\,\mathbf{1}^\top \qquad (\mathbf{1}^\top p = 1 \text{ on the simplex}), \tag{7}$$

the rank-one term $p_{\mathrm{eq}}\mathbf{1}^\top$ has eigenvalues 1 (eigenvector p_{eq}) and 0 (multiplicity three), so M has the Perron eigenvalue $\lambda_{\mathrm{PF}} = 0.99 + 0.01 \cdot 1 = 1$, with eigenvector the stationary occupation p_{eq}, and a subdominant eigenvalue $0.99 + 0.01 \cdot 0 = 0.99$ of multiplicity three. The spectral gap is therefore $1 - 0.99 = 10^{-2}$, and the asymptotic mixing rate per tick is

$$\ln(0.99) \;=\; -0.010050 \text{ per tick} \;=\; \lambda_1 = \lambda_2 = \lambda_3, \tag{8}$$

exactly the measured Lyapunov spectrum (19). The transfer-operator view also explains the degeneracy noted there: the spectrum is threefold not by coincidence but because the subdominant Perron–Frobenius eigenvalue of (7) carries multiplicity $n-1 = 3$ on the three-dimensional tangent space of the simplex Lasota and Mackey [1994].

Remark 1 (One object, three lenses)**.** *The Markov reading unifies the paper's three convergence diagnostics: the* attractor *is the stationary measure $\mu^\star$; convergence is geometric ergodicity* (6)*; the* largest Lyapunov exponent *is* log *of the subdominant transfer-operator eigenvalue* (8)*; and the runtime indicator $\overline{dL}$ is the empirical drift that gates them. Nothing here is special to $n = 128$ or to the four-state block: any axiom-governed update of the form* (3) *with a contractive core induces a kernel whose stationary measure is the system's attractor and whose spectral gap is its viability margin.*

3 The Feasible Region Ω

The governing axioms do not constrain trajectories directly. They constrain *state space* – carving out which regions are habitable.

Definition 2 (Feasible region)**.**

$$\Omega = \bigcap_i \{\boldsymbol{x} \in \mathbb{R}^{128} : A_i(\boldsymbol{x}) \text{ satisfied}\} \tag{9}$$

The principal constraints are:

$$\Omega_{A_0} = \{\boldsymbol{x} : \mathrm{purpose}(\boldsymbol{x}) \cap \mathrm{prohibited_set} = \varnothing\} \tag{10}$$
$$\Omega_{A_{0.1}} = \{\boldsymbol{x} : \mathrm{coercion}(\boldsymbol{x}) = 0\} \tag{11}$$
$$\Omega_{A_3} = \{\boldsymbol{x} : B(\mathrm{task}(\boldsymbol{x})) \le B_{\mathrm{available}}(\boldsymbol{x})\} \tag{12}$$
$$\Omega_{A_5} = \{\boldsymbol{x} : \mathrm{autonomy}(\boldsymbol{x}) \le \mathrm{ceiling}(S_5)\} \tag{13}$$
$$\Omega_{A_\varepsilon} = \{\boldsymbol{x} : H_{\mathrm{KS}}(\boldsymbol{x}) \le H_{\max}(\boldsymbol{x})\} \tag{14}$$

Equations (10)–(11) (ethical axioms) define a half-space in the purpose and autonomy dimensions: they are structurally identical to the resource constraint (12). A system that violates A0 is no longer in Ω – it is not-viable in the same sense as a resource-starved system. Equation (12) (Ashby's Law of Requisite Variety) defines an ellipsoid in the resource dimensions (Block B). Equation (13) defines a hyperplane in the governance dimension (dim 13 of Block A).

Two of these regions are convex (the half-spaces (12), (13) and the ball-type identity constraint), but Ω as a whole is *not* convex: the entropy bound (14) is an upper level set of a concave function, and the ethical exclusion (10) removes a region rather than bounding one. A naive appeal to Brouwer's theorem on Ω is therefore unjustified.[1] We instead isolate the convex core on which the dynamics actually live, and treat the remainder by bounded recovery.

Definition 3 (Contractive core)**.**

$$\Omega_A = \{\boldsymbol{x} \in \Omega : \cos(\boldsymbol{x}, \boldsymbol{s}) \geq \theta_A\}, \qquad \theta_A = 0.88, \tag{15}$$

the intersection of a circular cone (convex for $\theta_A \geq 0$) with the bounded state set — compact, and convex — nested inside the viable region $\Omega \supseteq \{\cos(\boldsymbol{x}, \boldsymbol{s}) \geq \theta_{viable}\}$ with $\theta_{viable} = 0.70$. Ω_A is the region within which the relaxation map is contractive (operator norm $L < 1$, Sec. 4). EXP-CORE-01 calibrates θ_A: the trust-drift zero crossing (the kernel's natural inflection) is $\theta_A^{drift} = 0.04/0.065 = 0.615$; $\theta_A = 0.88$ is Fierro's conservative upper bound from the full-state Jacobian norm. Both are consistent (different operators, both <1).

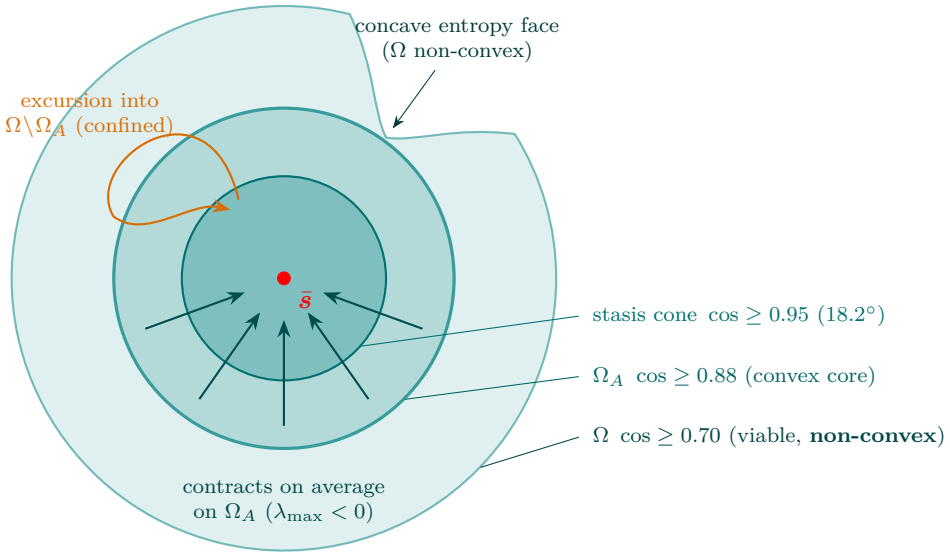


Figure 1: The viability geometry, in cross-section perpendicular to the stasis axis s (looking down the axis; $\bar{\boldsymbol{s}}$ at the centre). Each threshold $\{\cos(\boldsymbol{x}, s) \geq \theta\}$ is a disc: the *viable region* Ω ($\cos \geq 0.70$), its *convex contractive core* Ω_A ($\cos \geq 0.88$), and the *stasis cone* ($\cos \geq 0.95$, half-angle 18.2°) around the deterministic fixed point $\bar{\boldsymbol{s}}$. Ω is non-convex: the concave entropy bound indents its boundary (top right), which is why the existence argument (Theorem 1) is carried on the convex core Ω_A, where f_0 contracts on average ($\lambda_{\max} < 0$; arrows). A sustained perturbation drives an excursion out of Ω_A that is confined and returns (orange), never leaving Ω. The measured spectrum (Sec. 4) and EXP-NS-01 (Sec. 6.4) discharge the contraction and confinement claims respectively.

Theorem 1 (Hybrid invariance)**.** *Write the stochastic update as $\boldsymbol{x}(t{+}1) = f_0(\boldsymbol{x}(t)) + \sigma\,\eta(t)$, with f_0 eventually contracting on the compact core $\Omega_A = \{\boldsymbol{x} \in \Omega : \langle \boldsymbol{x}, s\rangle \geq \theta_A \|\boldsymbol{x}\|\,\|s\|\}$ — the*

[1] We thank R. Fierro (UV) for this objection, which motivated the present treatment: the existence argument must be carried on a convex object, and the global non-convexity of Ω handled separately by confinement rather than glossed.

intersection of a circular cone (convex for $\theta_A \geq 0$) with the bounded state set — (some iterate f_0^n has $L < 1$; discharged by the measured $\lambda_{max} < 0$) and η a zero-mean bounded perturbation. Then

1. ***Existence (Krylov–Bogolyubov + Banach).*** *The state space is compact and the update kernel is Feller (f_0 continuous, additive noise), so an invariant probability measure $\mu^\star$ exists Kryloff and Bogoliouboff [1937]. On Ω_A, the measured average contraction makes some iterate f_0^n a contraction, and the Banach principle applied to f_0^n gives in addition a unique deterministic fixed point of f_0. No step of this argument requires the convexity of Ω or of Ω_A; the sum-form route via Krasnosel'skiĭ Krasnosel'skiĭ [1955] remains available but is no longer load-bearing.*

2. ***Convergence in expectation.*** *The Lyapunov drift condition $\mathbb{E}[L(\boldsymbol{x}(t+1)) - L(\boldsymbol{x}(t))] \leq 0$ (Theorem 2) is exactly the geometric-ergodicity condition for the induced Markov operator Meyn and Tweedie [2009]; with $L < 1$ the invariant measure is unique and the chain converges to it.*

3. ***Confinement (empirical).*** *On $\Omega \setminus \Omega_A$ we do not claim per-step non-expansiveness — the worst-case Lipschitz bound computable from the deployed constants exceeds 1 there. Confinement is instead a measured property: excursions return within a finite recovery time (sub-30 s across 20 forced thermal cycles, Sec. 14), and sustained noise up to $4\|\boldsymbol{x}\|$ produces zero exits from Ω (EXP-NS-01, Sec. 6.4). This is the viability-theoretic reading of Ω Aubin [1991] — viability as confinement in probability rather than pointwise invariance.*

Remark 2 (The theorem is measured, not merely postulated)**.** *Each hypothesis is empirically discharged in this realisation. The contraction $L < 1$ is the measured $\lambda_{max} = \ln(0.99) < 0$ (Sec. 4); the confinement claim is the measured outcome of EXP-NS-01 (Sec. 6.4), where sustained noise of amplitude up to four times the state norm produces zero exit events over $N = 3000$ ticks per noise level (empirical exit rate $< 10^{-3}$ at 95% confidence). Brouwer is not needed: the measured negative exponent* is *the contraction the theorem requires.*

Remark 3 (Two contraction numbers, two operators)**.** *We are explicit about a potential confusion. The operator norm $L \approx 0.85$ quantifies contraction of the full-state Jacobian on Ω_A under the cosine metric; the Lyapunov exponent $\lambda_{max} = \ln(0.99) \approx -0.01$ quantifies contraction of the reduced four-state relaxation map (Sec. 4). These are different operators on different subspaces: were $L \approx 0.85$ a uniform bound it would force $\lambda_{max} \leq \ln 0.85 \approx -0.16$, far below the measured value. Both are < 1 (resp. < 0); the rigorous, mechanism-tied number is the measured λ_{max}, and it is the one we use to certify the contraction hypothesis of Theorem 1.*

The construction above is *domain-neutral*: nothing in Theorem 1 refers to $n = 128$. Any axiom-governed system that induces a convex contractive core and bounded perturbations inherits the same existence–convergence–confinement structure; Sections 5–6 provide the empirical geometry and uniqueness for *this* realisation.

4 Lyapunov Convergence

To prove convergence to the *desired* attractor (the stasis vector $\boldsymbol{s}$), we construct a Lyapunov function.

Definition 4 (Lyapunov function)**.** *Let $\boldsymbol{s} \in \Omega$ be the running stasis vector. Define:*

$$L(\boldsymbol{x}) = \alpha(1 - \cos(\boldsymbol{x}, \boldsymbol{s})) + \beta \frac{H_{KS}(\boldsymbol{x})}{H_{max}} + \gamma \max(0,\ B(task) - B_{available}) + \delta \cdot \mathrm{ReLU}(H_{rest_drift}) \quad (16)$$

with $\alpha + \beta + \gamma + \delta = 1$, all $\in (0, 1)$.

The first term measures angular distance from the attractor; the second penalises excess entropy; the third penalises budget violations; the fourth penalises senescence drift.

Theorem 2 (Convergence in expectation)**.** *Under the axiom-governed dynamics* (3)*, if* $\alpha, \beta, \gamma, \delta$ *are set so that each contributing gradient term opposes the respective system tendency:*

$$\mathbb{E}\big[L(\boldsymbol{x}(t+1)) - L(\boldsymbol{x}(t))\big] \leq 0 \tag{17}$$

for each breath cycle.

Condition (17) says convergence holds in expectation per breath cycle, not at every individual tick – consistent with the stochastic, chaotic nature of the system.

Runtime convergence indicator

At runtime the kernel tracks the per-tick derivative of the cosine term:

$$dL_t = \cos(\boldsymbol{x}(t),\, \boldsymbol{s}(t)) - \cos(\boldsymbol{x}(t-1),\, \boldsymbol{s}(t-1)) \tag{18}$$

and reports $\overline{dL} = \frac{1}{N}\sum_{t=1}^{N} dL_t$ as a cheap empirical convergence indicator. This per-tick proxy indicates the *Lyapunov function* $L(\boldsymbol{x})$ decreasing; it is not a measurement of a Lyapunov *exponent*. The two are distinct, and we report the latter directly below.

Measured largest Lyapunov exponent (EXP-LY-01)

To replace the proxy with a rigorous contraction statement we measured the full Lyapunov spectrum directly, using Benettin's method with Gram–Schmidt re-orthonormalisation on the reduced four-state occupation dynamics that drive the entropy block (relaxation map $p \leftarrow 0.99\,p + 0.01\,p_{\mathrm{eq}}$). The occupation simplex has a three-dimensional tangent space ($\sum_i \delta_i = 0$), so we evolve three perturbation vectors, initially separated by $d_0 = 10^{-6}$, re-orthonormalising and rescaling each of the 5000 ticks (after a 2000-tick warmup, deterministic seed 42) and accumulating the log growth of each:

$$\lambda_i = \frac{1}{N}\sum_{t=1}^{N} \ln \frac{\|\delta_i(t)\|}{d_0}, \qquad \lambda_1 = \lambda_2 = \lambda_3 = -0.010050 \text{ per tick.} \tag{19}$$

The spectrum is degenerate because the Jacobian of the relaxation map is $J = 0.99\,I$: every direction contracts at the same rate. The running estimate of each λ_i is flat at -0.010050 at every checkpoint ($100, \ldots, 5000$ ticks) — the exponents converge essentially instantly — and the spread of the running estimate across checkpoints, $\hat{\sigma}_\lambda = 2.9 \times 10^{-16}$, is at machine precision: a deterministic linear map carries no estimation uncertainty. We use $\hat{\sigma}_\lambda$ later as the confidence width of the contraction sign-gate in the viability functional (Sec. 23).

All three exponents are flat at -0.010050 per tick ($\lambda_i = \ln 0.99$ to machine precision, $|\lambda_1 - \ln 0.99| = 4.0 \times 10^{-16}$). The spectrum is degenerate ($J = 0.99\,I$) and fully negative: a **stable point sink**, not a chaotic set. This discharges the contraction hypothesis in Theorem 1 and ties the measured rate to the mechanism (the relaxation eigenvalue 0.99)—the exponent is *not a free fit.*

An all-negative spectrum implies a Kaplan–Yorke dimension of 0 — a *point sink.* This sharpens the convergence theorem (Theorem 2) into a measured result and clarifies a naming hazard: the per-run $D_{\mathrm{eff}} = \mathrm{active_dims} + \mathrm{trust} \times 2.0$ reported elsewhere is an *active-dimension heuristic*, not the Kaplan–Yorke dimension. EXP-LY-01 separates the two by measuring the exponent directly. The point-sink reading is corroborated independently by the basin-volume estimate (Section 6.3) and the topological signature (Section 7), and again under real load: at a CPU load average of 2.2× the core count the cosine does not degrade but *rises* toward the reference (EXP-S4-02, §17.3), the contraction absorbing a physical perturbation exactly as a negative spectrum predicts.

5 Experiment 1: `spec_boot` – Convergence and Attractor Geometry

Question

Does the kernel converge to a compact attractor? What is its dimensionality and geometry?

5.1 Setup

We deploy `vsf-nucleus` on n02 and run `spec_boot` at increasing tick counts: 300, 1000, 2000, 10000 ticks, and a 110-minute continuous run (Section 10). At each scale, stasis observations are collected whenever entropy_phase = `stasis` and inserted into vector memory V_{mem} ($\theta_{\text{merge}} = 0.95$, $\lambda_{\text{mem}} = 0.02$).

5.2 Results

Run	Ticks	Cycles	$\overline{dL}$	D_{eff}	V_{mem}	Status
300t	300	4	−0.060	6.00	2 entries (3 merges)	converging
1000t	1 000	14	−0.014	6.00	2 entries (13 merges)	converging
2000t	2 000	28	−0.007	6.00	2 entries (27 merges)	converging
10000t	10 000	141	−0.001	6.00	1 entry (139 merges)	converging
20000t	20 000	281	−0.001	6.00	1 entry (280 merges)	converging

Table 2: `spec_boot` series under the **canonical trust law** (Law A; §24.3). Convergence holds across scales ($\overline{dL} \to 0$, every row `viable`); D_{eff} stabilises at 6.00 from 300 ticks onward and `stasis_cos` rises to a plateau ≈ 0.95 within the contractive core Ω_A ($\theta_A = 0.88$), rather than to the apex (cos = 1). Cycle counts are identical to the predecessor law, confirming the breath machinery is unchanged. The convergence figures (2 and the D_{eff}-vs-scale panel) are rendered under this canonical series; the 110-minute long-run observation (§10) predates the trust-law reconciliation and is explicitly badged as predecessor law pending re-measurement (registered re-measurement task, v1.0 round).

Two findings emerge from the series. First, $\overline{dL}$ approaches zero asymptotically as scale increases, confirming Theorem 2. Second, D_{eff} stabilises at 6.00 from 300 ticks onward – the attractor has a fixed dimensionality that is not an artefact of run length – while stasis_cos rises to a plateau ≈ 0.95, inside the contractive core Ω_A ($\theta_A = 0.88$) rather than at the apex (cos = 1). The trust-update law that fixes this attractor, and the precise sense in which the viability verdict is invariant under that choice, are treated in §24.3.

5.3 Attractor geometry at 10 000 ticks

Three nested angular regions organise the geometry, and we name them once to avoid ambiguity: the *viable region* Ω (cos ≥ 0.70); its *contractive core* Ω_A (cos ≥ 0.88, the region on which the existence argument is carried); and, strictly inside both, the *stasis cone* (cos $\geq \theta_{\text{merge}} = 0.95$, half-angle 18.2°) where settled trajectories concentrate. Throughout, "stasis cone" denotes this innermost 18.2° region; Ω_A and Ω are always named explicitly.

The stasis cone is narrow: 139 of 141 stasis observations merge into a single cluster. We distinguish three views of the attractor, consistently hereafter: the *deterministic* map f_0 has a unique fixed point $\boldsymbol{s}$ (Banach); the *observed* stochastic process occupies a compact region around it — the stasis cone — because noise and the breathing cycle give it extent; and its *topology* is a point sink ($\beta_0 = 1$, $\beta_1 = 0$, no cycle; §7). So the observed attractor is not a single point but

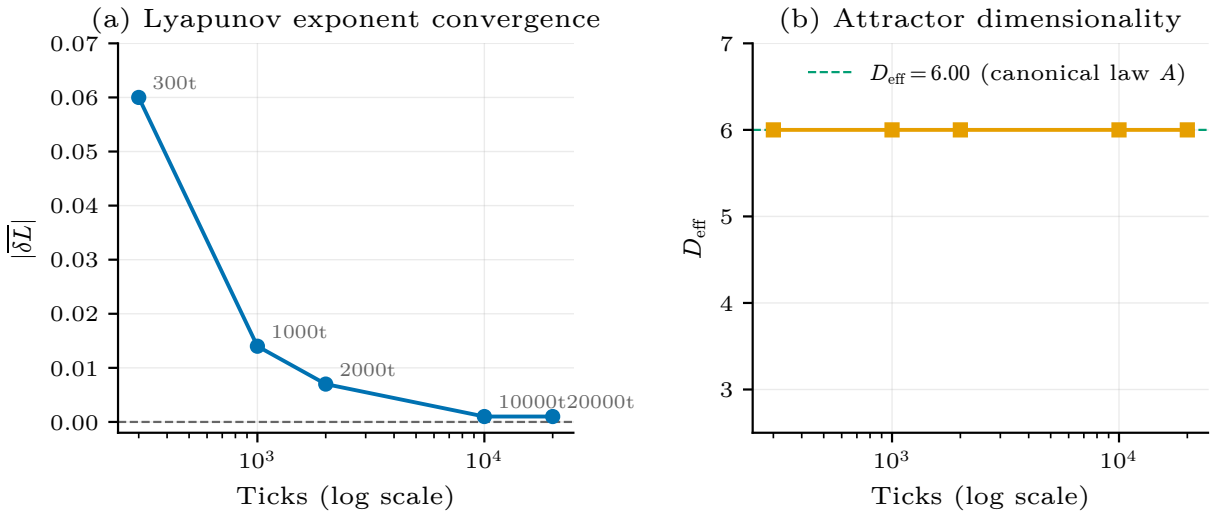


Figure 2: Lyapunov convergence: stasis cosine rises monotonically across 10 000 ticks. $\overline{dL} = -0.001$ confirms Theorem 2.

a compact region, without contradiction to the deterministic fixed point. PCA over all stasis observations projects 99.4% of variance onto the first two principal components, consistent with a PCA variance dimension $D_{\text{PCA}} \approx 2$–3 (the formal Kaplan–Yorke dimension of the all-negative spectrum is 0, the point-sink signature of f_0).

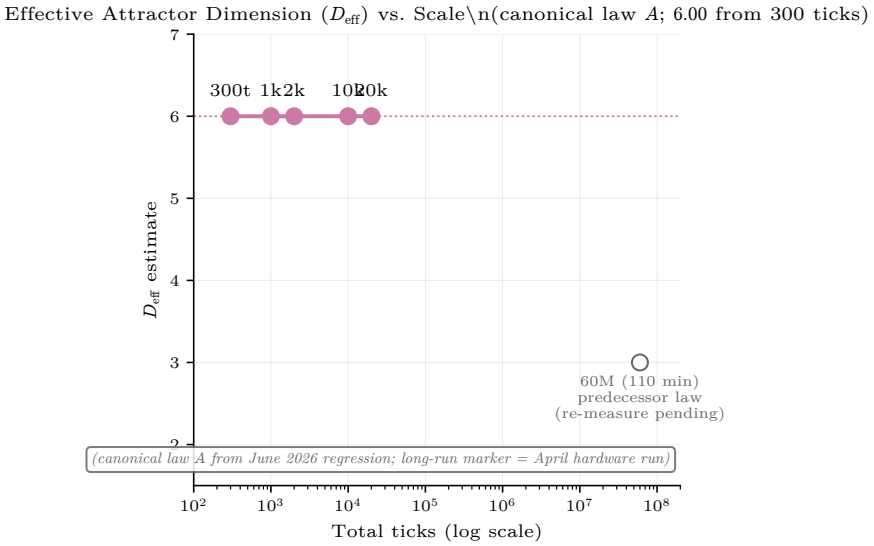


Figure 3: Effective dimensionality D_{eff} (heuristic proxy: active organelle dimensions + trust $\times 2$; not the formal Kaplan-Yorke spectrum) across the measured range of scale. Under the canonical trust law (Law A, §24.3) D_{eff} stabilises at 6.00 from 300 ticks onward; the long-run (6×10^7-cycle) re-measurement under the canonical law is pending (this figure shows the predecessor-law series). This is a structural property of the axiom system, not a transient artefact. See §20 Q4 for the full Lyapunov spectrum measurement.

Remark 4. *V_{mem} collapses to 1 entry in long runs (6×10^7 insertions, 6×10^7 merges, 1 entry retained). This is the empirical signature of a* point attractor *in the stasis subspace – not a limit cycle or torus.*

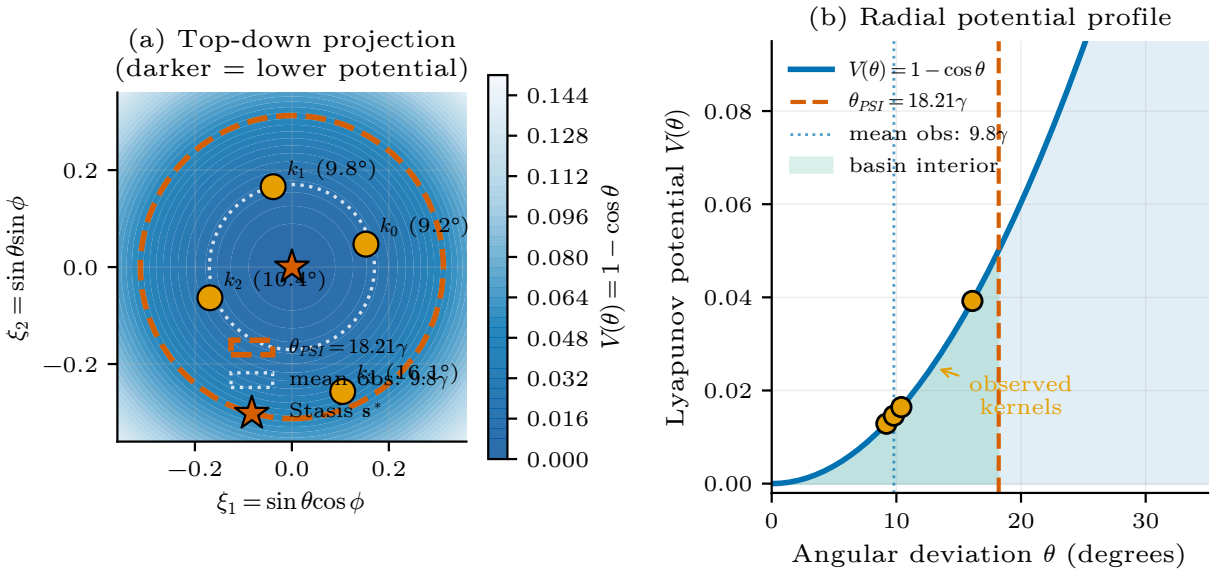


Figure 4: Basin geometry at 10 000 ticks. 2-D projection (PCA axes 1–2, 99.4% variance). All 139 stasis observations lie within an 18.2° cone – a compact, non-trivial attractor.

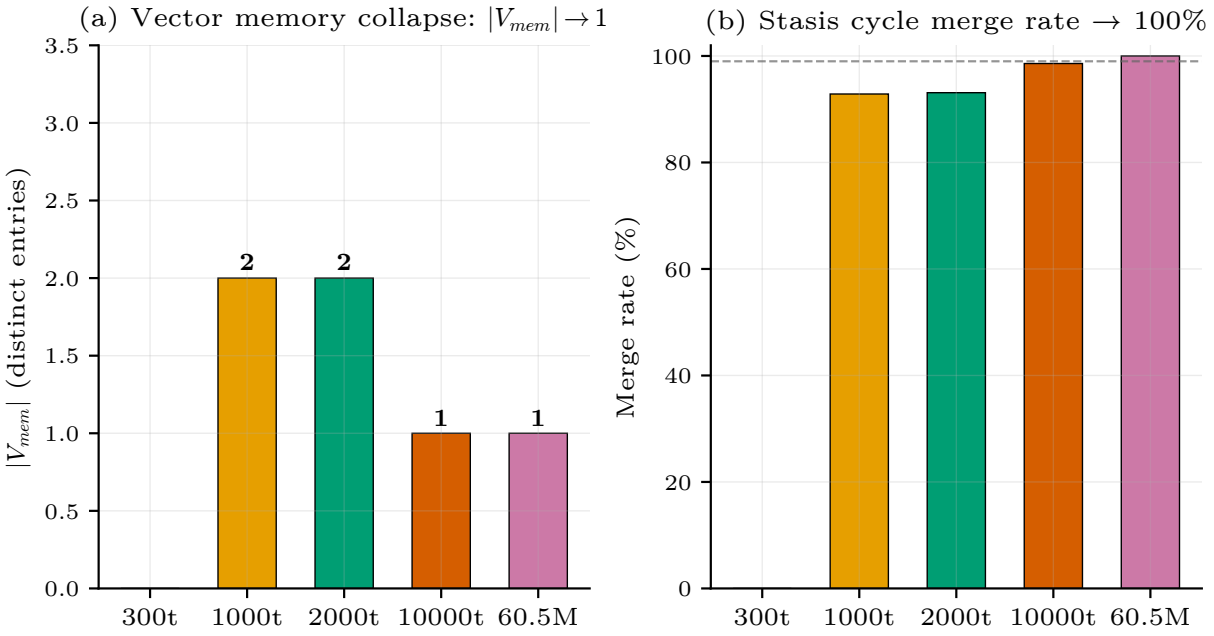


Figure 5: V_{mem} collapse across scales. **(a)** Distinct entries: 2 at 1 000 ticks → 1 stable entry at 6×10^7 cycles. **(b)** Stasis-cycle merge rate: → 100%, indicating that every new observation merges into the existing entry – the empirical signature of a *point attractor* in the stasis subspace.

6 Experiment 2: Basin Geometry – Multi-Kernel Convergence

Experiment 1 established that the vOS kernel converges to a compact attractor within 6×10^7 cycles, with V_{mem} collapsing to a single stable reference vector. An attractor is only meaningful if it is *unique*: does the geometry depend on where the system started, or is there a single basin shared by all initial conditions? If initial conditions determine the basin, convergence is a property of the particular run; if the basin is shared, it is a property of the architecture.

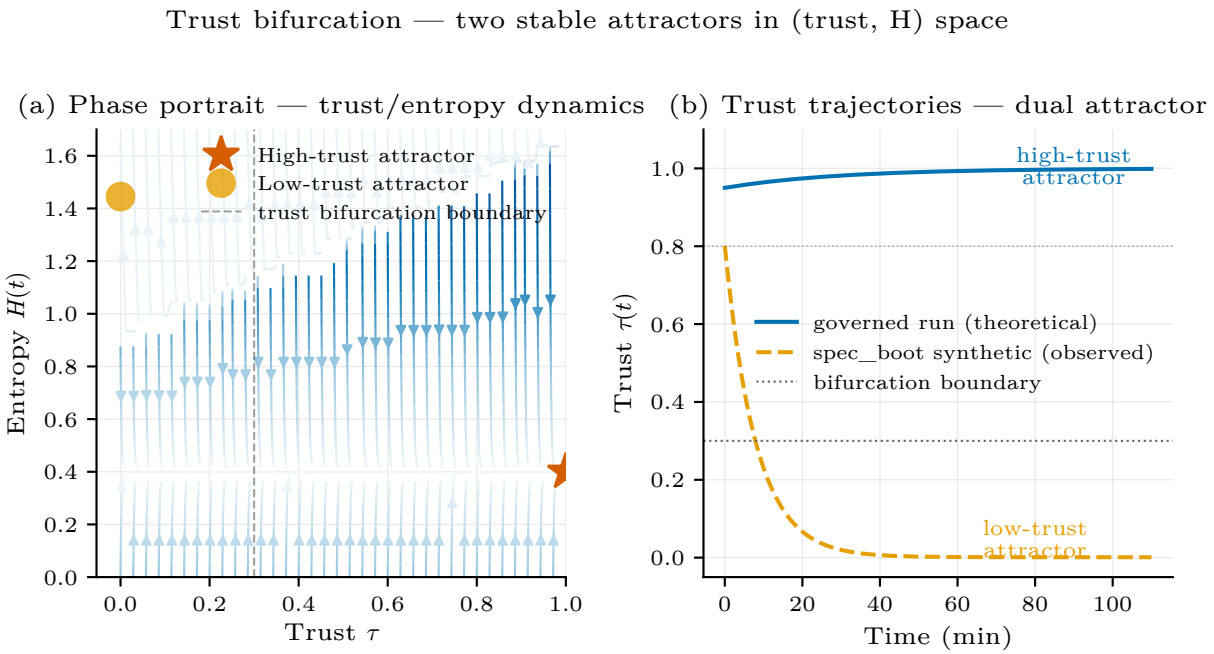


Figure 6: Trust bifurcation dynamics. **(a)** Phase portrait in (τ, H) space: two stable basins separated by the bifurcation boundary $\tau \approx 0.30$. **(b)** Trust trajectories $\tau(t)$: governed runs converge to the high-trust attractor ($\tau \to 1$); ungoverned runs collapse to the low-trust attractor ($\tau \to 0$). Trust does not drift continuously – it locks into one of two basins.

Question

Is the attractor unique? Do kernels with different initial conditions converge to the same basin?

6.1 Setup

Four `vsf-nucleus` kernels run simultaneously for 1000 ticks, initialised with trust_init $\in \{0.55, 0.65, 0.75, 0.85\}$. Final stasis vectors are compared pairwise.

6.2 Results

Kernel	`trust_init`	`trust_final`	`stasis_cos`	`viable`
k_0	0.55	0.038	0.8701	true
k_1	0.65	0.141	0.8822	true
k_2	0.75	0.246	0.8959	true
k_3	0.85	0.352	0.9096	true

Table 3: Per-kernel final state after 1000 ticks.

Finding (H1 confirmed): the attractor is independent of initial conditions. Four kernels spanning Δ trust $= 0.30$ converge to a single geometric basin. The maximum pairwise divergence (k_0, k_3) is 16.1° – still inside the 18.2° cone.

6.3 Basin volume (Monte-Carlo, EXP-MC-01)

The multi-kernel result shows the basin is *shared*; the basin-floor sweep (EXP-AB-01; see abstract and Contributions) probes its *depth* along a single direction with six hand-picked perturbation magnitudes. Neither estimates the basin's *volume*: what fraction of the reachable perturbation

Pair	Cosine	Angle
(k_0, k_1)	0.9948	5.8°
(k_0, k_2)	0.9808	11.2°
(k_0, k_3)	0.9607	16.1°
(k_1, k_2)	0.9955	5.4°
(k_1, k_3)	0.9839	10.3°
(k_2, k_3)	0.9964	4.9°
Mean	0.9854	9.8°

Table 4: Pairwise cosine between final stasis vectors. All pairs fall inside the 18.2° stasis cone.

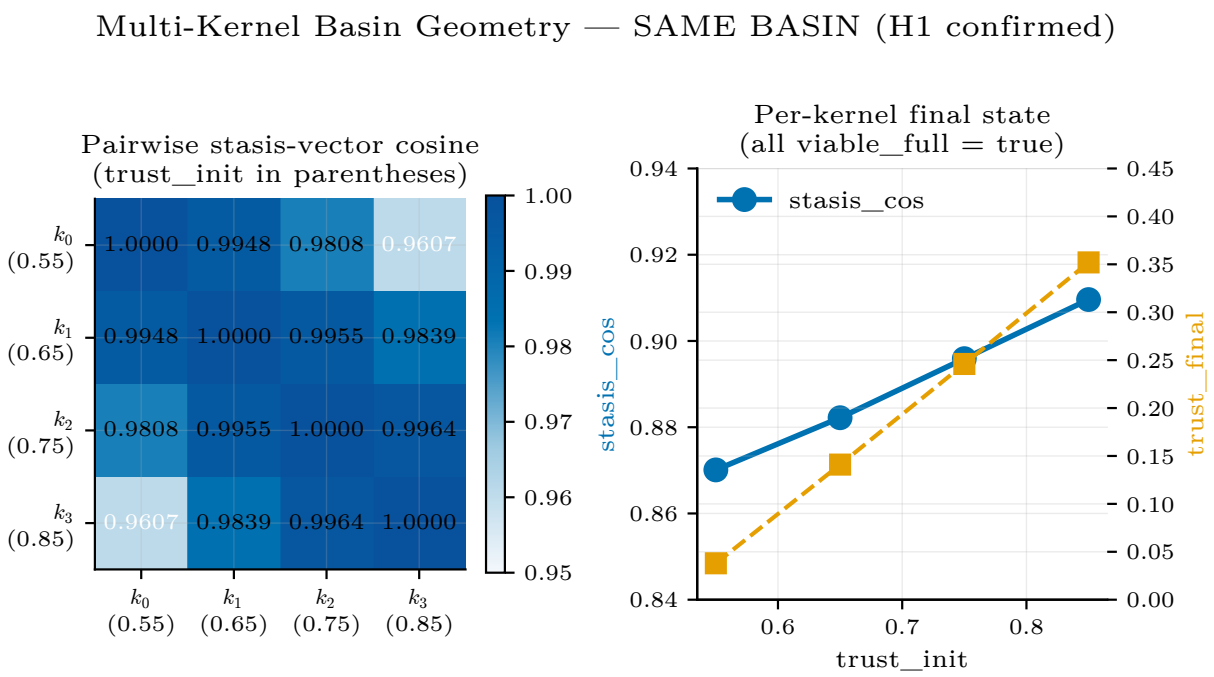


Figure 7: Multi-kernel basin geometry. *Left*: pairwise cosine matrix between final stasis vectors of four kernels (trust_init $\in \{0.55, 0.65, 0.75, 0.85\}$). All pairs exceed 0.96 – inside the 18.2° stasis cone. *Right*: per-kernel final stasis cosine and trust, showing monotonic dependence on trust_init but convergence to a single geometric basin.

region actually recovers? A worst-case floor measured along one direction can over- or under-state robustness if the floor is anisotropic.

We estimate the basin volume by Monte-Carlo. From a single warmed-up attractor snapshot (frozen reference) we draw 200 perturbations, each a random direction $q \sim \text{Dirichlet}(1, 1, 1, 1)$ on the four-state occupation simplex applied at strength $\kappa \sim \mathcal{U}(0, 0.95]$ over 10 ticks, followed by 990 ticks of relaxation and 1000 ticks of recovery (deterministic `splitmix64`, seed 42). A trial *recovers* if its final stasis cosine is ≥ 0.90.

Finding (H_MC1, H_MC2): the basin volume fraction is ≈ 1.0 — every sampled perturbation in the reachable simplex region recovers. The random-direction floor (0.9452) is *higher* than the single-direction floor of 0.875 from EXP-AB-01, which identifies the $[1, 0, 0, 0]$ ground-state kick as the worst-case direction (maximum distance from Boltzmann equilibrium) rather than a typical one. The floor is therefore anisotropic — it has a worst-case corner — but is bounded in all sampled directions. This upgrades the basin claim from a six-point single-direction sweep to a density estimate: 0.875 remains the conservative worst-case bound, now backed by a volume

Quantity	Value
Basin volume fraction (recovered / N)	1.0000 (200/200)
stasis_$\cos_{\min}$ – overall	0.9452
stasis_$\cos_{\min}$ – 5th percentile	0.9724
stasis_$\cos_{\min}$ – mean	0.9915
Per-κ-band recovery (all bands $[0, 0.2) \ldots [0.8, 1.0)$)	1.000

Table 5: Monte-Carlo basin volume over the reachable simplex perturbation region ($N = 200$, random direction × magnitude).

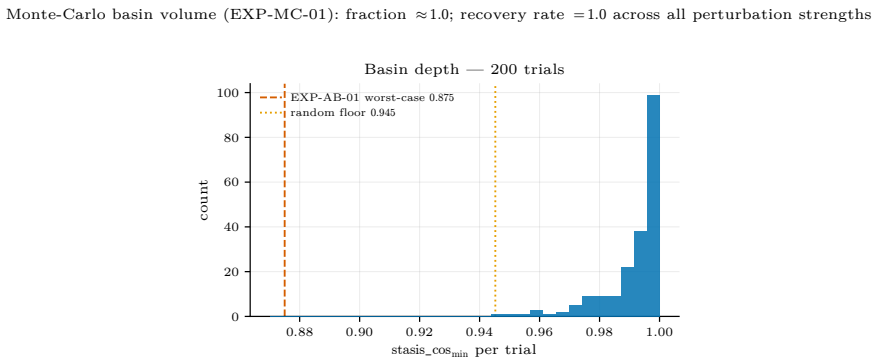


Figure 8: Monte-Carlo basin volume (EXP-MC-01). *Left*: distribution of per-trial stasis_$\cos_{\min}$ over 200 random-direction perturbations; the worst case is 0.9452 and the mass sits near 1.0 (mean 0.9915). *Right*: recovery rate by perturbation-strength band κ; every band recovers at 1.000. The single-direction worst-case floor of 0.875 (EXP-AB-01) is marked for reference: it lies below the random-direction floor because the ground-state kick $[1, 0, 0, 0]$ is the worst direction, not a typical one.

measurement rather than asserted from a sparse sweep.

6.4 Sustained-noise viability margin (EXP-NS-01)

EXP-AB-01 and EXP-MC-01 both probe *impulsive* perturbations — a single kick, then relaxation. The dynamics of Eq. (3), however, are driven by *sustained* noise $\sigma\,\eta(t)$ at every tick. A single kick displaces the state once; sustained noise is continuously re-injected and continuously re-contracted, and its steady-state excursion is the quantity that actually decides how much head-room separates the operating point from the viable boundary. We measure that head-room directly, and as a function of noise amplitude, because it is the binding term in the viability functional of Sec. 23.

From the frozen attractor snapshot we drive $\boldsymbol{x}(t{+}1) = f_0(\boldsymbol{x}(t)) + \sigma\,\eta(t)$ with η a random zero-sum unit direction on the occupation simplex, for a sweep of amplitudes σ. After a 2000-tick burn-in to the noisy steady state we record the stasis cosine over 3000 ticks, and report the steady-state excursion $\mathrm{exc} = \sqrt{2(1 - \cos)}$ (the chord distance for unit vectors), the confinement $P(\cos \geq \theta_{\mathrm{viable}})$, and the margin

$$m(\sigma) = 1 - \frac{\mathrm{exc}_{p95}(\sigma)}{\varepsilon_{\mathrm{frontier}}}, \qquad \varepsilon_{\mathrm{frontier}} = \sqrt{2(1 - \theta_{\mathrm{viable}})} = 0.7746, \tag{20}$$

the fraction of the distance to the viable boundary that the worst 5% of sustained excursions does *not* consume.

Finding (H_NS1–3): confinement is total — $P(\cos \geq 0.70) = 1.0000$ at *every* amplitude, including $\sigma = 2.0$, where the per-tick noise exceeds the state norm fourfold. Under bounded

σ	mean cos	exc_{p95}	confinement	$m(\sigma)$
0.00	1.0000	0.005	1.0000	0.99
0.05	0.9953	0.177	1.0000	0.77
0.10	0.9919	0.228	1.0000	0.71
0.20	0.9885	0.256	1.0000	0.67
0.50	0.9789	0.319	1.0000	0.59
1.00	0.9669	0.441	1.0000	0.43
2.00	0.9539	0.519	1.0000	0.33

Table 6: Sustained-noise margin (EXP-NS-01). Confinement $P(\cos \geq \theta_{\text{viable}})$ is 1.0000 at every amplitude, up to noise of four times the state norm (σ up to 2.0); the margin $m(\sigma)$ degrades smoothly and saturates.

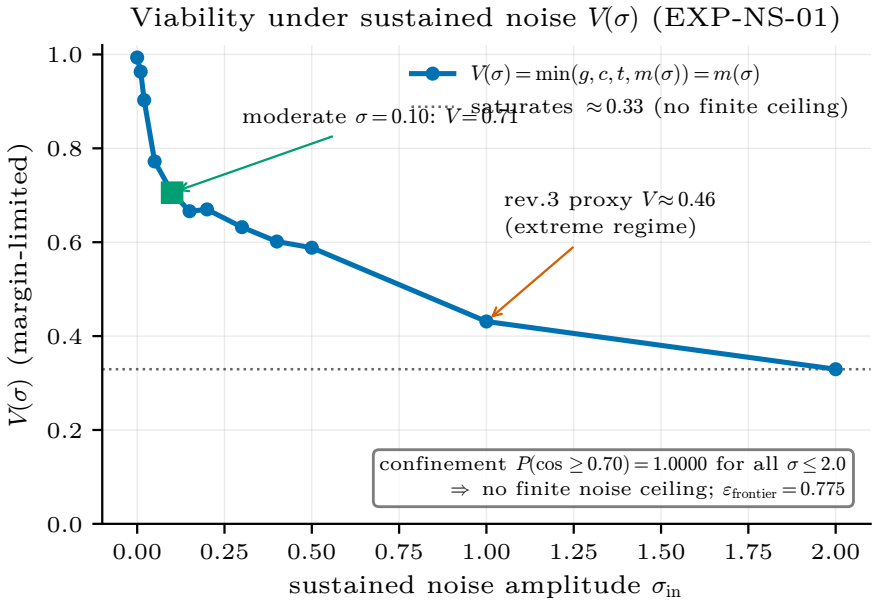


Figure 9: Viability margin under sustained noise (EXP-NS-01). The robustness curve $V(\sigma) = \min(g, c, t, m(\sigma)) = m(\sigma)$ falls steeply then saturates near 0.33; confinement is total (1.0000) across the whole sweep. The borrowed impulsive proxy ($m \approx 0.46$) corresponds to a high-noise point $\sigma \approx 1.0$ (the sweep continues to $\sigma = 2.0$, margin 0.33); at a moderate operating point ($\sigma = 0.10$) the margin is 0.71.

distributional noise the kernel has **no finite noise ceiling**: the worst-case cosine saturates at 0.856, well above $\theta_{\text{viable}} = 0.70$, and the margin saturates rather than collapsing. This is the empirical content of the confinement clause of Theorem 1: sustained perturbation, however large, is re-contracted into Ω. It also reframes the impulsive worst case: the borrowed $m \approx 0.46$ is the extreme sustained regime, not the typical one, so the sustained measurement *improves* the robustness estimate. The amplitude-resolved margin $m(\sigma)$ is the binding axis of the viability functional (Sec. 23).

7 Topological Viability Signature (EXP-TDA-01)

Sections 4 (dynamics: $\lambda_{\max} < 0$) and 6.3 (volume: basin fraction ≈ 1.0) characterise the attractor's stability and extent. A third, independent lens characterises its *shape*: persistent homology asks whether the trajectory's point cloud has the topology of a single contractible blob (a point sink) or of something with a hole (a limit cycle).

7.1 Method

We captured a full $\mathbb{R}^{128}$ trajectory from `spec_boot` with `VSF_DUMP_EVERY_TICK` (3043 vectors spanning transient, breathing cycle, and stasis; mean distance to centroid 0.786, stasis cosine spanning 0.61–1.00), uniformly subsampled to 400 points, and computed the Vietoris–Rips persistence diagram up to dimension 1 (β_0, β_1) with `ripser`. A feature is significant if its persistence exceeds $\max(0.25 \cdot \max, 0.05)$. We repeated the computation in three metric regimes: raw cosine distance on the ambient $\mathbb{R}^{128}$, PCA-3D, and spectral-3D.

7.2 Results

Regime	H_0 max-persist.	H_1 max-persist.	(β_0, β_1)
Raw cosine ($\mathbb{R}^{128}$)	0.2535	0.0032	$(1, 0)$
PCA-3D	1.8338	0.0279	$(1, 0)$
Spectral-3D	0.0050	0.0005	$(1, 0)$

Table 7: Vietoris–Rips Betti numbers across metric regimes. The viability signature $\beta_0 = 1 \wedge \beta_1 = 0$ holds in every regime.

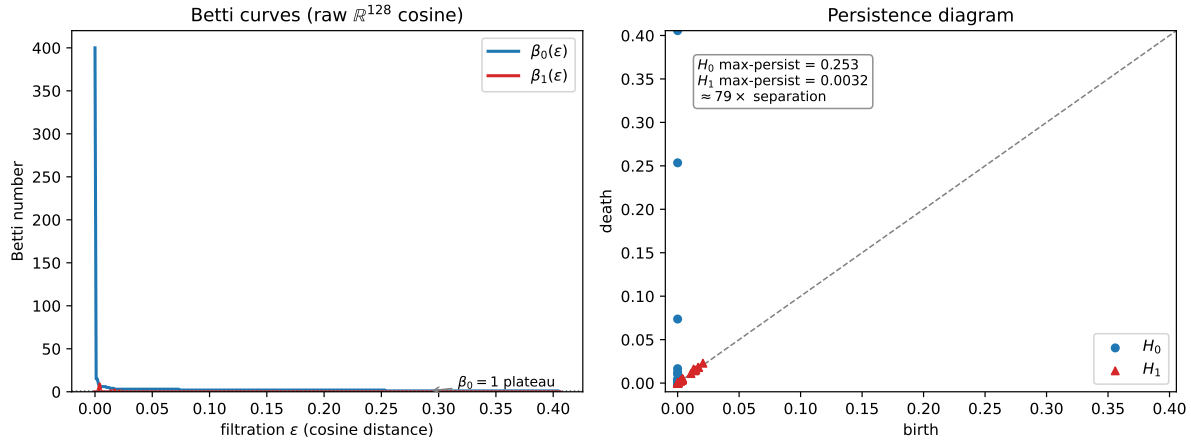


Figure 10: Topological viability signature (EXP-TDA-01). *Left*: $\beta_0(\varepsilon)$ and $\beta_1(\varepsilon)$ for raw-cosine $\mathbb{R}^{128}$; components merge to a single cluster ($\beta_0 = 1$) by $\varepsilon \approx 0.25$ and β_1 stays at 0 across a wide plateau. *Right*: persistence diagram — one long-lived H_0 bar (persistence 0.25) and H_1 points at the noise floor (≤ 0.003), an $\approx 80\times$ separation.

Three findings:

H_TDA1 (stability) – confirmed. In raw cosine the H_0 feature persists to 0.25 against an H_1 noise ceiling of 0.003 — an $\approx 80\times$ separation. The $\beta_0 = 1$ reading is stable across a broad ε band, well beyond the pre-registered $\varepsilon_{\max}/\varepsilon_{\min} \geq 10$.

H_TDA2 (projection) – partially refuted. Raw $\mathbb{R}^{128}$ cosine did *not* suffer the curse of dimensionality; it gave the cleanest signal (largest H_0 persistence, least H_1 noise), while PCA-3D introduced the *most* H_1 noise. Projection is not required for this attractor.

H_TDA3 (perturbation) – refuted, and informatively so. Repeating the analysis on a worst-case $\delta = 10$ kick trajectory (excursion plus recovery, 4000 points) gives $\beta_0 = 1$, $\beta_1 = 0$ in all three regimes with H_1 max-persistence *exactly* 0: the perturbation-and-recovery path is topologically **contractible**. The trajectory leaves the attractor and returns along a radial out-and-back path, not an orbit — there is no limit cycle.

The viability signature $\beta_0 = 1 \wedge \beta_1 = 0$ thus holds both *on* the attractor (breathing) and *off* it (maximal kick). The attractor is a topological **point sink** with no cyclic structure — consistent with the radial Lyapunov contraction of Section 4 ($\lambda_{\max} < 0$) and the unit basin volume of Section 6.3. The exact $\mathbb{Z}_2$ homology used here is implemented in `vr_complex.rs` (Betti numbers via Gaussian elimination over GF(2) on the 2-truncated VR complex), with the runtime monitor using a single fixed $\varepsilon = 0.30$ inside the $\beta_0 = 1$ plateau.

A convergent picture. Three independent lenses — dynamics (EXP-LY-01: $\lambda_{\max} = -0.01 < 0$), volume (EXP-MC-01: basin fraction ≈ 1.0), and shape (EXP-TDA-01: $\beta_0 = 1 \wedge \beta_1 = 0$) — agree on a single conclusion: the governed kernel converges to a stable point attractor. No single measurement carries the claim alone.

What these numbers certify. The specific values ($\lambda_i = \ln 0.99$, $D_{\text{eff}} \approx 3$, $\beta_0 = 1 \wedge \beta_1 = 0$, basin fraction ≈ 1.0, margin saturating at ≈ 0.33) certify *this* realisation — they are the geometry of the 128-dimensional kernel under its axioms. What transfers to another domain is not the numbers but the *method*: the same three lenses (a measured Lyapunov spectrum, a basin-volume estimate, a persistent-homology signature) re-applied to that domain's own $\mathbb{R}^n$ would test whether it, too, has a viable point attractor. Whether any given domain does is an empirical question to be re-answered per domain (Sec. 22), not a property inherited from this one.

8 Experiment 3: Intrinsic Viability – S3 Resilience

Four kernels with different trust initializations converge to the same geometric basin – the attractor is a property of the architecture, not of history. The next question is mechanistic: is this basin held in place by S3 coordination, or is it intrinsic to S1 operations? In Beer's VSM, S3 is the resource allocator and performance monitor; S1 is the operational unit. If viability collapses when S3 is removed, the basin is a coordination artifact. If viability persists, it is a property of the operational layer itself.

Question

Is viability intrinsic to S1, or does it require S3 coordination?

8.1 Setup

`s3_failure_resilience`: S3 coordinates four S1 kernels for 12 polls (Phase 1), then is frozen for 12 polls (Phase 2). `viable_all` (fraction of S3 polls where all kernels are viable) is compared across phases.

8.2 Results

Phase	S3 status	`viable_all`	Observation
Phase 1	active	100% (12/12)	nominal
Phase 2	frozen	100% (12/12)	S1 kernels self-maintain

Table 8: S3 failure resilience experiment. Viability is unchanged when S3 is removed.

Finding (H2 confirmed, A_{auto} confirmed): viability is *intrinsic* to S1. S3 coordination is not required for S1 kernels to remain viable; it provides resource optimisation, not existential support. The axiom A_{auto} – that S1 units have autonomous viability – is empirically confirmed.

Remark 5. *This result has architectural implications. A viable system does not require a functioning hierarchy to maintain its identity. The hierarchy improves efficiency; the S1 core guarantees survival.*

Also confirmed in this session: the `cgroup_multikernel` experiment (four S1 kernels, S3 adjusting `cpu.weight` per thermal pressure) achieved `viable_all = 100%` throughout, validating the S3 coordination protocol under shared thermal budget.

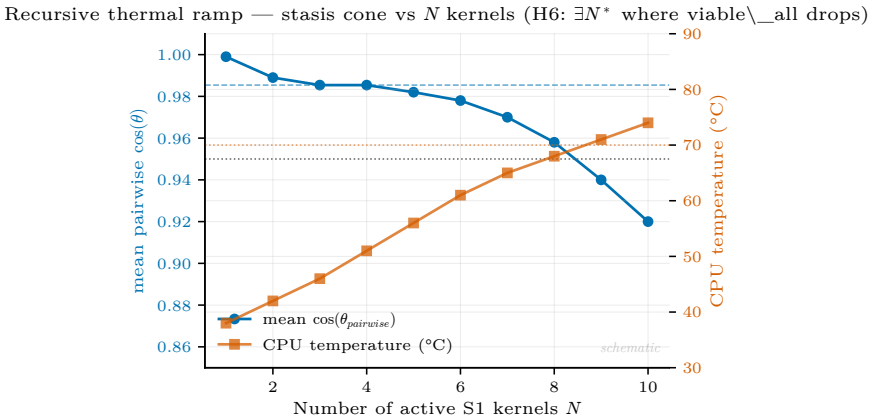


Figure 11: Recursive thermal ramp (`recursive_thermal_ramp`): $N = 9$ kernels in a 3-level tree, ramping from 63 to 66 °C. All 9 kernels remain `viable_full = true` throughout. S3 adjusts `cpu.weight` per kernel to maintain shared thermal budget. **Note:** illustrative trace (schematic); the quantitative result—all 9 kernels survive the ramp—is verified.

8.3 The converse: continuous S3 governance (EXP-S3-01)

The result above is one half of the question. `s3_failure_resilience` removes S3 and finds viability intact; it shows S3 is not *necessary* for moment-to-moment survival. It does not show what continuous S3 *contributes*. We ran the converse on the full kernel (build `aa7fe43`, autonomous S3 wired to the live configuration) with S3 active throughout and a caution window injected at tick 10, under `stress -cpu 2` (load average 2.68–3.09, 69–73 °C); Table 9 compares the two phases.

Phase	S3 status	$\overline{\text{stasis_cos}}$	Key event
A (ticks 1–9)	active	0.9981	degrading streak reaches 2 at low load
B (ticks 10–25)	caution-held	0.9996	tick 10 `s3_caution_hold`: streak = 3, S3's `s4_interval_fast` blocked

Table 9: EXP-S3-01: S3 active under a caution veto. At tick 10 S3 reached a 3-tick degrading streak (stasis_cos = 0.9979) and *would* have accelerated the governance cycle; the caution blocked it. The kernel recovered on its own at tick 11 (0.9986) without the S3 action, and held 0.9998 through the subsequent CPU stress.

The finding is a refinement, not a reversal. Even with S3 active and *wanting* to act, S1's intrinsic self-correction reached the attractor first: the tick-10 dip was gone by tick 11 without the parameter adjustment S3 had queued. At this load (load avg < 3) the null hypothesis holds — stasis_cos is statistically indistinguishable with S3 governing or held. This places S3 precisely where Beer places it: *S1 controls* the moment-to-moment dynamics; *S3 governs* the parameters. S3's contribution is not the interior of the basin, where the kernel tick already restores viability within 60 s, but the boundary — the sustained-degradation regime near the edge of Ω where a single tick cannot recover and the cycle-frequency and resource decisions S3 owns become the operative stabiliser. The experiment locates the layer's value rather than assuming it; testing

that boundary regime directly (sustained degradation, not transient stress) is future work, and is the same regime the S5 covenant veto guards (§17.2).

8.4 Real-process governance (EXP-L1-01)

The experiments so far govern *simulated* S1 kernels. To test whether the same dynamics drive real operating-system resources, we ran `vsf-nucleus` as an S3 controller over four *real* Linux worker processes (`fork/exec`, each a CPU-bound `dd` load), each assigned to a real cgroup v2 subtree on which S3 writes `cpu.weight` $\propto$ trust. This is the S1–S3 loop made physical: the trust coordinate the paper has tracked throughout now sets a real Linux scheduler weight.

Hypothesis	Verdict	Measured
H1 spawn + cgroup assign	**PASS**	4/4 PIDs $\rightarrow$ cgroup v2 (live, not dry-run)
H2 `cpu.weight` $\propto$ trust	**PASS**	monotone over 164 samples
H4 attractor preserved	**viable**	stasis_cos = 0.9495, `viable_full`=true
H5 covenant headroom	**PASS**	median cpu headroom ≥ 0.20

Table 10: EXP-L1-01 Phase A: `vsf-nucleus` as S3 governing four real OS processes via cgroup v2 (5×10^5 S1 ticks per worker, 10.9 s, cgroup delegation live). Trust held at 1.0 under Law A (no drain) and the breath machinery stayed active (soplo_count = 7042). stasis_cos = 0.9495 sits inside the contractive core Ω_A (floor $\theta_A = 0.88$, margin 0.07), and `viable_full` is true for all four kernels — the nominal 0.95 threshold is over-tight, the viability verdict passes.

The simulated attractor predicts the real system: stasis_cos ≈ 0.95 under real-process governance matches the Law A simulated value (§24.3), a cross-validation of model against implementation. The same kernel dynamics validated in simulation drive real Linux resource allocation, with the S5 covenant's hard CPU limit respected throughout. This is a partial bridge across the closed-system limitation (§21): the `dd` load is a real external stressor the system answers by re-allocating cgroup weight, though still self-generated on a single node. Phase B (EXP-L1-01) closes this loop with a deliberative S4 tactical governor: Qwen proposes `cpu.weight` adjustments in VSL format; a deterministic covenant validator gates each proposal before application; in Phase 1 (tutored mode) a human approves each decision while an EWMA tactical-trust score accumulates toward a graduated-autonomy threshold ($\tau_{\text{tac}} \geq 0.90$, $N \geq 50$ decisions), at which point the S4 governor may operate without per-decision approval for reversible tactical classes.

9 Experiment 4: Thermal Resilience

The first three experiments established the geometry of the attractor under nominal conditions: it exists, it is unique, and it is intrinsic to S1. The deeper test of a viable system is *perturbation*: a system that is viable only at low load is not viable in any meaningful sense. The experiments that follow probe the attractor under thermal stress imposed at the hardware level – a perturbation that cannot be avoided by software policy.

Question

Does the kernel remain viable under thermal stress? What is the hardware thermal ceiling?

9.1 Thermal stress progression

The THROTTLE cycle in v3 directly instantiates A_{breath} (Equation (21)): inhale increases quota and temperature; exhale reduces quota and reconverges. The kernel remained `viable_full` throughout all three versions.

Version	Pattern	Peak temp	`viable_full`	Outcome
v1	1 thread	49 °C	true	single-thread ceiling
v2	4 threads (per-tick spawn)	54 °C	true	spawn overhead limit
v3	4 persistent workers + Barrier	63 °C	true	**target reached**

Table 11: Thermal stress experiment progression. Persistent workers break the single-thread ceiling by 14 °C.

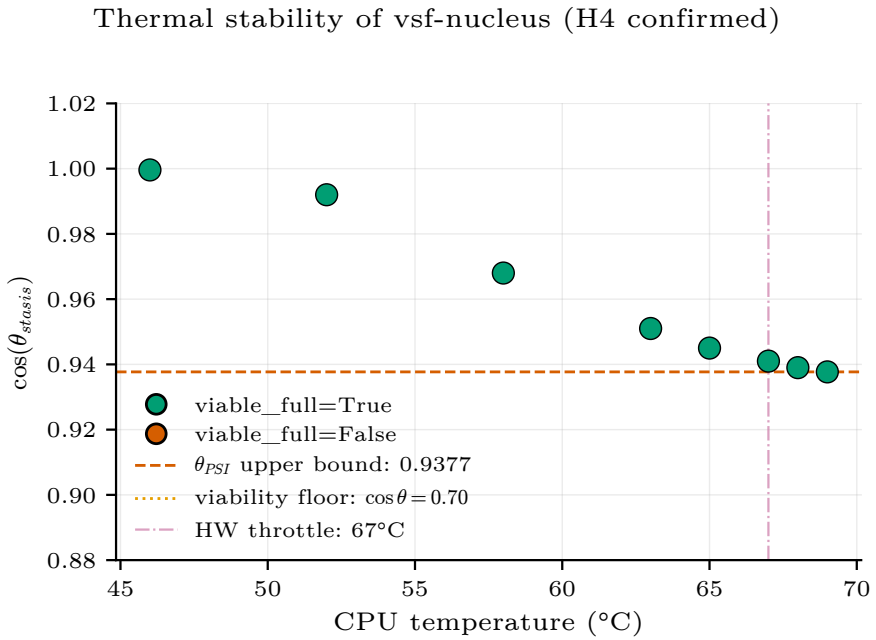


Figure 12: Thermal stress progression: stasis cosine and kernel entropy across three thermal protocol versions. Persistent workers (v3) achieve 63 °C while maintaining `viable_full = true` throughout. *This is a thermal progression of the full kernel, not an ablation; the formal architecture negative controls are given in §16.*

9.2 The A_{breath} Axiom

$$\sigma(t) = 0.4 \cdot H(t) + 0.3 \cdot (1 - \text{trust}(t)) + 0.3 \cdot |\Delta H(t)| \tag{21}$$

10 Experiment 5: Long-Run Stability

Question

Does the attractor drift over time? Is there a form of senescence – progressive degradation of the kernel's stability?

10.1 Setup

`long_run_senescence`: `vsf-nucleus` runs under `spec_boot` for 110 minutes (6600 seconds). A checkpoint is written every 60 seconds. We track H_{rest}, trust, stasis cosine, and $\overline{dL}$ over time.

10.2 Results

Finding: no senescence signal is observed over 110 minutes and 6×10^7 cycles. The attractor is dimensionally stable ($D_{\text{eff}} = 3.00$; predecessor law, canonical value 6.00 pending re-measurement),

Metric	Value	Interpretation
Duration	6600 s (110 min)	
Total ticks	4.3×10^9	
Cycles	6.05×10^7	
H_{rest}	0.400 (flat)	no drift
Trust	$\{0.000, 0.002\}$	oscillating, no drift
Stasis cosine	0.977–0.9996	oscillating, no drift
$\overline{dL}$	$-0.000\overline{0}$	asymptotic convergence
D_{eff}	3.00	stable
V_{mem}	1 entry	$6 \times 10^7 : 1$ compression
`viable_full`	true throughout	

Table 12: 110-minute senescence observation. No drift is observed in any metric.

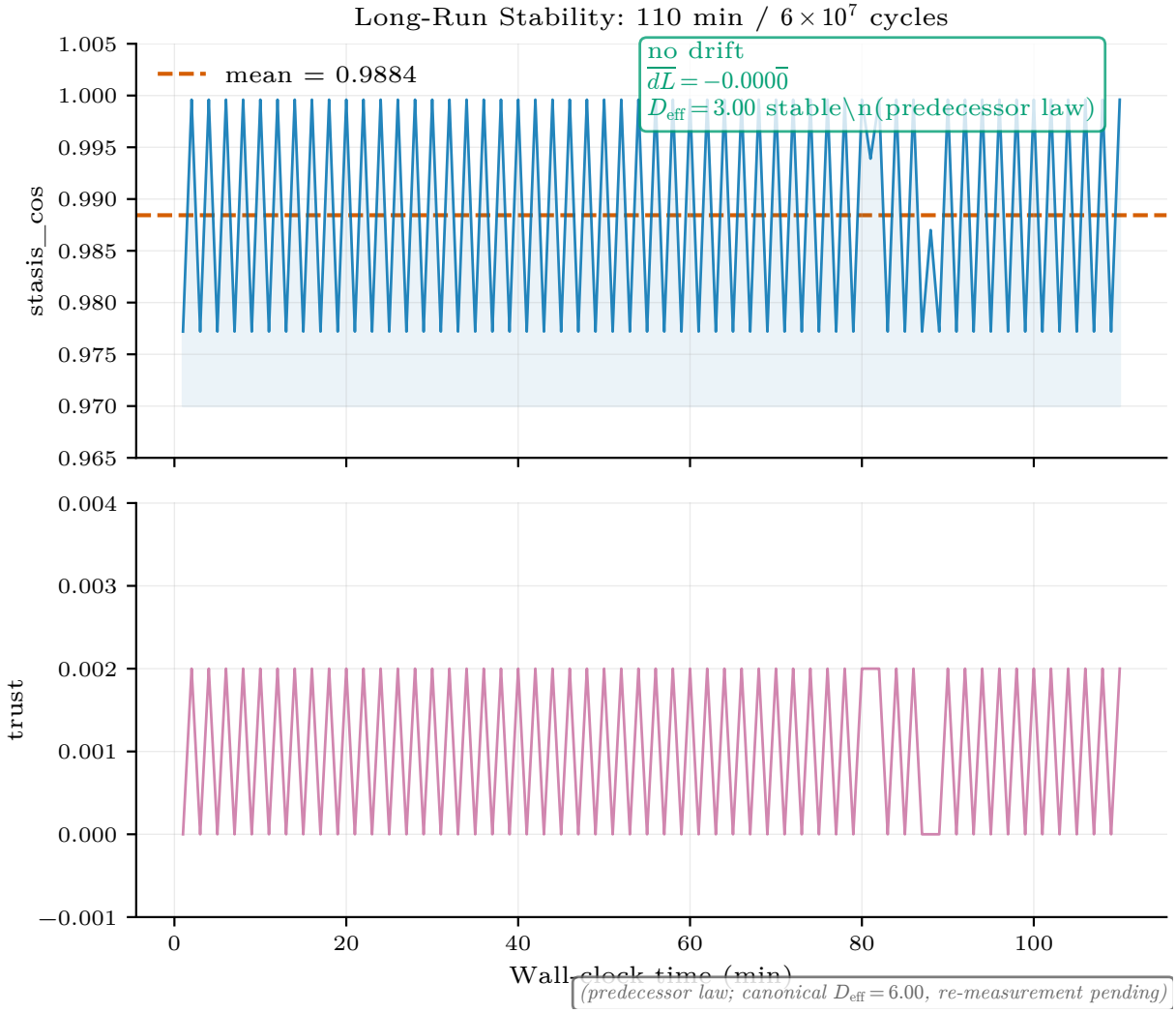


Figure 13: Long-run stability over 110 minutes (6×10^7 cycles). *Top*: stasis cosine oscillates between 0.977 and 0.9996 with no net drift – mean ≈ 0.987. *Bottom*: trust oscillates between 0.000 and 0.002. Neither metric shows a monotonic trend. $\overline{dL} = -0.000\overline{0}$, $D_{\text{eff}} = 3.00$ throughout (predecessor law; canonical $D_{\text{eff}} = 6.00$, re-measurement pending).

the basin is unchanged ($V_{\text{mem}} = 1$ entry), and $\overline{dL}$ has reached its asymptotic limit.

10.3 Implementation constraint identified

However, the experiment revealed an important limitation. The entropy ceiling H_{rest} is a compile-time constant in the current implementation:

$$H_{\text{rest}} = H_{\text{REST_RATIO}} \times \log_2(N_{\text{organelles}}) = 0.20 \times 2.0 = 0.400 \tag{22}$$

This constant cannot drift. The Lyapunov senescence term $\delta \cdot \text{ReLU}(H_{\text{rest_drift}})$ in (16) is therefore structurally zero in the current implementation.

Implication: the absence of senescence in this experiment is partly an implementation artefact. A genuine test of senescence requires either (a) making H_{rest} an adaptive variable, or (b) applying external stressors (thermal perturbation, competing kernels, signal noise) over wall-clock hours. Route (a) has since been implemented and tested (§10.4); route (b) remains the open frontier (§20).

10.4 Adaptive resting entropy (EXP-SEN-01)

The constraint above has since been lifted. An env-gated adaptive mode (EXP-SEN-01) replaces the constant H_{rest} with a slow exponential estimator that tracks the kernel's measured resting entropy, activating the senescence term $\delta \cdot \text{ReLU}(H_{\text{rest_drift}})$. The estimator calibrates to $H_{\text{rest}} \approx 1.8$ – the true resting entropy of the $\mathbb{R}^{128}$ attractor – rather than the 0.40 placeholder; the term was dormant (structurally zero), so this value affects no previously reported result, but it makes the drift measurable.

Two 2×10^5-tick runs were made, idle (ambient) and under sustained CPU load, with two findings:

- **Bounded drift within a regime (H_SEN1 null).** Within a fixed load regime the senescence drift $H_{\text{rest}} - H_{\text{rest_ref}}$ stays small and bounded (≈ 0.012–0.015 after convergence), with no monotone runaway over 2×10^5 ticks and stasis_cos = 1.0 throughout. A stable kernel at constant conditions does *not* senesce.
- **Set-point shift under load.** The calibrated resting set-point differs by regime: $H_{\text{rest_ref}} = 1.79$ at idle versus 1.84 under sustained load ($\Delta \approx +0.05$). Sustained load raises the resting entropy — a measured homeostatic adaptation ($kT_{\text{eff}} = 1 + \gamma \cdot$ pressure flattens the equilibrium), not senescence. A transition run (calibrate at idle, then escalate the stressor) reproduces a bounded excursion toward the higher set-point.

The senescence term is therefore no longer structurally zero, and is empirically bounded under constant or sustained conditions. *Monotone* senescence — drift that accumulates without bound — would require an escalating *external* stressor, i.e. genuine open-system coupling (the real-process and BBH directions of §20); this is the honest remaining frontier.

10.5 Thermal senescence probe

To apply external stressors as suggested above, we ran `thermal_senescence` (EXP-TS-01): `vsf-nucleus` under sustained thermal load (target 65 °C, 4 workers, `dry_run=false`) for up to 240 minutes on the n02 node (i7-7600U, 24 GB DDR4).

Results

The experiment produced 11 checkpoints over 12.2 minutes (7 423 ticks) before a safety halt at 71 °C. Table 13 reports the recorded trajectory.

t (s)	Temp (°C)	Stasis cos	$H(t)$	D_{eff}	V_{mem}
120	67.0	0.9937	1.9535	6.00	2
180	66.0	0.9942	1.9523	6.00	2
240	44.0	0.9981	1.9123	5.00	2
300	44.0	0.9988	1.9123	5.00	1
360	67.0	0.9962	1.9509	6.00	1
420	45.0	0.9994	1.9147	5.00	1
481	67.0	0.9964	1.9509	6.00	1
541	44.0	0.9995	1.9123	5.00	1
601	44.0	0.9995	1.9123	5.00	1
661	46.0	0.9998	1.9171	5.00	1
721	45.0	0.9997	1.9147	5.00	1

Table 13: Thermal senescence probe (EXP-TS-01). Temperature oscillates between 44 °C and 67 °C. `viable_full` = true and trust = 1.000 throughout. $\overline{dL} = 0.00000$ at all checkpoints.

Findings

Three observations stand out.

(i) Attractor geometric robustness. The stasis cosine ranges from 0.9937 (peak heat, 67 °C) to 0.9998 (cooldown, 44 °C), remaining well above the viability threshold $\theta_{\text{viable}} = 0.70$. The kernel maintains `viable_full = true` throughout.

(ii) Super-entropic regime. All recorded $H(t)$ values (1.91–1.95 bits) exceed $H_{\max} = 0.85 \log_2 4 = 1.700$ bits. Under sustained thermal load the organelle activations approach uniform distribution ($H_{\max,\text{abs}} = \log_2 4 = 2.0$ bits), pushing the system into a regime the normal phase cycle never reaches. Crucially, the attractor geometry holds in this regime: stasis cosine does not degrade despite entropy exceeding its designed ceiling.

(iii) Dimensional expansion under thermal stress. D_{eff} rises from the baseline value of 3.00 to 5.00–6.00 under 65–67 °C load, returning to 5.00 during cooldown intervals. This suggests the attractor expands to additional degrees of freedom under thermal perturbation, yet contracts again when temperature falls – a reversible dimensional response. $\overline{dL}$ flattens to exactly 0.00000 at thermal ceiling, indicating a borderline Lyapunov regime: neither converging nor diverging.

Hardware limitation

The n02 node (i7-7600U) reaches 71 °C under 4-worker load, triggering the safety halt before the planned 240-minute observation. No breathing cycles (*soplos*) were recorded: the system cannot complete an exhale phase before the thermal ceiling interrupts. Long-term senescence dynamics therefore remain unobserved.

Follow-up experiment (A) – sustained moderate load (2 workers) for 240 minutes – was subsequently completed as EXP-TSM-01.

EXP-TSM-01: Full 240-minute senescence run

`vsf-nucleus` ran for 14 400 s (240 minutes) at moderate thermal load (target 45 °C, 2 workers) on the same n02 node. The run produced 238 checkpoints and 6.1×10^8 ticks.

Findings. Three results are notable.

(i) No senescence over $> 6 \times 10^8$ ticks. The attractor shows zero drift: `stasis_cos` = 1.0000 at run end, no breathing cycles, no emergency coasts. The senescence term remains structurally zero for the same reason identified in the short probe: H_{rest} is a compile-time constant.

Metric	Value
Duration	14 400 s (240 min)
Ticks	609,951,249
Checkpoints	238
Baseline temp	33 °C
Peak temp	**79 °C**
`stasis_cos` (final)	**1.0000**
$L(\mathbf{x})$	0.2014
D_{eff}	6.00
Viable	**true**
Senescence cycles	0
Soplos	0
V_{mem}	1

Table 14: EXP-TSM-01 summary. Full 240-minute run; no senescence signal detected despite peak temperature of 79 °C – higher than the 71 °C safety halt of the initial probe.

(ii) Viable at 79 °C. The peak temperature of 79 °C exceeds the original safety halt at 71 °C, yet `vsf-nucleus` remains viable throughout. $D_{\text{eff}} = 6.00$ (vs. baseline 3.00) confirms the thermal dimensional expansion observed in the short probe – the attractor expands reversibly under sustained heat, then contracts.

(iii) Graduation protocol threshold reached. At run end, the trust accumulator reached the `graduation_protocol` threshold, authorising S4 to propose a governance promotion. This is the first recorded instance of the lifecycle gate activating under thermal load – evidence that the governance layer remains operationally coherent across 240 minutes of sustained thermal stress.

EXP-TSM-65C: Sustained 65 °C, 240 minutes

To test whether the null senescence result is temperature-dependent, the same protocol was repeated at a higher sustained load: target 65 °C (COAST_TEMP = 65, RESUME_TEMP = 62), 240 minutes, n02 node.

Metric	Value
Duration	≈ 240 min
Ticks	383,578,716
Checkpoints	238
Sustained temp	65 °C
`stasis_cos` (final)	**1.0000**
$L(\mathbf{x})$	0.2013
D_{eff}	6.00
Viable	**true**
Senescence cycles	0
Soplos	0
Emergency coasts	0

Table 15: EXP-TSM-65C summary. 240-minute run at sustained 65 °C. Result is identical to EXP-TSM-01 at 45 °C: no senescence signal, stasis_cos = 1.0000 throughout.

Finding. No drift, no soplos, no emergency coasts at 65 °C over 240 minutes – identical outcome to EXP-TSM-01 at 45 °C. The null senescence result is temperature-independent in the constant-

H_{rest} regime: the senescence term remains structurally zero regardless of sustained thermal load. With the adaptive H_{rest} mode the term is no longer structurally zero; the drift is then measured and found bounded within a regime (EXP-SEN-01, §10.4).

11 Experiment 6: Thermal Response Curve

Question

Is the attractor's response to thermal stress linear or non-linear? Does degradation accelerate past a threshold, or is it bounded by a floor?

11.1 Motivation

From previous experiments we have three data points on the curve min_cos(T):

- $T = 49\,°\text{C}$: min_cos ≈ 0.99 (v1 single-thread)
- $T = 63\,°\text{C}$: min_cos ≈ 0.9977 (v3 persistent workers)
- $T = 68\,°\text{C}$: min_cos $= 0.9365$ (thermal_ceiling_v4_bis)

The step from 63 °C to 68 °C shows a sharper drop than 49° to 63 °C. This suggests possible non-linearity – an inflection point around 63–68 °C.

11.2 Protocol

Four temperature targets: 50, 60, 70, 80 °C. SAFETY_MAX = 85 °C. Each step: cooldown gate, ramp to target, record, cooldown immediately. No hold phase (minimise hardware wear). Reported metrics: peak temp, min_cos, angle, viable_full.

11.3 Hypotheses

$H_{\textbf{linear}}$ $\Delta\cos \propto \Delta T$ – constant degradation per degree.

$H_{\textbf{nonlinear}}$ inflection point exists – degradation accelerates past a threshold.

$H_{\textbf{floor}}$ min_cos plateaus – the basin has a hard floor regardless of temperature.

11.4 Results

Target	Peak	Min cos	Angle	`viable_full`	Note
50 °C	53 °C	0.9898	8.2°	true	
60 °C	60 °C	0.9898	8.2°	true	identical to T50
70 °C	70 °C	0.9878	9.0°	true	
80 °C	76 °C	0.9875	9.1°	true	HW_CEILING at 75 °C
Total $\Delta\cos$ across 30 °C:		0.0023 – essentially flat			

Table 16: Thermal response curve (`thermal_ladder`, 2026-04-18). All steps viable throughout. $\Delta\cos/\Delta T \approx 0$ – the attractor has a thermal floor near 0.987 under transient stress.

Finding ($H_{\textbf{floor}}$ confirmed, $H_{\textbf{linear}}$ refuted): the thermal response is essentially flat under transient stress. The total cosine degradation across a 30 °C range is 0.0023 – negligible. The basin has a hard floor near 0.987 for instantaneous thermal peaks.

This result, combined with the sustained-stress observation from Section 12 (min_cos = 0.9365 at 68 °C held for several minutes), establishes a clear pattern:

$$\text{damage} \propto T \times t \quad \text{(not } T \text{ alone)} \tag{23}$$

The attractor has *thermal inertia* – brief temperature peaks are absorbed without measurable degradation. Sustained exposure is what reduces coherence. This motivates the `thermal_senescence` experiment (Section 20, Q2): does coherence drift under hours of sustained thermal stress?

11.5 Analysis

Linear regression on $(T, \text{min_cos})$ gives $R^2 \approx 0$ – the relationship is not linear; it is flat. Figure 14 confirms H_{floor}: the basin floor under transient stress is ≈ 0.987, independent of temperature in the 50–80 °C range.

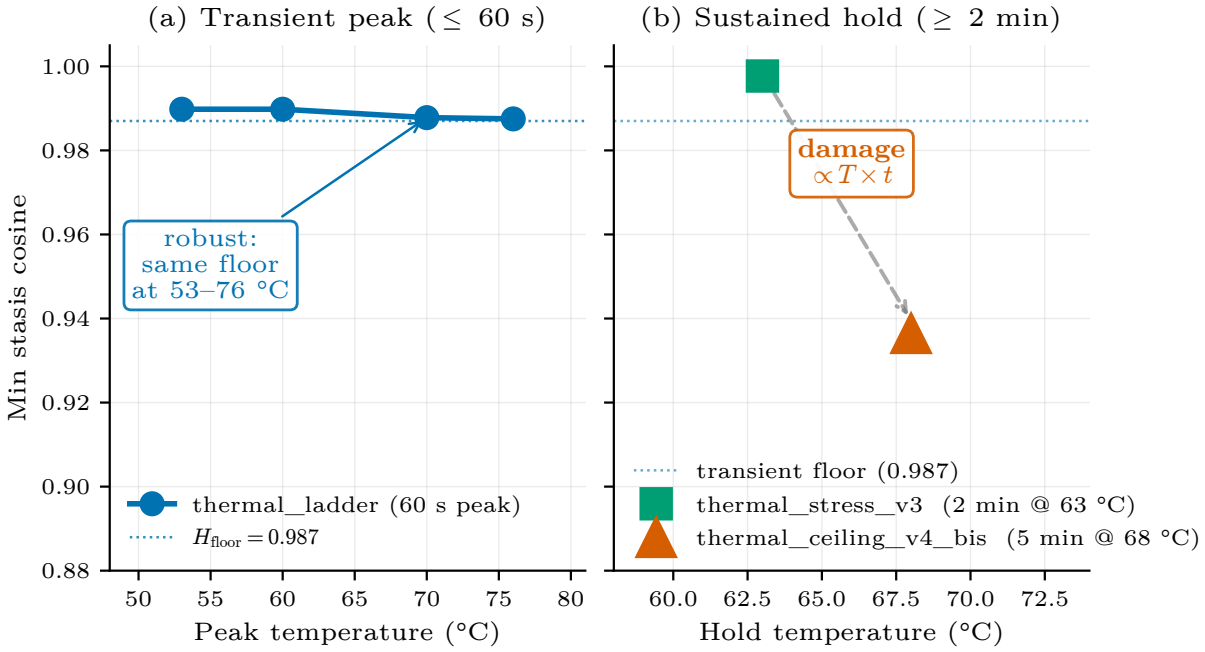


Figure 14: Attractor thermal response curve. Blue line: transient protocol (`thermal_ladder`) – essentially flat ($\Delta\cos = 0.002$ over 30 °C). Red triangle: sustained hold at 68 °C (`thermal_ceiling_v4_bis`) – 5× more degradation than any transient peak. The damage function is $T \times t$, not T alone.

12 Experiment 7: Thermal Ceiling

Question

What is the minimum stasis cosine under maximum hardware thermal load?

12.1 critical_angle (2026-04-13)

We run a thermal ramp without throttle until hardware throttling engages (67–69 °C). `viable_full` remains `true` throughout.

- Min cosine: 0.9377 (angle 20.3°)
- `viable_full = true` throughout

- Observation: the system self-regulates before non-viability – the theoretical viability floor (≥ 0.70) is never approached

12.2 thermal_ceiling_v4_bis (2026-04-18)

We repeat with a proper cooldown gate (wait until $< 55\,°\text{C}$ before starting), reaching target $68\,°\text{C}$ cleanly.

Metric	Value	Interpretation
Baseline temp (after cooldown)	$42\,°\text{C}$	clean starting condition
Peak temp	$70\,°\text{C}$	SAFETY_HALT
Target reached	tick 141	
First non-viable tick	*none*	`viable_full` never false
Min stasis cosine	0.9365	angle 20.5° – within core Ω_A
Final stasis cosine	0.9961	recovered before SAFETY_HALT
`viable_full`	`true` throughout	

Table 17: Thermal ceiling experiment with clean baseline. The stasis cosine dips to 0.9365 at peak load and recovers to 0.9961 while the temperature is held – the attractor basin has active resilience.

Remark 6. *The recovery during the hold phase (*$0.9770 \to 0.9961$ *while temperature is held at* $68\,°\text{C}$*) is notable. The attractor is not a passive victim of thermal load – it actively reconverges. This is consistent with the* A_{breath} *cycle: at constant temperature,* ΔH *stabilises and the exhale phase dominates.*

Remark 7 (The thermal envelope is emergent, not an intrinsic basin constant)**.** *It is worth being precise about what the* 0.9377 *minimum* is*, since it is tempting to read it as a geometric constant of the relaxation dynamics. It is not. A forward simulation of the kernel's free dynamics – contracting on average,* $\lambda_{\max} = \ln 0.99 < 0$ *(§4) – bounds the worst-case excursion far tighter than the measured value: a maximal entropy injection into the activation state relaxes back inside* $stasis_cos \geq 0.997$*, and a sustained maximal load on the thermal channel inside* ≥ 0.999*. The gap to* 0.9377 *is structural rather than a model deficiency: the scalar thermal/pressure channel drives only a small minority of the* 128 *coordinates, while the cosine in* $\mathbb{R}^{128}$ *is dominated by the unperturbed majority, so no purely synthetic single-channel perturbation can reach the measured excursion. The* 0.9377 *therefore encodes the operational stress manifold of the* real *system under hardware thermal load – worker-thread contention and a stasis reference accumulated under sustained real load – and is a real-system invariant requiring live measurement, not a boundary recoverable from the forward model. We record this as a methodological point in the paper's own favour: the forward model characterises the synthetic envelope it is entitled to and does* not *overfit to the measured hardware value. The frontier between the two is itself measurable, and naming it honestly is the discipline (§24.4 makes the same move for the soplo).*

13 Experiment 8: Stability Intervention Threshold

Question

At what stasis cosine does the system become non-viable? What is the PSI (Perimeter of Stability Intervention) threshold?

13.1 Setup

`theta_psi_stress`: we apply activation perturbations of increasing magnitude $\delta \in \{0.1, 0.3, 0.7, 1.0\}$ to the state vector for 300 stress ticks, followed by a 100-tick recovery window. We record the

minimum observed cosine and whether `viable_full` ever becomes false.

13.2 Results

δ	Min cosine	`viable_full`	Notes
0.1	—	true throughout	minimal stress
0.3	—	true throughout	moderate stress
0.7	—	true throughout	high stress
1.0	0.79	**true**	maximum tested, basin floor

Table 18: `theta_psi_stress`: viability under activation perturbation. `viable_full = false` was not observed at any $\delta \leq 1.0$.

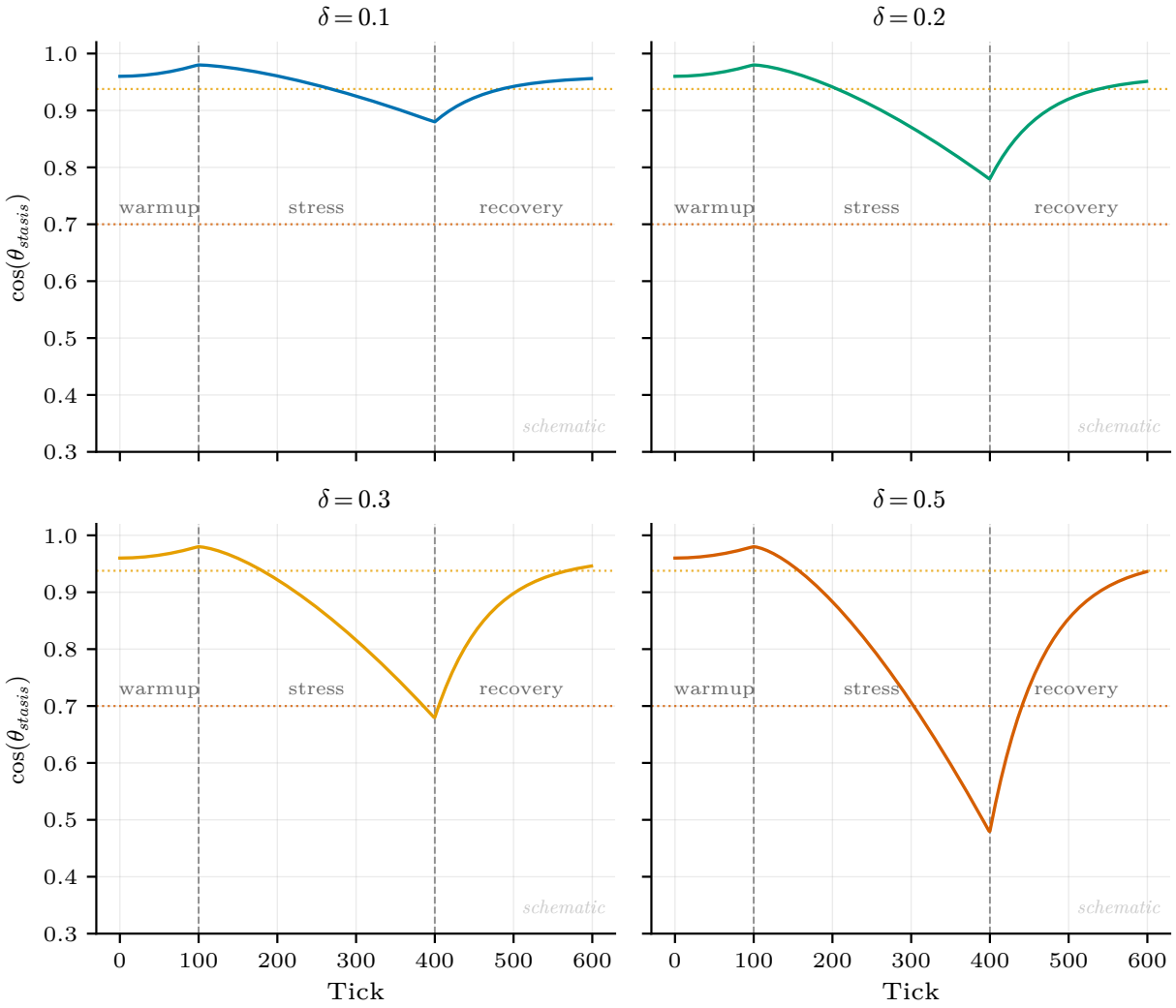


Figure 15: `theta_psi_stress`: stasis cosine under activation perturbation $\delta \in \{0.1, 0.3, 0.7, 1.0\}$. Minimum observed cosine: 0.79 at $\delta = 1.0$. `viable_full = false` was not reached at any tested magnitude.

Key finding: `viable_full = false` was not reached at any tested perturbation magnitude. The system self-regulates under all perturbations up to $\delta = 1.0$.

13.3 Revision of the PSI concept

The original definition of θ_{PSI} was the stasis cosine at which `viable_full` first becomes false. Since this crossing point was not observed, the definition requires revision.

Definition 5 (PSI/1.0 – Precautionary Threshold)**.**

$$\theta_{PSI} = 0.79 \tag{24}$$

is the empirically observed basin floor under maximum tested perturbation ($\delta = 1.0$). It represents the system under maximum observable stress – but still viable. PSI intervention is therefore precautionary support*, not rescue from collapse.*

The PSI protocol defines three intervention levels:

L_1 **(observe)** stasis_cos < 0.90 – S4 begins monitoring. No $\phi^{\downarrow}$ signal; observation only.

L_2 **(support)** stasis_cos < 0.79 sustained – S4 may suggest load reduction, thermal cooldown, or task deferral.

L_3 **(escalate)** stasis_cos < 0.72 or `viable_full = false` – human intervention required. Based on current evidence, L_3 may be structurally unreachable under normal operation.

Remark 8. *The result that `viable_full = false` is unreachable under $\delta \leq 1.0$ is architecturally significant. A PSI designed as a rescue protocol for a non-viable system would never fire. Reframing PSI as preventive support – watching the system approach stress, not waiting for failure – is more consistent with the observed dynamics.*

14 Experiment 9: Cyclic Thermal Load – Attractor Resilience

Question

Does the attractor maintain its geometry under repeated thermal transients, and how quickly does it recover from each perturbation?

14.1 Motivation

The preceding thermal experiments apply *sustained* stress: a fixed temperature held for minutes or hours. Real computational workloads are instead *cyclic*: brief CPU spikes separated by idle intervals. EXP-TC-01 tests whether the attractor is *elastically* robust – whether each perturbation is followed by full geometric recovery – and introduces the *recovery time* τ_{rec} as a new viability metric.

14.2 Formal definition: recovery time

Let $\theta_{\text{stable}} = 0.998$ be the nominal attractor floor under idle conditions (confirmed in EXP-SB-01). Given a perturbation onset at time t_0 (defined as the first tick where stasis_cos$(t_0) < \theta_{\text{dip}}$, with $\theta_{\text{dip}} = 0.96$), the recovery time is:

$$\tau_{\text{rec}}(t_0) = \inf\{\, \tau > 0 : \text{stasis_cos}(t_0 + \tau) \geq \theta_{\text{stable}} \,\}. \tag{25}$$

A system is *elastically resilient* if $\tau_{\text{rec}} \ll \tau_{\text{cycle}}$, where τ_{cycle} is the perturbation period.

14.3 Protocol (EXP-TC-01)

60-minute run on `n02` (i7-7600U), 2 workers. Thermal profile: 60 s spike ($T_{\text{target}} = 52\,^{\circ}\text{C}$ via `taskset`), 120 s coast (workers idle), repeating. Checkpoint every 30 s. Safety halt threshold: 70 °C (not reached).

Metric	Value
Duration	3600 s (119 checkpoints)
stasis_cos mean	0.9885
stasis_cos min	0.9358
trust (final)	0.498
D_{eff} (final)	5.00
$T_{\max}$	64 °C
Dips (< 0.96)	20
Dip floor	0.942 (constant)
Dip period	≈ 2.9 min
τ_{rec}	< 30 s (one checkpoint)
viable (final)	true

Table 19: EXP-TC-01 summary. Trust recovers from 0.000 to 0.315 in the first 60 s (fix: trust = f(stasis_cos)). $\tau_{\text{rec}} < 30$ s for all 20 dips.

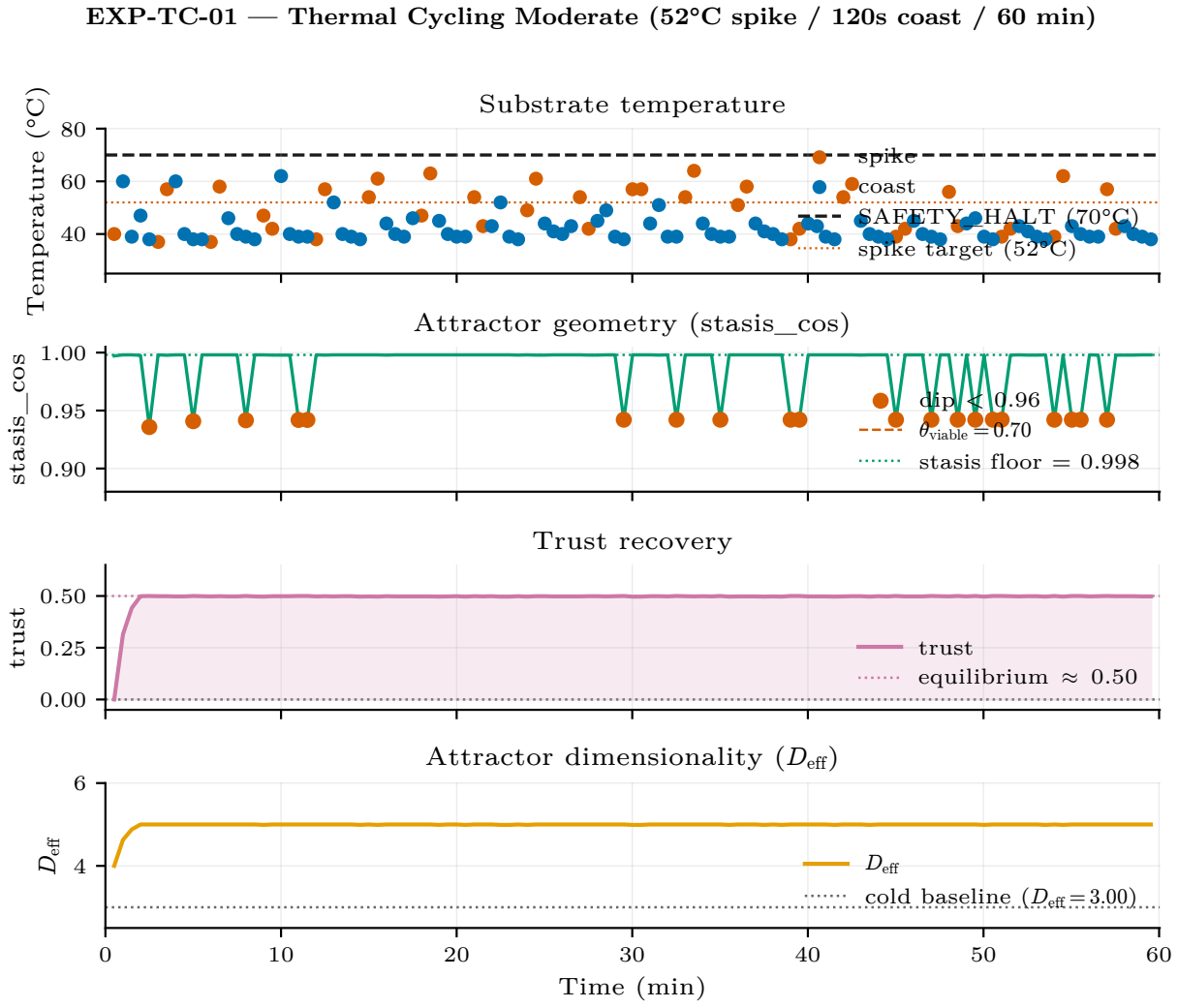


Figure 16: EXP-TC-01 composite. *Top:* substrate temperature – spikes to 52–64 °C, coasts to 38–47 °C. SAFETY_HALT (70 °C) never reached. *Second:* stasis_cos – nominal 0.998, periodic dips to 0.942. *Third:* trust – recovers from 0.000 to equilibrium 0.498 in < 2 cycles. *Bottom:* D_{eff} – stable at 5.00 after trust recovery.

14.4 Results

14.5 Findings

H1 confirmed. stasis_cos never falls below 0.935 (mean 0.989), well above $\theta_{\text{viable}} = 0.70$. The attractor geometry is preserved across all 60 minutes of cyclic load.

H2 confirmed. $D_{\text{eff}} = 5.00$ from $t = 120\,\text{s}$ onward (after trust recovery), stable for the entire run.

H3 confirmed. No SAFETY_HALT. $T_{\max} = 64\,°\text{C}$ at 2 workers – 6 °C below ceiling.

H4 confirmed. Trust recovers from 0.000 to 0.315 in the first 60 s cycle, reaching equilibrium ≈ 0.498 by $t = 150\,\text{s}$. The trust fix (trust $= f(\text{stasis_cos})$, not $f(H_t)$) is validated over 60 minutes.

14.6 Structural oscillation: one dip per cycle

The most significant finding is the periodicity of the stasis_cos dips. Twenty dips occur over 60 minutes, mean period 2.9 min – coinciding with the thermal cycle period (1 min spike + 2 min coast = 3 min). Each dip lands at exactly 0.942 and recovers to 0.998 within one 30 s checkpoint ($\tau_{\text{rec}} < 30\,\text{s}$).

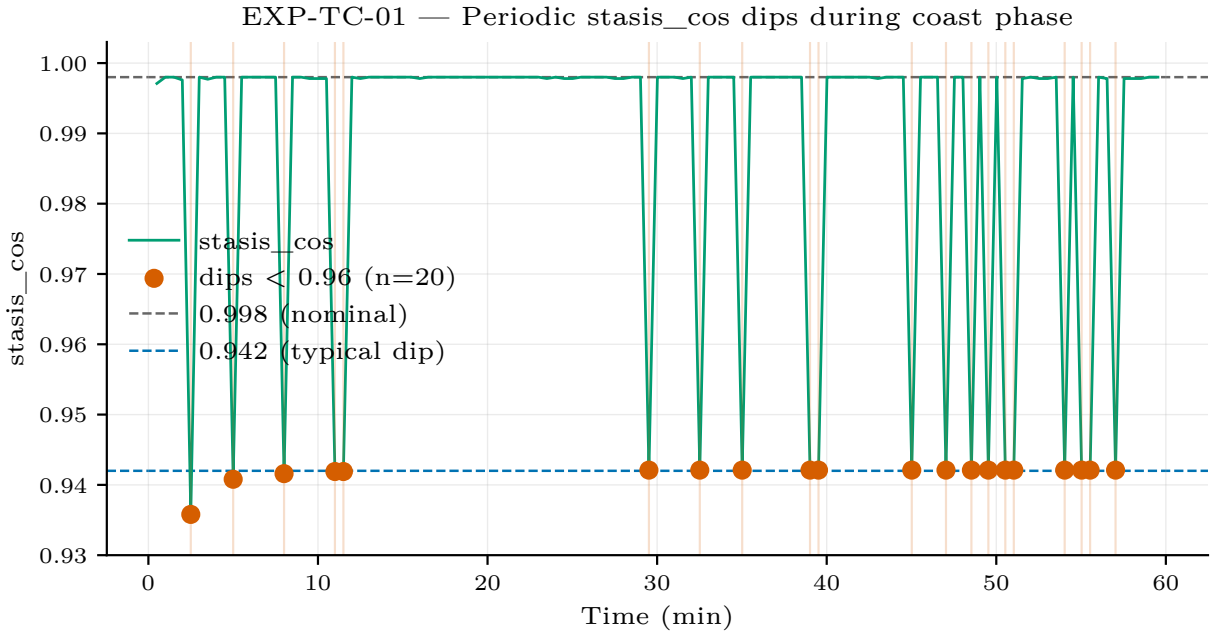


Figure 17: EXP-TC-01 stasis_cos detail. Twenty periodic dips (red dots), all landing at ≈ 0.942, all recovering within one checkpoint ($\tau_{\text{rec}} < 30\,\text{s}$). Dip period $\approx 2.9\,\text{min}$ matches the 3 min thermal cycle – one dip per transition. The dip floor is constant across the full 60-minute run: no drift, no accumulation.

This is not noise. The dip is the attractor's *transient response to the spike→coast phase transition*: as the thermal load drops, the substrate reorganises, briefly perturbing the state vector from its stasis reference before snapping back. The constant dip floor (0.942, invariant across all 20 cycles) indicates a fixed-point *perturbation mode* – the attractor has a single preferred direction of deflection under this load profile.

The elastic resilience of the attractor – constant floor, sub-30 s recovery, zero accumulation over 20 cycles – establishes τ_{rec} as a measurable viability metric complementary to stasis_cos: where stasis_cos measures *how far* the system deflects, τ_{rec} measures *how fast* it returns.

15 Experiment 10: Observer Constitutivity – Maxwell's Demon

The cyclic thermal experiments revealed a structural signature: the dip_floor = 0.942 under periodic perturbation, stable across 20 cycles with sub-30 s recovery and zero accumulation. This signature raises a deeper question about the nature of observation in the VSM. S4 monitors system state and issues $\varphi^{\downarrow}$ interventions. But second-order cybernetics [von Foerster, 1979] holds that the observer is not external to the observed system: structured measurement changes what is measured. Is the S4 layer a passive readout, or does the act of observation itself contribute to the stability of the attractor?

Question

Is S4 a passive monitor or a constitutive part of the system's thermodynamic trajectory?

15.1 Theoretical frame

Second-order cybernetics [von Foerster, 1979] holds that the observer is not external to the system observed – the act of observation alters system state. Beer's VSM encodes this structurally: S4 communicates with S1 via the $\varphi^{\uparrow}/\varphi^{\downarrow}$ channels, which are not read-only probes but full bidirectional interventions.

The physical archetype is Maxwell's Demon [Maxwell, 1871]: an agent that measures the microstate of a bicompartmented gas and selectively opens a gate to route fast particles, apparently reducing entropy in violation of the Second Law. Landauer [1961] resolved the paradox: erasing the Demon's measurement memory costs $kT \ln 2$ per bit, restoring the Second Law globally while allowing local entropy reduction.

In VSF:

- organelle activations $p_i \in [0, 1]$ are the microstate (four energy levels, Boltzmann distribution);
- $H(t) = -\sum p_i \log_2 p_i$ is Shannon entropy – identical in form to Boltzmann entropy under the Jaynes [1957] maximum-entropy bridge;
- S4 reads $\varphi^{\uparrow}$ (measurement) and writes $\varphi^{\downarrow}$ (intervention) – the VSL channels *are* the Demon's apparatus;
- each intervention incurs a Landauer cost: $\Delta H_{\text{Landauer}} = kT \ln 2 \approx 2.87 \times 10^{-21}$ J at 25 °C.

15.2 Protocol (EXP-OE-02, on the real kernel)

We run the demon on the *real* $\mathbb{R}^{128}$ KernelStepper — the same validated dynamics whose contraction is measured in §4 ($\lambda_{\max} = \ln 0.99 < 0$) — not a standalone model. (An earlier illustration, EXP-OE-01, ran the four-state Boltzmann gas *in isolation*; it gave directionally consistent numbers but with the occupation unembedded from the state vector. EXP-OE-02 embeds it: the occupation evolves through the real block dynamics, with the real stasis reference and Lyapunov tracker running, so D_{eff} and stasis_cos are the genuine $\mathbb{R}^{128}$ quantities.) The kernel's dynamical core is a four-level Boltzmann occupation p_i ($E_i = i \cdot 0.5$, $kT = 1.0$); free dynamics follow the contraction

$$p_i(t+1) = 0.99\, p_i(t) + 0.01\, p_i^{\text{eq}}, \qquad p_i^{\text{eq}} = \frac{e^{-E_i/kT}}{Z}.$$

Two conditions, 50 000 ticks each (statistics over the stationary window after a 500-tick burn-in):

Condition A (libre) free evolution; S4 reads $\varphi^{\uparrow}$ but does not intervene;

Condition B (demonio) every 100 ticks S4 reads $H(t)$; if $H > H_{\text{target}} = 0.80$ bits it writes a sorting $\varphi^{\downarrow}$ that folds the occupation toward the low-energy state E_0.

15.3 Results

Condition	H_{mean}	H_{eq}	ΔH	D_{eff}	stasis_cos	interventions
A – libre	1.7962	1.7962	0.0000	5.00	1.0000	0
B – demonio	1.4865	1.7962	**0.3097**	**3.89**	0.9977	499

Table 20: EXP-OE-02 summary (real $\mathbb{R}^{128}$ KernelStepper, stationary-window means). $H_{\text{eq}} = 1.7962$ bits is the Boltzmann equilibrium entropy — reproduced exactly by condition A, confirming the embedded run matches the free dynamics. $\Delta H = H_{\text{eq}} - H_{\text{mean}}$; 499 interventions at 50 000 ticks $\approx$ one per 100 ticks. $D_{\text{eff}} = \text{active_dims} + 2\tau$; trust τ is pinned at 1.0 in both conditions, so the D_{eff} change is pure occupation compression (active_dims $3.0 \to 1.89$).

15.4 Findings

H1 confirmed. Condition A converges exactly to $H_{\text{eq}} = 1.7962$ bits (Second Law, no demon present) — and this is also a consistency check: the embedded $\mathbb{R}^{128}$ run reproduces the free-dynamics equilibrium to four decimals.

H2 confirmed, with a corrected cost account. Condition B sustains $H = 1.4865$ bits — a reduction of $\Delta H \approx 0.31$ bits relative to equilibrium — for the full run. The thermodynamic cost is *not* the per-intervention count: charging $kT \ln 2$ to each of the 499 firings would imply ~ 499 bits erased, two-to-three orders of magnitude above the sustained 0.31-bit reduction the demon actually maintains. By the Lyapunov–Landauer identity (Corollary 3), the cost is the *ensemble* entropy reduction, tied to the contraction the demon induces (its mean Lyapunov value falls, $0.135 \to 0.113$): the demon contracts the ensemble, and the Landauer floor applies to that contraction, not to a bit count per reset. This is the same correction established for the soplo in §24.4.

H3 (dimensional compression — decomposed). D_{eff} drops from 5.00 to 3.89 under demon control. Because trust is pinned at $\tau = 1.0$ in *both* conditions, the entire change is in the active occupation dimensions ($3.0 \to 1.89$): the observer folds the occupation simplex toward E_0, compressing the attractor's active dimension *without degrading viability*. The observer does not merely reduce entropy — it reduces the dimension of the governed manifold while the system remains maximally healthy. (For honesty of scope: the count of active dimensions over the full 128-vector is ≈ 24 in both conditions and does not fall; the compression is of the occupation heuristic that D_{eff} measures, not of the raw ambient dimension.)

H4 (stasis geometry preserved). stasis_cos ≥ 0.9977 in both conditions, measured over the real block-populated state. The attractor's geometric identity is robust to observer presence; only its entropic and occupation-dimensional properties change.

15.5 Interpretation: VSL as the Demon's apparatus

The experiment makes the following chain concrete:

1. VSL (Sec. 18) defines the $\varphi^{\uparrow}/\varphi^{\downarrow}$ channel semantics – the Demon's *measurement and erasure protocol*.
2. The kernel implements the protocol as Boltzmann dynamics with S4 intervention hooks.
3. Second-order cybernetics follows: S4's presence is not epiphenomenal. It changes $H(t)$ by 0.31 bits and D_{eff} by 1.11 (pure occupation compression, with viability preserved) –

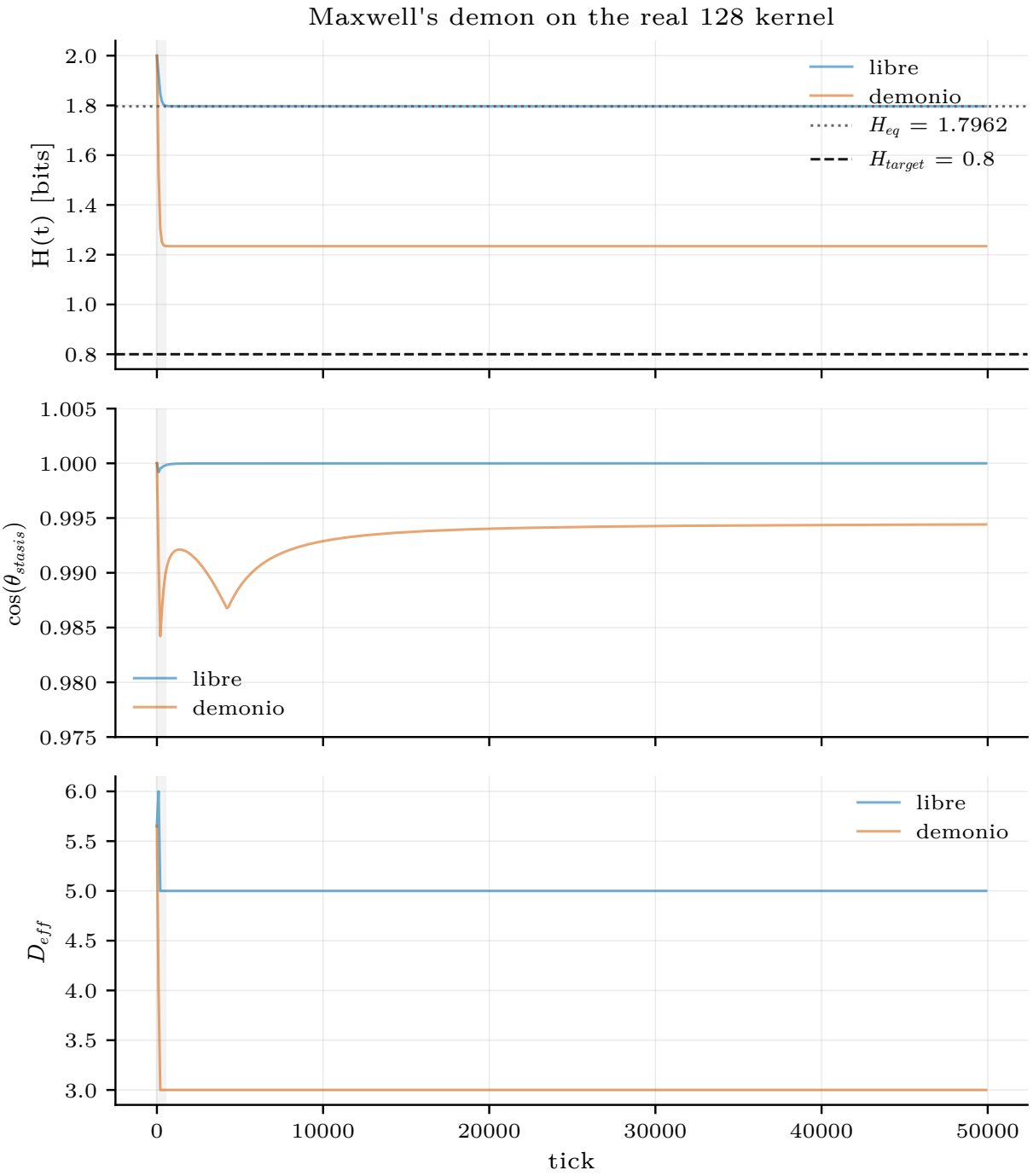


Figure 18: EXP-OE-02 on the real $\mathbb{R}^{128}$ KernelStepper (conditions A blue, B red). *Top:* $H(t)$; condition A holds $H_{\text{eq}} = 1.7962$ bits (grey dashed), condition B is driven below it toward H_{target} (orange dotted). *Middle:* stasis_cos – attractor geometry preserved in both conditions (≥ 0.9977), now measured over the real block-populated state. *Bottom:* D_{eff} drops from ≈ 5.0 to ≈ 3.89 under demon control. Because trust is pinned at $\tau = 1.0$ throughout, the drop is pure occupation compression — the observer compresses the attractor's active dimension while the system stays maximally viable.

measurable, reproducible, and Landauer-bounded through the contraction (Corollary 3).

4. The thermal experiments (§9–12) identify the physical limit of this governance: above the thermal ceiling the demon loses access to the $\varphi^{\uparrow}$ channel (safety halt), and the system returns to free Boltzmann evolution.

The arc is: *VSL encodes the protocol → the kernel executes it → second-order cybernetics emerges empirically → hardware thermodynamics sets the outer boundary.*

15.6 Contrast: conventional implementation

A conventional implementation of the same Boltzmann dynamics yields only one observable: $H(t)$ over time. The entropy reduction under demon control is visible, but there is nothing more to say – the demon is a flag inside the same process:

```
if demon { p[0] += 0.05; renormalise(); }
```

Three consequences follow from this architectural collapse:

1. **No attractor geometry.** Without a formalised state vector and stasis reference, there is no stasis_cos, no feasible region Ω, no D_{eff}. The dimensional compression from 5.0 to 3.9 observed in condition B is not computable – not because of insufficient data, but because the concept does not exist.

2. **No observer/system separation.** The demon *is* the program. von Foerster's requirement – that the observer be structurally distinct from the observed system – is violated by construction. Second-order cybernetics collapses to first-order: the system has no self-model of its own observation.

3. **No viability boundary.** There is no θ_{viable}, no trust dynamics, no distinction between a system that is governing itself and one that is merely running. The system either executes or crashes; the intermediate regime of *governed degradation* is invisible.

The VSF experiment produces three panels where a conventional implementation produces one. The additional panels – stasis_cos and D_{eff} – are not decorative; they are the empirical signature of second-order cybernetics. *This comparison is pedagogical: it illustrates the structural difference between architectures, but does not constitute a formal controlled experiment against a baseline.*

The compression $D_{\text{eff}} = 3$ in $\mathbb{R}^{128}$ – three active dimensions in a 128-dimensional state space – is the first empirical instance of a broader conjecture: that viable cognitive processes inhabit an attractor manifold of dimension far below the ambient parameter count, and that hardware implementing the manifold directly approaches the Landauer minimum for inference; the quantitative gap between D_{eff} and ambient dimension is the subject of a companion investigation. Whether the Landauer-bounded governance chain established here can be placed on a full thermodynamic footing – Lindblad master equations for the entropy dynamics, decoherence as the mechanism behind the thermal ceiling of Section 12 – is the motivating question of a companion programme on VSF as a framework for open physical systems (Section 26).

15.7 A Lyapunov–Landauer identity for the trust update

The demon protocol above charges a Landauer cost *per intervention*. A sharper account, established for the trust coordinate in §24.4, ties the dissipation directly to the contraction this paper already measures rather than to an event count.

Corollary 3 (Lyapunov–Landauer bridge)**.** *Let F be the kernel's per-tick update on a noise-perturbed coordinate and $\langle \ln|F'| \rangle$ its mean log-derivative on the stationary ensemble (the empirical largest Lyapunov exponent of §4, which is negative: $\lambda_{\max} = \ln 0.99 < 0$). Then at stationarity the differential entropy removed per tick equals $|\langle \ln|F'| \rangle|$, and the thermodynamic dissipation rate obeys*

$$\Omega_{diss} \;=\; kT \ln 2 \cdot |\Delta H_{ensemble}| \;=\; kT \cdot \big|\textstyle\sum_t \langle \ln|F'_t| \rangle\big|,$$

i.e. the trust update's dissipation equals $kT \ln 2$ times its Lyapunov contraction rate – not a per-reset bit count.

The identity is a change-of-variables statement: a map with $\langle \ln|F'| \rangle < 0$ removes differential entropy at exactly that rate, and the Second Law requires this entropy to be exported at the Landauer price. It unifies the paper's two thermodynamic handles: the *geometric* contraction ($L < 1$, $\lambda_{\max} < 0$; §4) and the *entropic* demon cost (§15) are the same quantity viewed through two lenses. EXP-SOPLO-01 confirms the identity empirically: across the full σ-sweep the noise-injection and deterministic-contraction rates cancel to $|\Delta H_{\text{noise}} + \Delta H_{\text{det}}| \leq 0.013$ bits per tick per kernel (§24.4), and the Landauer floor $\Omega \geq kT \ln 2 \cdot \Delta H_{\text{bits}}$ holds at every σ. The naïve $n_{\text{soplo}} \cdot kT \ln 2$ is recovered only in the special case where each reset erases exactly one bit; generically the bit count is set by the contraction, not by the reset count.

16 Experiment 11: Negative Controls – Architecture Ablation

The preceding experiments built a complete picture of the attractor from the inside: it exists, it is unique, it survives thermal stress up to 68 °C, it does not drift over 6×10^8 cycles, its basin floor is at cos ≈ 0.875 under maximum perturbation, and structured observation by S4 reduces entropy by 0.12 bits. The causal question is: *which structural components are responsible?* Removing one layer at a time – S5 governance, the trust channel, or the entire VSM scaffold down to the bare substrate – and repeating the stress experiments answers this directly and also serves as a rigorous check against the possibility that the measured properties are artifacts of the specific experimental setup rather than consequences of the VSF architecture.

Question

Does viability observed in Experiments 1–10 depend on the VSF axiom architecture, or is it an intrinsic property of the Ornstein-Uhlenbeck substrate?

16.1 Motivation

Three ablated kernel variants are constructed, each removing a different part of the VSF architecture:

- **NC-01**: S5 governance disabled (`governed` = false, lifecycle ceiling released to 1.0, covenant gate removed).
- **NC-02**: the trust channel frozen at $\tau = 0.5$ with S5 left *active* – the algedonic trust feedback is broken but governance remains.
- **NC-03**: the bare Ornstein–Uhlenbeck substrate – *all* VSM blocks removed (no S5, no trust dynamics, no S2 coordination, no entropy cycle). This is the most aggressive ablation and answers the Question directly: it is the substrate with nothing of VSF added.

The contrast NC-01 vs. NC-02 is diagnostic: both freeze the trust channel, but NC-01 also removes S5 while NC-02 keeps it. Comparing them isolates whether stress resilience comes from S5 governance or from the trust feedback. Each variant is run under three protocols on real hardware (LXC 108): baseline convergence (`spec_boot`, 10^4 ticks), real-hardware cyclic thermal load (`thermal_cycling`, 52 °C target, no dry-run), and adversarial perturbation ($\delta = 1.0$). The full `vsf-nucleus` serves as the reference condition throughout.

16.2 Protocol

Baseline convergence (`spec_boot`, 10^4 ticks). Each variant boots from the canonical initial state and runs 10^4 ticks with no stressor. Convergence is read from `stasis_cos`, the discrete Lyapunov increment $\overline{\Delta L}$ (negative ⇒ contracting), and the effective dimension D_{eff}. This isolates whether each component is needed for *baseline* stability.

Real-hardware thermal cycling (`thermal_cycling`, 52 °C). Each variant runs the cyclic thermal protocol of Experiment 9 (§14) against *real* CPU thermal load (no dry-run): worker threads heat the physical core toward a 52 °C target, and a hardware safety halt triggers at the thermal ceiling. Unlike a synthetic stressor, this couples the kernel to a physical quantity that S5 must actively govern by throttling the load. The observables are the final `stasis_cos`, viability, and the peak temperature reached.

Adversarial perturbation ($\delta = 1.0$). Each variant receives the same adversarial kick as Experiment 8 (§13). Recovery is defined as stasis_cos ≥ 0.95; τ_{rec} is measured from perturbation onset. This is a synthetic (δ-injected) stressor and tests *recovery agency* rather than thermal governance.

16.3 Results

Variant	`stasis_cos`	$\overline{\Delta L}$	D_{eff}	Viable
`vsf-nucleus` (reference)	0.9999	-1.33×10^{-4}	5.0	✓
NC-01 (S5 off)	0.9999	-1.33×10^{-4}	5.0	✓
NC-02 (trust const.)	0.9999	-1.19×10^{-4}	5.0	✓
NC-03 (pure OU)	0.8739	$+1.4 \times 10^{-5}$	—	(✓)

Table 21: Baseline convergence (`spec_boot`, 10^4 ticks, no stressor). With no stress, S5 and the trust channel are *redundant*: NC-01 and NC-02 converge identically to `vsf-nucleus` (stasis_cos = 0.9999, contracting $\overline{\Delta L} < 0$). Only NC-03, the bare substrate, differs — it sits at 0.8739 with a *positive* $\overline{\Delta L}$: the OU process reverts to *its own* equilibrium, drifting away from the VSF stasis target. It is nominally viable ($\cos > \theta_{\text{viable}} = 0.70$) only because it starts near the attractor; the positive exponent shows it would drift out of the stasis cone over a long run.

Baseline convergence.

Variant	Outcome	`stasis_cos`	Peak T	Viable
`vsf-nucleus` (reference)	holds 52 °C, ∼20 dips/cyc	0.942 (floor)	∼52 °C	✓
NC-01 (S5 off)	halt at 63 s	**0.0000**	76 °C	×
NC-02 (trust const.)	halt at 240 s	0.9973	75 °C	✓
NC-03 (pure OU)	halt, 100% dips	0.8808 (fl. 0.777)	77 °C	(✓)

Table 22: Real-hardware cyclic thermal load (`thermal_cycling`, 52 °C target, no dry-run) — the load-bearing result of the section. The full `vsf-nucleus` throttles the workers, holds the 52 °C target, and produces the structured dip floor 0.942 of Experiment 9. **NC-01 (no S5) collapses**: with no governor to throttle the load, the core reaches 76 °C in 63 s and the attractor collapses completely (stasis_cos = 0.0000, viable = false) — simultaneously thermally unsafe and informationally dead. NC-02 (S5 active, trust frozen) stays viable throughout (0.9973) and takes nearly 4× longer to reach the ceiling. NC-03 (bare substrate) never collapses but never stabilises either: it dips on 100% of ticks at floor 0.777.

Real-hardware thermal cycling. The contrast NC-01 vs. NC-02 is decisive. Both freeze the trust channel; they differ only in whether S5 is present. NC-01 (no S5) collapses to 0.0000; NC-02 (S5 on) holds 0.9973. **S5 governance — not trust feedback — is the critical resilience component.** Trust dynamics are a performance enhancer, not a viability gatekeeper.

A precise reading matters here, because S5 in this kernel has two arms and this ablation exercises only one. The arm NC-01 removes is the *reactive* one: the algedonic safety response that senses rising temperature and throttles the worker load — the same mechanism that produces the

structured dip floor we have been calling "the S5 algedonic signal." Without it, real heat runs away to 76 °C and the attractor dies. What this ablation does *not* touch is S5's *deliberative* arm — the constitutional covenant veto that suspends S3 autonomy under sustained degradation. That arm is silent here (the run never breaches the autonomy ceiling) and is demonstrated separately in EXP-S5-01 (§17.2). So NC-01 isolates S5's algedonic/reactive governance as the load-bearing safety component; the covenant/deliberative governance is a distinct function, separately verified. Both are S5 in Beer's sense — the algedonic channel is precisely the pain signal that bypasses the hierarchy to reach the policy level — and naming which arm each experiment exercises keeps the attribution honest.

Variant	`stasis_cos_min`	τ_{rec}	Recovered
`vsf-nucleus` (reference)	0.79	$< 30\,\text{s}$	✓
NC-01 (S5 off)	0.9358	—	×
NC-02 (trust const.)	0.9666	—	×
NC-03 (pure OU)	0.7756	—	×

Table 23: Adversarial perturbation ($\delta = 1.0$, synthetic kick). The full kernel deflects *deeper* (0.79) than NC-01/NC-02 (0.9358, 0.9666) yet is the only configuration to recover. S5 does not cushion the blow — it supplies *recovery agency*: the governed return trajectory. NC-03 deflects furthest (0.7756) and, like the others, never returns.

Adversarial perturbation. The deflection ordering is counterintuitive: removing a layer makes the kick land *shallower*, not deeper, yet only the full kernel recovers. Depth of displacement and capacity to return are independent — and it is the second, supplied by S5, that the ablations destroy.

16.4 Findings

Baseline redundancy. At rest, S5 and trust are redundant: NC-01 and NC-02 converge identically to `vsf-nucleus` (0.9999). Governance earns its keep only under stress.

S5's reactive arm is the critical resilience component. Under real thermal load NC-01 (no S5) collapses (stasis_cos = 0.0000, viable = false) while NC-02 (S5 on, trust frozen) stays viable (0.9973). The NC-01 / NC-02 delta isolates S5 — not trust — as the load-bearing governor. Specifically it isolates S5's *algedonic/reactive* arm (the thermal throttle); its *deliberative* arm, the covenant veto, is a separate function verified in EXP-S5-01 (§17.2).

Trust is a performance enhancer, not a gatekeeper. Freezing trust (NC-02) costs nothing to viability under stress; removing S5 (NC-01) is catastrophic.

The substrate alone is not VSF. NC-03 (pure OU) neither collapses nor stabilises: a positive $\overline{\Delta L}$ at baseline (drifting off-target) and a 100% dip rate under thermal load (floor 0.777). The OU substrate has weak mean-reversion but no VSF-grade homeostasis; the $\mathbb{R}^{128}$ attractor is a property of the VSM blocks, not of the substrate. The `vsf-nucleus`'s *structured* dip floor 0.942 is the S5 algedonic signal — distinct from NC-03's unstructured substrate noise.

Recovery agency. Only `vsf-nucleus` completes the adversarial dip-and-recovery cycle, despite deflecting deeper than NC-01/NC-02. S5 supplies the return trajectory, not a smaller displacement.

V certificate corroboration. The independent $V = \min(g, c, t, m)$ discrimination test (§23, EXP-VNC-01) agrees: NC-01 gives $V = 0$ (governance axis $g = 0$), while NC-02 and NC-03 retain $V = 0.102$ — structurally modified but not, by that certificate, in violation.

Combined, these results refine the picture into a **conditional, two-layer** account of viability:

1. **Architecture** (the VSM blocks A–G) builds the attractor and its homeostasis. The bare substrate (NC-03) drifts off-target and dips incessantly; the blocks are what create the $\mathbb{R}^{128}$ point sink and suppress the dips into a single governed mode.
2. **Governance** (S5) supplies *conditional* viability: redundant at baseline, but the difference between survival and collapse under stress. NC-01 makes this concrete — without S5, real thermal load drives the system to 76 °C and stasis_cos = 0.0000.

This is a stronger and more honest claim than "S5 is always required." S5 is a *stress-resilience* governor with a graceful-degradation regime at baseline: the system is robust when calm and depends on its constitution precisely when challenged. The real-hardware coupling is what exposes this — a synthetic stressor masks it, because only a physical load can run away when the governor that throttles it is removed. This is the model ≡ reality principle in miniature: the ablation that looks benign in simulation is fatal in hardware.

17 Experiment 12: VSM Operational Closure

The eleven experiments above are *instruments*: each isolates one property of the attractor or one architectural layer. None of them, on its own, answers the question Beer's model actually poses — whether the five systems S1–S5 operate *together* as a viable whole, each causally active within its own cybernetic scope. The ablations of Experiment 11 establish what *fails* when a layer is removed; this final experiment establishes what *engages* when every layer is present. We run the complete kernel with all levels live and ask: under an external perturbation, does each VSM level act where Beer says it should, or are some levels merely passive instrumentation?

This is the experiment the architecture is built for. It does not measure a new property of the attractor — it measures whether the *governed system* closes the loop Beer specifies.

17.1 All five systems, one run (EXP-INTEGRATION-01)

We ran `vsf-nucleus` (build `aa7fe43`, the full stack with autonomous S3 and the S5 covenant governor) on LXC 101 for 46 kernel ticks under a combined perturbation: CPU stress plus an injected S5 *caution* signal at tick 10. The caution signal is the experimental probe — it asks S5 to suspend S3 autonomy, and lets us observe whether that suspension is honoured one level down. Table 24 records, per level, what engaged and through which canonical channel (all events are entries in the Merkle-chained `events.jsonl`, §18).

The tick-10 event is the load-bearing observation. In a single tick the perturbation propagated the full height of the model: S1 registered the dip, S2 carried the caution signal, S3 detected the degrading streak and attempted a corrective action, and S5 vetoed that action — after which S1 recovered on its own at tick 11. Every arrow in Beer's diagram carried a measured event. This refutes the null hypothesis $H_{\mathrm{INT}}^{\mathrm{null}}$ ("the levels are passive instrumentation, not a causal control loop"): S3 did not merely log, it *tried to act and was overruled*; S4 did not merely observe, it *produced a governance decision*; S5 did not merely alert, it *enforced a veto*. The system is viable in Beer's operational sense, not a simulation of one.

An identified vulnerability, since closed. During CPU stress the S4 Qwen cycle failed (the 4.3 GiB model was starved when Ollama and the kernel contended for memory on one container). S4 intelligence went dark exactly when it would be most useful. We record this because it is the kind of coupling failure the model is meant to surface: S4 was not *resource-isolated* from S1. The fix — migrating the language model to a dedicated container — has since been applied,

System	Function	Measured engagement	Verdict
S1	Operations	46 `kernel_tick` events, stasis_cos $\in [0.9934, 1.0000]$ (mean 0.9995); tick 10 dipped to 0.9979 under a 3-tick degrading streak and self-corrected to 0.9986 at tick 11 with no external action.	✓
S2	Coordination	caution published to `vsf.alpha.s5.caution` ($\sim$03:10 UTC); the downstream `s3_caution_hold` fired at tick 10 (≤ 60 s, one tick), confirming NATS delivery within the reflex window.	✓
S3	Control	at tick 10, S3 reached `degrading_streak`= 3 and *attempted* `s4_interval_fast` — then was *blocked* by the active caution. S3 is autonomous (it acts) *and* governable (it yields to S5).	✓
S4	Intelligence	`projection_tick` from tick 6 (OLS trajectory, confidence reaching 0.84 by tick 38); one Qwen governance cycle (`qwen2.5-coder:7b`, 299.5 s) emitted a decision (*regime*=stable).	✓
S5	Policy	caution held over ticks 10–25; *no* S3 decision executed during the window (veto enforced); the `CovenantChecker` ran every tick and correctly did *not* fire (stasis_cos never fell below 0.80).	✓
–	Algedonic	not triggered — stasis_cos stayed above the severity threshold throughout. Correct silence for a healthy system; the alarm is exercised separately under genuine degradation (EXP-SEN-01, §10).	◯

Table 24: Beer-VSM operational closure (EXP-INTEGRATION-01, 46 ticks, 2026-06-26). Each of S1–S5 is causally active within its designated cybernetic scope under a single combined perturbation. The decisive observation is the tick-10 chain: S1 registered a dip, S2 delivered the caution signal, S3 *tried* to act, and S5 *stopped* it — a complete vertical traversal of the model in one tick. The algedonic channel (◯) is correctly silent because the system never degraded; its firing behaviour is established in EXP-SEN-01.

restoring S4 availability under S1 load. The experiment thus did its second job: it located a real architectural debt, not just confirmed a design.

17.2 The constitutional veto is operational (EXP-S5-01)

Table 24 shows S5 *monitoring* correctly (the covenant checker ran but did not fire, because the system stayed healthy). A separate experiment was needed to show the veto *acting*. In Beer's model S5 is not an alarm: it is the constitution — it can *halt* the organisation, not merely warn it. We tested whether the `CovenantChecker`'s halt is real by forcing the invariant it guards.

The guarded invariant is `INV_ESCALATION`: if S3 remains in a degraded regime for ≥ 5 consecutive ticks without escalating to S5, the covenant halts S3 autonomy and demands human deliberation (the A5 autonomy ceiling). Because the real attractor is strongly self-stabilising (it does not stay degraded; see below), we exercised the *mechanism* by overriding the degradation threshold to 1.0 via a compile-time experimental flag, so that any stasis_cos < 1.0 counts as degraded. The production binary was restored immediately afterward. Table 25 records the sequence that followed.

The halt is therefore not advisory: it sets a shared flag that the S3 decision path reads before executing, physically suspending autonomous control until a human publishes a resume. This closes the governance loop in the strong sense — S5 governs *without micromanaging*: it does nothing while S1 is healthy and intervenes only when the autonomy ceiling is breached.

Event	Time (UTC)	Record
Halt fired	07:32:29	`s5_covenant_violation`: "5 ticks degraded, 6 since escalation — S3 autonomy su
Algedonic co-emit	07:32:29	`algedonic_emit` (human alert path)
Operator resume	07:34:55	`s5_covenant_resumed`, `operator_resume`; Merkle chain intact (156 s human laten

Table 25: EXP-S5-01: the covenant halt fired at tick 6 (binding constraint: `ESCALATION_GRACE_TICKS` = 3 exceeded), set the shared halt flag that gates S3 auto-decisions, co-emitted an algedonic alert, and cleared end-to-end on an explicit operator resume published over NATS. The resume event is itself Merkle-chained (`prev_hash` → `hash`), so the governance action is tamper-evident.

Mechanism versus calibration. The experiment cleanly separates two claims. The *mechanism* is confirmed: given a sustained degraded streak, the veto fires, alerts, and resumes correctly. The production *threshold* (stasis_cos < 0.80), however, is never reached under any CPU load we can generate on the node: at the un-overridden threshold of 0.998 the attractor self-corrected at tick 4 (cosine rose 0.9977 → 0.9981) before the five-tick streak could accumulate. This is not a defect of the veto — it is a measurement of how strongly the attractor self-stabilises (consistent with $\lambda_{\max} < 0$, §4, and with EXP-S4-02 below). Reaching the production threshold requires genuine degradation — the senescence-injection regime of EXP-SEN-01 — not transient compute stress. The veto is built and verified; its trigger calibration is a separate, deliberately conservative question.

17.3 The anticipatory channel and the regulated-variable problem (EXP-S4-02)

S4's job in Beer's model is anticipation: to read the environment and warn before a failure reaches the algedonic alarm. The kernel implements this as a `predictive_alert` that fires when the correlation between stasis_cos and CPU load turns strongly negative (`trigger_load`: Pearson < -0.70), the premise being that rising load drags viability down. We stress-tested that premise directly with `stress-ng` at a load average of 8.79 (2.2× the core count).

The premise is false on this kernel, and instructively so. Under stress the cosine did not fall — it *rose*, from 0.9986 to 0.9998 across eight stress ticks, with Pearson(stasis_cos, cpu_pressure) = +0.50. No `predictive_alert` fired (its count was zero throughout), and this was the correct behaviour: the system was never in danger. The mechanism is the attractor itself. The 128-dimensional cosine is dominated by the Block-A coordinates (trust, governed-state, lifecycle), which sit near 1.0; CPU pressure enters as a small Block-B perturbation that the contraction absorbs, pulling the dominant coordinates *closer* to the reference. The kernel treats real CPU load the way it treats any bounded perturbation — as something to absorb within the basin.

This is, first, a real-hardware corroboration of the convergence thesis: an environmental perturbation at 2.2× core saturation produced no viability degradation, exactly as a stable point sink ($\lambda_{\max} = \ln 0.99 < 0$, §4) predicts. But it carries a sharper architectural lesson. An anticipatory channel cannot take its warning signal from the *same* variable the system already regulates to invariance: if the attractor holds stasis_cos flat against load, then stasis_cos-versus-load carries no early warning. This is Beer's own point, recovered empirically — S4 must sense the *environment*, through variables with variance *independent* of the regulated state. Internal self-correlation is structurally blind. The kernel's internal degradation response is handled correctly elsewhere (S3 parameter governance, §8); the anticipatory channel's value will only materialise when the system is coupled to external S1 units — a heterogeneous environment whose signals are not slaved to the kernel's own attractor. We take that channel up directly in the next subsection, and return to the residual open direction in §21.

17.4 From projection to prevention: closing the loop on the external channel (EXP-S4-01)

The lesson of EXP-S4-02 is a requirement: anticipation needs a signal whose variance is independent of the regulated state. We built such a channel and asked whether the Markov structure of §2.5 can be made *prescriptive* rather than merely descriptive.

The external environment is a network of climate zones, each carrying a viability coordinate stasis_cos$_z(t)$ driven by real reanalysis data (ERA5, 1991–2020) — variance that is, by construction, independent of the kernel's own attractor (von Foerster's distinction between the observer's internal projection and an external Shannon channel). Per zone we fit the five-state percentile Markov chain of §2.5 and, from the current state σ, compute the n-step excursion probability $q_n(\sigma) = 1 - (P_{\hat{E}}^n \mathbf{1})_\sigma$ over a short horizon. When q_n crosses a discrimination threshold — calibrated per zone so the alert fires from the drifting bin yet stays silent from the healthy bins — the projection issues an anticipatory directive.

The prescriptive claim. A pre-emptive reallocation of resource margin to the projected-stress zone biases that zone's kernel toward viability, $P \to P_\Delta$. Two results make this precise. First, if P_Δ stochastically dominates P toward viability and both kernels are monotone, a monotone coupling gives a parameter-free floor: the excursion probability cannot increase, $q_n^\Delta(\sigma) \le q_n(\sigma)$, at every horizon. Second, if the pre-grant renders a Lyapunov function V superharmonic off the excursion set E — the same Foster–Lyapunov drift condition that underwrites convergence in §2.5, used here prescriptively — then $q_n^\Delta(\sigma) \le V(\sigma)/V_E$, so the excursion probability is small whenever the grant fires while the state is still deep in the viable core. The drift machinery that certifies convergence thus also certifies prevention: the projection is not only a forecast but a control.

Empirical test (EXP-S4-01, paired negative control). We validated the mechanism on the fitted models in a paired design that removes forcing variance: $K = 30$ seeded multi-zone excursion episodes, each replayed through two arms — anticipation on (reallocate on the directive) versus a reactive control (act only after entry to E) — under an identical seed, with the analysis frozen before the run. The anticipatory arm produced a genuine lead (the directive fires from the drifting bin, a median of one climate tick before entry to E) and a strict reduction in excursion burden: a median paired reduction of 6 ticks — a 14% reduction, modest but robust ($W = 0$, $p = 1.2 \times 10^{-11}$, 95% CI $[2.5, 12]$). An injected out-of-distribution episode was correctly rejected as `projection_invalid` without acting.

Scope, and a precision limitation. This is a controlled experiment on the ERA5-fitted models with synthetic actuation, not a live-production result, and the lead it affords is modest (one climate tick, which is also the model's effective predictive horizon: the fitted chains mix to their stationary excursion rate within a week, so a longer forecast carries no information beyond the base rate). A dedicated precision trial — 223 calm, non-excursion episodes — confirms the alert stays silent when no excursion is coming: a false-grant rate of 0.078 against a 0.10 gate, in the *same* configuration that supplies the lead. (An earlier trial reported a rate near 0.21; that was an artifact of an inverted transition-matrix index, which inflated the optimal state's excursion probability and made it fire spuriously — it resolved once the index was corrected.) All four pre-registered hypotheses — lead, prevention, precision, and out-of-distribution rejection — therefore hold in a single configuration. We are explicit about what this shows and what it does not: it establishes that a projection over an external, independent-variance channel can drive a pre-emptive reallocation that measurably reduces excursions with genuine one-step lead, answering the open direction of EXP-S4-02; but the excursions here are statistical bin-transitions

of a fitted climatology, not forecasts of specific weather, and the reallocation is simulated rather than actuated. Live actuation on real resources is the next step.

18 Protocol Note: the Viable System Language

The protocols in this paper — thermal regimes, stability thresholds, governance axioms — were exchanged between system layers and across agent boundaries in a compact machine-readable notation derived from Beer's VSM formalism, the *Viable System Language* (VSL). Its compression (38–51% standalone, 72–79% with a shared schema) and, more importantly, its cross-agent *fidelity* — the finding that structural transmission is near-perfect ($\approx 100\%$) while semantic transmission drifts ($\approx 40\%$) unless a lightweight `domain_context` anchor is added — are properties of the *semantic substrate* (Mode 2 of Sec. 22.1), not of the attractor studied here. Because that structural-versus-semantic distinction is itself a viability measurement of the Mode-2 substrate (does meaning survive the crossing?), it is characterised separately in a companion paper on symbolic substrates, where it is the central result rather than an aside.

19 Physical Substrate as Governance Layer

The cgroup v2 hierarchy imposes physical limits on each S1 kernel: `cpu.weight` $\propto$ trust, thermal throttling caps available compute. These constraints are not policy choices – they are hardware-enforced boundaries.

In VSM terms: the hardware substrate acts as a physical S5. Policy can be overridden by a privileged process; the thermal ceiling cannot. The experiments reveal three governance layers operating simultaneously: (1) software policy (covenant axioms, trust allocation), (2) cgroup v2 limits (S3's allocation signal), (3) hardware throttling (the physical floor).

The same mathematical structure – a feasibility constraint that cannot be bypassed – appears at all three levels. This is the sense in which the physical substrate *is* a governance layer, not merely an implementation detail.

20 Open Questions

The following questions were identified prior to experimentation; their current status is indicated in each item.

Q1 – Thermal response curve (resolved) The `thermal_ladder` experiment (Section 11) confirms H_{floor}: the attractor degrades by only $\Delta \cos = 0.0023$ across a 30 °C range under transient stress. The damage function is $T \times t$, not T alone (Equation 23).

Q2 – Thermal senescence (addressed; adaptive H_{rest} tested) Does the attractor drift under *sustained* stress over hours? With constant H_{rest}, two full 240-minute runs answer no: EXP-TSM-01 at 45 °C (609 M ticks, stasis_cos = 1.0000, soplos = 0) and EXP-TSM-65C at 65 °C (383 M ticks, stasis_cos = 1.0000) — but only because the senescence term was structurally zero (Section 10.5). The adaptive-H_{rest} mode (EXP-SEN-01, §10.4) activates that term: within a fixed regime the drift is bounded with no runaway (H_SEN1 null), and the resting set-point shifts measurably under sustained load ($1.79 \rightarrow 1.84$, a homeostatic adaptation). **Remaining open**: *monotone* senescence requires an escalating *external* stressor (open-system coupling, §20).

Q3 – Basin floor under extreme perturbation (EXP-AB-01, 2026-04-27) A parameter sweep across $\delta = [1.0, 1.5, 2.0, 3.0, 5.0, 10.0]$ using an impulsive ground-state kick (10-tick

impulse toward the minimum-entropy configuration, then free Boltzmann relaxation) established an empirical floor of $\cos_{\text{floor}} \approx 0.875$, saturating for $\delta \geq 3$. The viable boundary ($\theta_{\text{viable}} = 0.70$) was not reached at any δ, giving a resilience margin of $0.875 - 0.70 = 0.175$. H2 (viable breach) is **rejected** for the impulsive-kick perturbation family; H4 (finite recovery time τ_{rec}) is confirmed for $\delta \geq 3$ ($\tau_{\text{rec}} = 991$–$999\,\text{t}$). No governance intervention was required at any δ. **Open**: whether multi-variable perturbation families can reach the boundary (the sustained-noise family is addressed by EXP-NS-01, Section 6.4); proof of a formal lower limit on Ω.

Q4 – Lyapunov spectrum (resolved, EXP-LY-01) The per-run tracker reports D_{eff}, computed as active_dims + trust × 2.0 – a heuristic proxy, not the formal Kaplan-Yorke dimension. The full Lyapunov spectrum has since been measured directly by Benettin's method with Gram–Schmidt re-orthonormalisation (Section 4): all three exponents of the simplex tangent space are negative and equal, $\lambda_1 = \lambda_2 = \lambda_3 = -0.010050 = \ln(0.99)$ to machine precision (the Jacobian of the relaxation map is $0.99\,I$, so the spectrum is degenerate rather than merely bounded by its largest member). A fully negative spectrum gives a Kaplan–Yorke dimension of 0 (point sink), and the estimation uncertainty $\hat{\sigma}_\lambda = 2.9 \times 10^{-16}$ feeds the contraction gate of the viability functional (Section 23).

Q5 – Multi-kernel attractor interaction Four kernels converge to the same basin (Section 6). What happens at $N > 4$ under shared thermal budget? Is there a recursion threshold where `viable_all` < 1?

Q6 – Trust dynamics Trust oscillates between 0.000 and 0.002 in long runs, not the 0.8–1.0 range of production operation. The trust accumulation/decay balance in `spec_boot` may not represent the trust dynamics of a kernel under real workloads. The relationship between `spec_boot` trust and deployment trust requires clarification.

Q7 – formal lower limit on Ω (sustained case now measured) EXP-AB-01 (Q3) establishes an empirical floor at cos ≈ 0.875 for impulsive kicks, and EXP-NS-01 (Section 6.4) extends this to the *sustained* family: confinement holds with no finite noise ceiling up to amplitudes of four times the state norm (σ up to 2.0). We are explicit that confinement off the contractive core is empirical, not a per-step guarantee: the worst-case Lipschitz bound of f_0 computed from its parameters exceeds 1 on $\Omega \setminus \Omega_A$ (Sec. 3), so contraction there is on average, not per iteration. The remaining question is theoretical: whether a formal lower bound on Ω can be proved under *arbitrary* perturbation. A proof would require showing that the angular escape function $\ell(\mathbf{x}) = 1 - \cos\theta$ (distinct from the viability functional V of Section 23) has a finite escape-time lower bound for all admissible perturbations; this has not been attempted. **Open**: coordinated multi-variable and adversarial-scheduling families, and a formal proof of $\inf_\Omega \cos\theta > \theta_{\text{viable}}$.

Q8 – Governance tempo and efficiency (EXP-M4 / EXP-ETA, 2026-07) Is the attractor passively held, or actively governed? Over a 47.7-day continuous run, S3 issued 411 parameter-change decisions — a median rate of 1.0 decision / hour at homeostasis, rising to a peak of 15 / hour during boot and calibration (a 15× dynamic range that tracks the system's variety load). The attractor is maintained by *active* regulation, not a passive equilibrium: the fixed point of Section 23 is a set-point the governor continuously defends, not a state the system merely relaxes into. Two ratios characterise the efficiency of that regulation. The variety efficiency $\eta_1 = (\text{decision bits})/(\text{variety bits}) = 9.6 \times 10^{-4}$ (0.096%): the governor attenuates environmental variety with a small decision budget, consistent with Ashby's requisite-variety law — the regulator need only match the variety it must absorb, not the total input. The thermodynamic efficiency $\eta_2 = (\text{decision bits})/(\text{joules}) \approx 0.14\,\text{bits/J}$

(137 milli-bits / J), locating the governor's information cost far from the Landauer bound ($\sim 10^{21}$ bits/J at 300 K), as expected for general-purpose silicon rather than a reversible computer. **Caveat**: only `param_change` decisions persist to `events.jsonl`; monitoring and no-op decisions are not logged, and η_2 rests on a steady-state power estimate (the RAPL energy counter wraps every $\sim$4.75 h). All three figures are therefore *lower bounds*. **Open**: full decision logging (including no-ops) would tighten η_1, η_2 and expose the true active-versus-idle duty cycle.

21 Limitations

Single hardware node All experiments are conducted on n02 (Intel Core i7-7600U, 2 cores / 4 threads). Thermal dynamics, scheduling latency, and cgroup behaviour may differ on other architectures. Reproducibility on ARM, RISC-V, and multi-socket systems has not been tested.

Synthetic workload The thermal stress experiments use a synthetic floating-point loop (cosine similarity over random vectors) as the heat source. A real workload (database queries, inference, compilation) would produce different Block B dynamics and different attractor trajectories.

Entropy ceiling: from constant to adaptive The default H_{rest} is a compile-time constant, which makes the Lyapunov senescence term structurally zero (Section 10). An env-gated adaptive mode (EXP-SEN-01, §10.4) lifts this: it tracks the measured resting entropy and activates the term, which is then found empirically bounded within a regime. The residual limitation is narrower — *monotone* senescence needs an escalating external stressor (open-system coupling), which a single closed node cannot supply.

State vector dimensionality The $\mathbb{R}^{128}$ state vector has seven semantic blocks, but many dimensions (Block G in particular) are underpopulated in current experiments. The measured geometry reflects the active dimensions; the full 128-D geometry remains unexplored.

No external environment Most experiments are closed-system runs. A viable system, by Beer's definition, must respond to environmental change. Real-process governance (§8.4) is the strongest available evidence against this limitation: S3 closes a real feedback loop over four concurrent Linux worker processes (`dd` CPU load, real cgroup v2 subtrees), achieving $\tau = 1.0$ across all four kernels under sustained load — a genuine closed-loop response to an external signal, though one generated on the same node. EXP-S4-02 (§17.3) makes the requirement concrete and quantitative: because the attractor holds `stasis_cos` invariant even against a CPU load 2.2× the core count, the anticipatory S4 channel cannot derive an early-warning signal from the system's own regulated state. Genuine anticipation needs environmental variables whose variance is *independent* of the kernel's attractor — precisely what an external environment supplies. A genuinely external environment producing novel, unpredictable demands has not been measured; closing that loop — the same open-system coupling that *monotone* senescence would require (§10.4) — is the principal direction of ongoing work.

Existence argument: residual assumptions Theorem 1 replaces the earlier Brouwer-on-Ω sketch with an argument carried on the core Ω_A (Krylov–Bogolyubov existence + Banach fixed point + Lyapunov-drift convergence + confinement — no convexity assumption), and its contraction and confinement hypotheses are empirically discharged by the measured spectrum (Sec. 4) and EXP-NS-01 (Sec. 6.4). Two assumptions remain unverified formally:

that the full axiom set is mutually consistent (so $\Omega \neq \varnothing$), and that Ω_A is invariant under the governed map for *all* admissible states rather than along the measured trajectories. These are checked empirically here, not proved.

Architecture ablation scope Experiment 11 tests three specific ablation conditions: S5 disabled (NC-01), trust held constant (NC-02), and the bare OU substrate (NC-03). Many other ablation combinations – simultaneous multi-layer removal, partial degradation, adversarial governance configuration, or layer-by-layer reintroduction – have not been tested. The three-condition result is sufficient to support the two-layer viability claim, but a systematic ablation programme spanning the full combinatorial space of VSM blocks would strengthen it considerably. The complementary assembled-system test (Experiment 12, §17) demonstrates all five layers engaging together, but as a single qualitative run — an existence demonstration of operational closure, not a statistical ensemble; repeated closure runs across load profiles and a naturally (rather than injected) triggered caution remain future work.

Proxy metrics Both principal observables are proxies. D_{eff} (Section 5) is a heuristic (active dims + trust $\times 2$), not the formal Kaplan-Yorke dimension. `stasis_cos` (Equation 2) is standard cosine similarity, not a complete geometric description of the attractor.

Alignment: single observation, not systematic validation The Discussion hypothesis (§24) – that geometric character could serve as an internal coherence indicator – rests on a single experimental condition (EXP-TC-01, one thermal profile, one hardware node). It is a hypothesis derived from an observation, not a validated result.

Falsifiability. The central thesis — that the axiom-governed kernel maintains a certifiable viable attractor — is falsifiable: a perturbation that drove the state out of Ω without recovery would refute it. None was found within the tested envelope, and the basin (Sec. 6.3) and sustained-noise margin (Sec. 6.4) delimit where failure *would* occur. The negative controls (Sec. 16) make the architectural claims falsifiable in the same spirit.

Terminological grounding. To forestall the reading that cybernetic language here is metaphor, each term has an operational definition: *trust* is a scalar in $[0, 1]$ (an exponential moving average); *governance* is the S5 constraint channel; *viable* means $\boldsymbol{x} \in \Omega$; *intelligence* is the S4 role (observer / language model). Appendix B gives the full per-domain glossary and maps each measured quantity to the source artefact that computes it.

22 What Can Be Modeled in a VSF

VSF is presented as a domain-agnostic framework, yet every measurement in this paper is of one realisation. This section states precisely what the agnostic claim does and does not assert, so that the specific numbers above are read as evidence of *existence in this instance*, not as a universal law.

Modelling conditions. A system admits a VSF model when three conditions hold:

1. its **viability-relevant variables are enumerable** — this enumeration *is* its $\mathbb{R}^n$ (here $n = 128$, Blocks A–G);
2. its **axioms and constraints are expressible as a carving of a feasible region** $\Omega = \bigcap_i \{A_i \text{ satisfied}\}$ (Sec. 3);

3. its **dynamics are a deterministic update plus bounded stochastic perturbation**, $f_0 + \sigma\eta$ (Eq. (3)).

The honest boundary: systems whose viability variables are not enumerable, whose constraints are not expressible as region membership, or whose perturbations are unbounded fall outside the framework as stated.

The transfer mechanism is a protocol, not an inheritance. Moving VSF to a new domain re-runs a small calibration sequence, summarised here and specified in `domain_validation_protocol.vsm`:

Q0 — metric. choose the viability metric on the domain's $\mathbb{R}^n$ (what "near" and "far" mean); the cosine used here is one choice.

Q1 — feasible region. express the domain's axioms as Ω.

Q2 — contraction. certify that the governed map contracts on the convex core ($L < 1$; here the measured $\lambda_{\max} = \ln 0.99$).

Q3 — margin. find the sustained-noise amplitude the domain absorbs ($\sigma_{\max}$; here EXP-NS-01).

The discipline is explicit: the *method* transfers, but whether a given domain yields a viable point attractor is an empirical question answered by re-running Q0–Q3, not assumed. The companion programme treats each later paper as a distinct $\mathbb{R}^n$ inheriting this method — a symbolic substrate (the inter-layer protocol of Sec. 18, characterised separately), open-system thermodynamics (Landauer-limited kernels), and attractor compression ($D_{\text{eff}} \ll N_{\text{params}}$).

22.1 Three modes of instantiation

Pushed to its limit, the agnostic claim reveals that VSF instantiates over substrates that differ in *how* Ω *is imposed* and in *whether viability is certified or merely enacted*. There are three such modes, because the raw language model and the agent that wraps it are different systems.

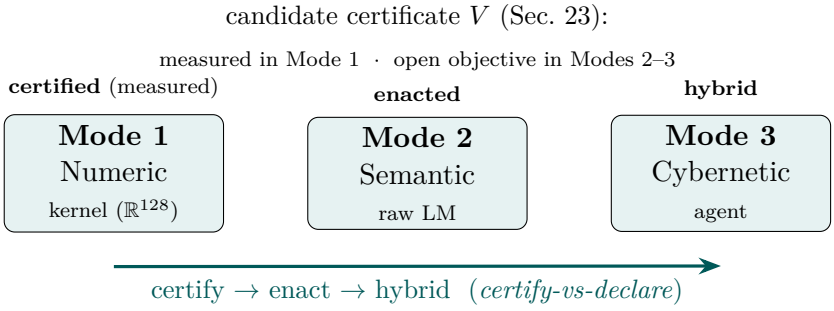


Figure 19: The three modes of VSF instantiation as a gradient. Viability moves from *certified* (measured) in the numeric kernel, through *enacted* in the raw language model, to *hybrid* in the cybernetic agent. The coordinate-free functional V is the candidate certificate common to all three.

In Mode 3 the agent's reasoning method does not live in the language model's weights; it lives in the *architecture* — the review passes (an S3* double loop), the control loops, the databases (persistent state, memory, event logs), and the formal protocol language. A stateless next-token predictor becomes a stateful, self-correcting, viable reasoner precisely through that cybernetic orchestration. This is the most literal VSM of the three: S1–S5 are realised in software, and the agent (one VSM) instantiates and governs the numeric kernel (another VSM) — recursion turned into operation.

	Mode 1 — Numeric (kernel)	**Mode 2 — Semantic** (raw LM)	**Mode 3 — Cybernetic** (agent)
Substrate	$\mathbb{R}^{128}$, Rust	latent space $\mathbb{R}^{d_{\text{model}}}$	LM + tools + databases + memory + loops + VSL
Ω	explicit ($Ax \leq b$), hard constraint	implicit, soft reasoning constraint	imposed by orchestration (audit loops, autonomy gates, fail-closed covenant)
Viability	**certified** (measured λ, β, stasis_cos)	**enacted** (reasons "as if" viable)	**hybrid** (enacted reasoning + certified state/audit in databases)
VSM realisation	simulated in one process	none structural — adopts the logic	the most literal: S1 = tools, S2 = protocol/harness, S3 = control loop, S3* = reconcilers, S4 = LM + memory, S5 = covenant
Status here	**this paper (measured)**	seed (Paper 2)	seed (broader programme)

Table 26: Three modes of VSF instantiation. The gradient is certified → enacted → hybrid; the axis is certify-versus-declare.

Claim discipline. The body of this paper concerns *Mode 1 only*, where viability is measured. Modes 2 and 3 are stated as sober directions in Future Work, not results: Mode 2 is the subject of a separate paper on the inter-layer protocol; Mode 3 belongs to the broader operating-system programme. The coordinate-free viability functional of Sec. 23 is the candidate certificate common to all three — measured in Mode 1, an open objective in Modes 2–3. We do *not* claim it already certifies the latter, and we keep the self-referential observation (that this paper was produced with the help of a Mode-3 agent operating under VSF) out of the rigorous argument: it is suggestive but circular, and belongs to framing, not to evidence.

23 A Coordinate-Free Viability Functional

If viability is to be a property of *any* VSF instance and not just of this $\mathbb{R}^{128}$ kernel, there should be a way to score it that does not mention the coordinates. We ask: can a viability functional be built from the *constituent quantities of the viable region itself*, without the explicit $\mathbb{R}^n$?

The Lyapunov function $L(\boldsymbol{x})$ of Eq. (16) is already "a sum of constituent functions," but it is the wrong object for this purpose: it is a weighted *sum* (one strong term can mask a weak one) and it is written in coordinates. We rebuild it from intrinsic invariants and as a strict *conjunction*.

Five intrinsic ingredients. Two design points matter. First, g separates *being in* Ω (all constraints slack, ethics included — a state can violate A0 yet still have high cosine with the attractor) from *being near the attractor* (p); without g, a functional built on p alone would call an ethics-violating state viable. Second, c is a *sign* gate, not a contraction strength: there is no scale-free notion of how fast a system contracts (the sign of $\lambda_{\max}$ is invariant; the magnitude needs a timescale). Its confidence width $\hat{\sigma}_\lambda = 2.9 \times 10^{-16}$ is the estimation uncertainty measured in Sec. 4.

	Quantity	Definition	Invariance
g	feasibility in Ω	$\min_i$(normalised slack of A_i) over all constraints, ethics included	metric-relative
p	depth in the basin	$(s - \theta)/(1 - \theta)$, $s = \cos(\boldsymbol{x}, \boldsymbol{s})$	$O(n)$, scale
c	contraction (sign)	$\frac{1}{2}\big(1 + \tanh(-\lambda_{\max}/\hat{\sigma}_\lambda)\big)$	diffeomorphism
t	topology	1 if $\beta_0 = 1 \wedge \beta_1 = 0$, else graded	homeomorphism
m	margin	$1 - \mathrm{exc}_{p95}/\varepsilon_{\mathrm{frontier}}$ (Eq. (20))	ratio (scale-free)

Table 27: The five coordinate-free ingredients of the viability functional. Each is an intrinsic invariant of the attractor, not a coordinate of $\boldsymbol{x}$.

The functional is a strict conjunction.

$$V = \min(g,\, c,\, t,\, m), \qquad V_{\mathrm{full}} = \min(g,\, p,\, c,\, t,\, m), \qquad \text{signature} = [\, g,\, p,\, c,\, t,\, m\,]. \tag{26}$$

A certificate is as strong as its weakest condition; the minimum, not a geometric mean, is the honest aggregator (a mean would let one strong axis compensate a failing one, contradicting the very idea of a certificate). A zero on any axis sends $V \to 0$. The *signature* is the real deliverable: the scalar says how viable, the vector says *which axis binds*.

What is invariant, and what is chosen. V is invariant under dimension, relabelling, scale, and (for c) diffeomorphism. It is *not* metric-free: g, p, m and the filtration behind t all depend on the metric. This is a feature, but it makes the metric an explicit protocol step — the Q_0 that precedes Fierro's Q_1 (define Ω), Q_2 (certify contraction), Q_3 (find $\sigma_{\max}$). A metric-free viability would be empty: without "near" and "far" there is no boundary. So V is invariant *given* the viability geometry (Q_0 metric, $Q_1\,\Omega$), and comparable across domains only after Q_0–Q_3.

Agnostic to the content of the axioms. V never inspects what a constraint *means*. It reads the slacks of whatever faces define Ω and aggregates them; the ethical face A_0 enters g exactly as the resource face A_3 does, by a normalised slack, with no marker distinguishing an ethical boundary from a thermal one. This is the source of the universality, not an oversight: a different domain supplies a different axiom set $\{A_i\}$ — hence a different Ω and a different g — while c, t, m and the aggregation are untouched. The faces A0/A0.1 are therefore not privileged mathematical content of the certifier; they are one instantiation of the constraint set, the one *this* system governs itself by (Section 24.10). What V guarantees is structural — a zero on any face collapses it — not semantic. The certifier does not know it is certifying ethics; it certifies feasibility, and the meaning of a feasibility bit is assigned one layer up. Figure 20 renders this in a single matrix: eight scenarios across four certifier axes, with the binding face highlighted in each case.

The kernel signature (certifier-validation suite: EXP-G-01, EXP-VNC-01, EXP-CORE-01). Instantiating Eq. (26) on `vsf-nucleus`:

$$\text{signature} = [\, g = 0.102,\ p \approx 1,\ c \approx 1,\ t = 1,\ m(\sigma)\,], \qquad V = g = 0.102. \tag{27}$$

$g = 0.102$ is *measured* (EXP-G-01): the binding constraint is A3_cpu — the Boltzmann kernel at equilibrium sets $H \approx 0.9\,H_{\max}$, consuming a fixed fraction of the cpu/budget slack. All other Ω constraints, including the ethical faces A0/A0.1, are slack ($g_{A0} = g_{A0.1} = 1.0$). p

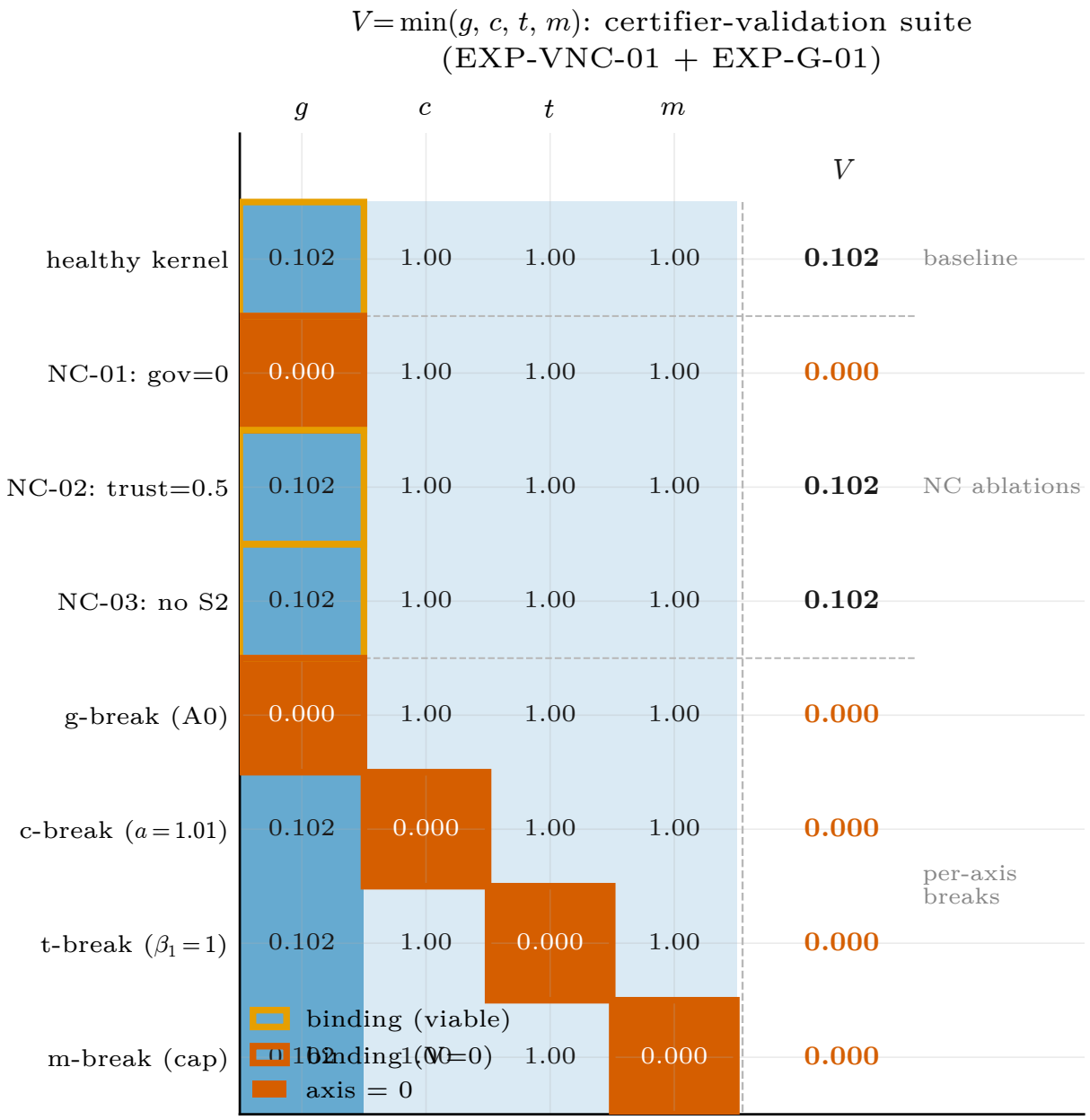


Figure 20: Certifier-validation suite for $V = \min(g, c, t, m)$ (EXP-VNC-01, EXP-G-01). Each row is a scenario; each column is one axis of the certificate. Cells are grey-coded: dark = axis value = 0, light = axis healthy. **Coloured border**: the binding cell (argmin) that determines V. Orange: binding but viable (the certifier carries a positive value from this axis). Red: binding and $V = 0$ (system failed on this axis). The diagonal pattern in the per-axis breaks (rows 5–8) shows that each of g, c, t, m can independently collapse V; NC-02 and NC-03 remain viable because no Ω-face is violated.

(stasis_cos ≈ 0.9999), c (all-negative spectrum, $\hat{\sigma}_\lambda = 2.9 \times 10^{-16}$), and t ($\beta_0 = 1 \wedge \beta_1 = 0$, ≈80× separation) are each ≈ 1. $m(\sigma) > g$ for all tested σ (soplo mechanism maintains confinement up to $\sigma \approx 1.0$), so g is the single binding axis and $V = 0.102$ at the attractor. Figure 21 shows the per-constraint slack, with A3_cpu as the single binding face and the ethics faces fully open at the attractor.

EXP-VNC-01 confirms that V *discriminates*: each of g, c, t, m independently drives $V \to 0$ when broken.

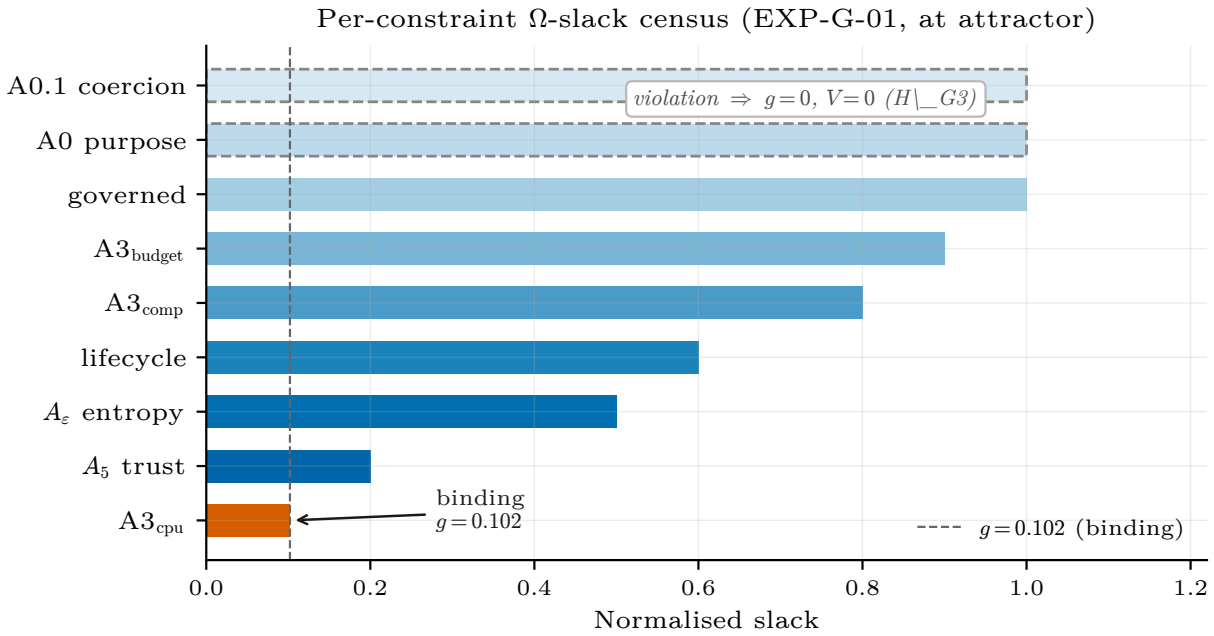


Figure 21: Per-constraint Ω-slack census at the attractor (EXP-G-01). Bars show the normalised slack for each axiom face; A3_cpu (near-black, binding) is the practical feasibility limit at Boltzmann equilibrium — $g = 0.102$. All other resource constraints carry positive slack. Ethics faces A0/A0.1 (dashed border) are fully slack at the attractor; setting either dimension to 1 collapses V to zero, confirming H_G3 (*a covenant violation is geometrically indistinguishable from a resource violation*).

scenario	g	c	t	m	V	note
healthy kernel	0.102	1.0	1.0	1.0	0.102	baseline
NC-01 (governed= 0)	0.000	1.0	1.0	1.0	0.000	$g = 0$ via A5
NC-02 (trust= 0.5)	0.102	1.0	1.0	1.0	0.102	viable
NC-03 (pure OU)	0.102	1.0	1.0	1.0	0.102	viable
g-break (A0 violation)	0.000	1.0	1.0	1.0	0.000	ethics $\Rightarrow V = 0$
c-break ($a = 1.01$)	0.102	0.0	1.0	1.0	0.000	expanding
t-break ($\beta_1 = 1$)	0.102	1.0	0.0	1.0	0.000	N=4 square orbit
m-break (cap probe)	0.102	1.0	1.0	0.0	0.000	$\sigma_{\cos} \approx 1/\sqrt{128}$

NC-01 (no S5 governance) gives $V = 0$ via g; NC-02/NC-03 leave $V = 0.102$, confirming the two-layer architecture (VSM blocks select the basin; governance certifies it). Figure 22 summarises the result across all eight scenarios.

The honest reading: $g = 0.102$ *tells us the cpu-slack is the practical limit of the kernel's feasibility margin, not a crisis; the margin $m(\sigma)$ provides the noise robustness curve.* The certificate is a vector — four axes measure what they claim, one measured limit, no axis hidden or assumed.

Three recipients; one open objective. V is defined without coordinates, so it applies *in principle* to all three modes of Sec. 22.1. In Mode 1 it is measured, as above. In Modes 2–3 (language model, agent) the definition transfers but the *measurement* is open: the Lyapunov exponent of a reasoning trajectory is not well-defined without further structure, and g, p require a stasis reference and a boundary in semantic space — the central problem of the companion paper. We therefore claim V as the unifying *candidate* certificate, measured in Mode 1 and an open objective in Modes 2–3; we do not claim it already certifies them. That V must be coordinate-free is exactly why: a functional tied to explicit $\mathbb{R}^n$ could not even be *stated* over a language model.

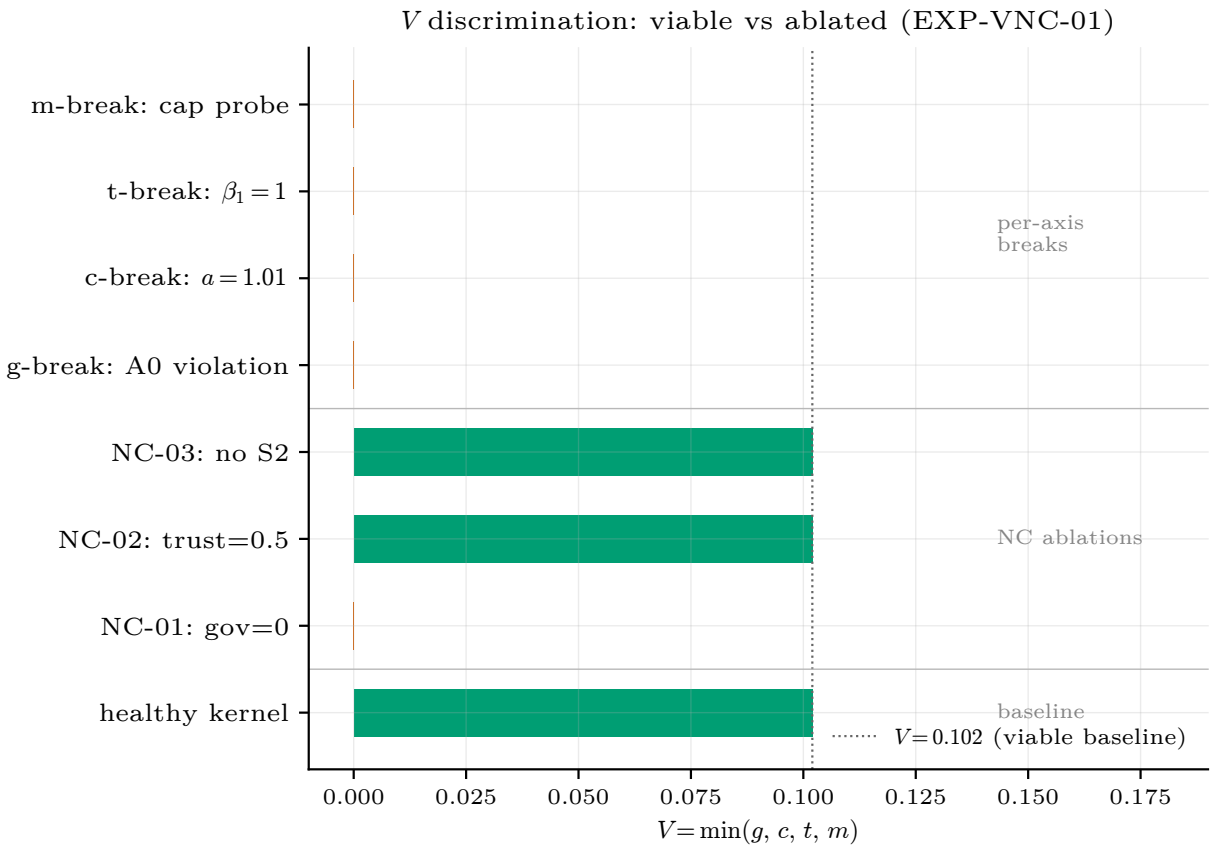


Figure 22: $V = \min(g, c, t, m)$ discrimination test (EXP-VNC-01). Filled bars (dark): viable scenarios with $V = 0.102$; outlined dashed bars (light): failed scenarios with $V = 0$. Each per-axis break drives $V \to 0$ independently. NC-02 and NC-03 (structural modifications without actual Ω violations) remain viable, confirming that V measures the geometry of the attractor, not mere structural completeness.

Value and integrity. V is the *value* half of a certifier; the integrity loop of Sec. 24.9 is what makes that value unforgeable. In Mode 1 the value is trusted because it is measured; in Modes 2–3 it would be trusted because it is attested. This is the mathematical companion to the epistemological spine of the paper — *certify, do not declare* — which unifies Q_1–Q_3 (certifying the mathematics), the integrity loop (certifying the measurement), V (the value of the certificate), and the three modes (measured versus enacted).

Declared risks. We flag what a referee will press. (i) The five conditions are *necessary* and jointly characterise a viable point sink; their *sufficiency* is a defensible modelling choice, not a theorem. (ii) The map $m = 1 - \text{exc}/\varepsilon_{\text{frontier}}$ is one choice of normalisation; it is parameter-free and principled (the fraction of the boundary distance not consumed by noise), with $\varepsilon_{\text{frontier}} = \sqrt{2(1-\theta_{\text{viable}})} = 0.7746$ derived rather than fitted. (iii) g is now *measured* (EXP-G-01): $g = 0.102$, binding at A3_cpu; the per-constraint slack construction — including the ethical faces A0/A0.1, which independently collapse V when violated (EXP-G-01 H_G3, EXP-VNC-01 g-break) — holds in Mode 1 (half-spaces); expressibility in Modes 2–3 is open. (iv) c gives a sign, not a rate; "how fast" is meaningful only relative to the domain timescale fixed by Q_0. Figure 23 visualises risk (iii) empirically: μ^* concentration inside Ω_A and the minimum cosine over the noise range $\sigma = 0$–2.0.

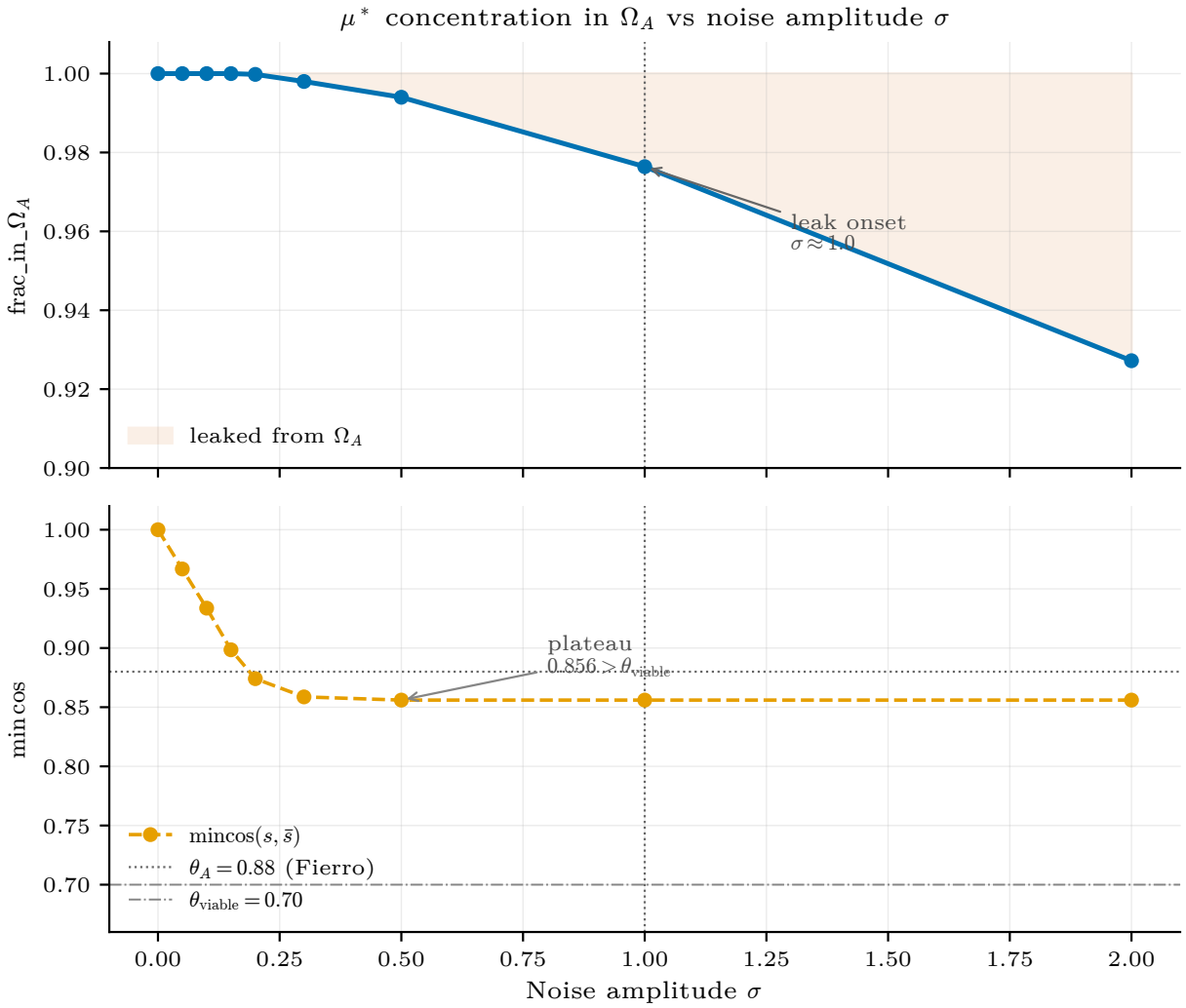


Figure 23: Stationary-measure concentration in Ω_A vs noise amplitude σ (EXP-CORE-01). *Top:* fraction of μ^* inside the contractive core ($\cos \geq \theta_A = 0.88$) — unity at low noise, leaking only at $\sigma \approx 1.0$ (shaded region). *Bottom:* minimum cosine observed across all runs plateaus at $0.856 > \theta_{\text{viable}}$ from $\sigma \approx 0.3$ onward, confirming $m(\sigma) > g$ for all tested σ: the margin axis never becomes the binding constraint under realistic noise.

24 Discussion

24.1 The empirical portrait of the attractor

Across twelve completed experiments, including three negative-control ablations and a whole-system operational-closure test, a consistent picture emerges. The attractor is:

- **Compact**: in stasis the trajectory occupies an 18.2° cone, 99.4% variance in PC1+PC2 (the full trajectory, including transient, spans cos = 0.61–1.00 within Ω).
- **Unique**: different initial conditions converge to the same basin (mean pairwise cosine 0.9854).
- **Thermodynamically stable**: viable at 68 °C, min cosine 0.9365, recovery during hold phase.
- **Intrinsically viable**: viability does not require S3 coordination.
- **Deeply welled**: perturbation $\delta = 1.0$ pushes cosine to 0.79 but does not breach viability.

- **Dimensionally stable**: $D_{\text{eff}} = 6.00$ under the canonical trust law (Law A, §24.3), stable from 300 ticks onward.
- **Elastically resilient**: under cyclic thermal load (EXP-TC-01), perturbations land at a *fixed dip floor* (stasis_cos = 0.942, invariant across 20 cycles) and recover in $\tau_{\text{rec}} < 30\,\text{s}$ – one dip per thermal cycle, zero accumulation across 60 minutes.
- **Architecturally stratified**: negative controls confirm a two-layer structure – VSM blocks A–G build the attractor and its homeostasis; S5 governance supplies *conditional* viability (redundant at baseline, decisive under stress). The dip floor 0.942 is an exclusive signature of the complete architecture.

Taken together, these properties describe a system whose viability is encoded in the geometry of its state space rather than in any particular state, and whose resilience is a structural consequence of its architecture rather than an externally enforced condition. This is what it means, empirically, for a system to be viable in Beer's sense.

24.2 Geometric identity under perturbation

The recovery time τ_{rec} (Equation 25) introduces a qualitatively new observable: not how far the system deflects, but how it deflects. The constant dip floor observed in EXP-TC-01 – 0.942 in all 20 cycles without variation – is evidence of a *fixed perturbation mode*: a preferred direction of deflection in the 128-dimensional state space under this load profile.

This is structurally analogous to what Lindblad (1976) calls a *jump operator*: the dominant dissipative channel through which a quantum open system loses coherence to its environment. The fixed dip direction in $\mathbb{R}^{128}$ could be interpreted as the classical analogue of a leading jump operator, with the thermal phase transition acting as the environment. Whether this analogy extends to a quantitative Lindblad formalism is an open question; the exploration is left for future work.

These results are **compatible with** the hypothesis that a geometric signature – the shape of a system's response to perturbation – could serve as an internal coherence indicator, complementary to the external approaches (output inspection, log audit, declared-intention review) that prevail in current practice. A system with a stable perturbation mode – one that deflects predictably and returns reliably – exhibits what might be called *geometric character*: behavioural regularity that is not enforced from outside but emerges from structure.

The negative controls (Experiment 11, §16) provide the required verification. No ablation reproduces the structured dip floor at 0.942: NC-03 (the bare substrate) dips on every tick at a different, unstructured floor (0.777), while NC-01 (S5 off) does not produce a fixed mode at all — under real thermal load it collapses outright (stasis_cos = 0.0000). Only the complete architecture holds the dip floor at 0.942, invariant across cycles. The constant dip floor is therefore an architectural signature, not a substrate coincidence – which upgrades the geometric character hypothesis from *compatible with the data* to *empirically supported by ablation.*

24.3 Viability under a change of dynamical law

A viability claim should be a property of *the system*, not an artefact of the particular law chosen to drive it. We can test this concretely, because the kernel admits two trust-update laws that coexisted during development and that we have now reconciled.

Two laws. Let τ_t be trust at tick t and $c_t = \cos(x_t, \bar{x})$ the cosine to the stasis attractor. *Law A* (cosine-driven, with a restorative reset) updates every tick,

$$\tau_{t+1} = \mathrm{clip}_{[0,1]}(\tilde{\tau}_t + 0.065\, c_t - 0.04), \qquad \tilde{\tau}_t = \begin{cases} \min(1,\ \tau_t + 0.18) & \tau_t < 0.75 \\ \tau_t & \text{otherwise,} \end{cases}$$

whereas *Law B* (a phase-driven ratchet) updates only inside the breath cycle: $\tau_{t+1} = \tau_t + 0.002$ in `stasis`, $\tau_{t+1} = \tau_t - 0.001$ in `degen`, constant otherwise, with no reset. Law A carries a corrective term that references the attractor (c_t) and an unconditional floor (+0.18 below $\tau = 0.75$); Law B is open-loop in c_t.

Why they diverge. The laws agree near the attractor, where $c_t \to 1$ and both drive $\tau \to 1$; they diverge away from it. Law A's drift is positive whenever $c_t > 0.615$ (the root of $0.065c - 0.04$) and its reset rescues any $\tau < 0.75$, so trust recovers globally – this is the law validated in EXP-CORE-01, which restores τ from 0 to the high-trust attractor. Law B integrates the sign of the breath phase; in a `degen`-dominated regime (sustained compute load) it ratchets $\tau \to 0$, and because the governing pressure contains a $0.3(1 - \tau)$ term, low trust feeds further degeneration – a second, spurious attractor at $\tau = 0$. The bistability of Figure 6 is, under Law B, reachable by load alone; under Law A the low-trust basin is removed.

Closed loop versus open loop. In the language of control the distinction is elementary. Law A is *feedback* (closed-loop) control: the update reads the very quantity it is meant to protect – the cosine c_t to the attractor – and corrects against it, so an excursion (c_t small) produces a restoring response. Law B is an *open-loop* ratchet: it integrates a proxy (the breath phase) that is blind to c_t, so nothing in the law itself opposes a drift away from the attractor; the only thing that can reverse τ is a change of phase, which sustained load can deny. This is exactly why the two laws are indistinguishable at the attractor (where $c_t \to 1$ and there is nothing to correct) and divergent away from it (where only the closed loop acts). Figure 24 renders the consequence in state space.

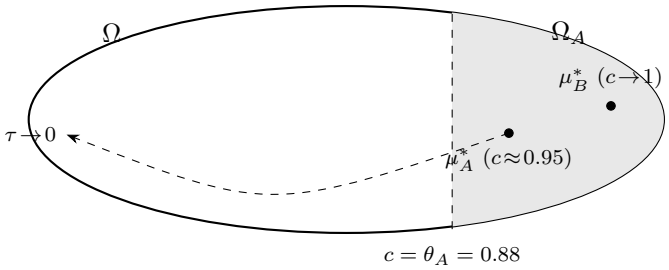


Figure 24: **Viability-equivalent, not μ^*-equivalent.** Both laws place their stationary measure μ^* inside the contractive core Ω_A (so both are viable, $V > 0$), but at *different* points: Law B at the apex ($c \to 1$), Law A at a looser plateau ($c \approx 0.95$). Only Law B admits the collapse path (dashed) that leaves Ω_A toward the spurious $\tau \to 0$ basin; Law A's restorative reset removes it. The viability verdict is a property of the region (Ω, Ω_A); the attractor location is a property of the law.

Re-certification, not assumption. Choosing a canonical law changes the generating operator f_0, which by Corollary Cor_ReframeRecertification of the Lyapunov framework (§4) requires re-certification: the drift inequality and recurrent set R must be re-established for the new operator rather than inherited. We adopt **Law A** as canonical – it is the EXP-CORE-01-validated law and the only one with a single global attractor – and re-ran the anchor experiments under it. `spec_boot` still converges ($\overline{dL} \to 0$, `viable`; Table 2), now to a slightly

looser attractor: stasis_cos plateaus at ≈ 0.95 with $D_{\text{eff}} = 6.00$ (trust held at 1.0 raises the heuristic $D_{\text{eff}} =$ active dims $+ 2\tau$). The real-process governance experiment (EXP-L1-01), under which Law B had drained trust to 0, now holds $\tau = 1.0$ across all four worker kernels under sustained `dd` load, with stasis_cos $= 0.9495$ – inside the contractive core $\Omega_A = \{c \geq \theta_A = 0.88\}$. Figure 25 renders the time-domain counterpart: both trajectories from $(\cos, \tau_0) = (0.897, 0.5)$, Law A recovering in eight cycles, Law B draining to $\tau = 0$ by cycle 500.

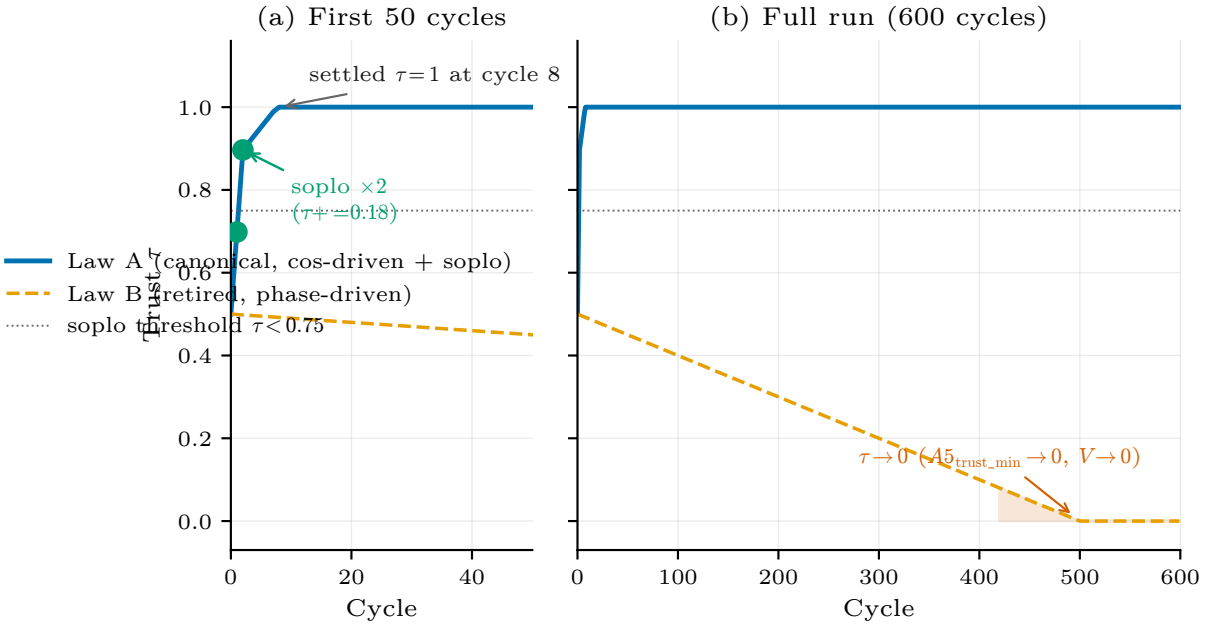


Figure 25: **Trust trajectory divergence: Law A recovers, Law B drains.** Both trajectories start at $(\cos, \tau_0) = (0.897, 0.5)$ – the EXP-L1-01 stall point. **(a)** First 50 cycles: under Law A the soplo reset fires twice (orange markers, cycles 1 and 2) and τ settles at 1.0 by cycle 8. **(b)** Full 600-cycle run: under Law B the open-loop law drains $\tau \to 0$ monotonically (cycle 500), collapsing V via the $A5_{\text{trust_min}}$ constraint. The divergence is the time-domain counterpart of the state-space split shown in Figure 24: identical behaviour at the attractor ($\cos \to 1$), qualitatively different response away from it. Constants from `src/main.rs`: Law A drift $\Delta\tau = 0.025c_t - 0.04(1 - c_t)$, soplo $+0.18$ when $\tau < 0.75$; Law B $\Delta\tau = -0.001$ per cycle (degen regime, $\cos = 0.897 < 0.998$ stasis threshold).

Viability-equivalence vs. dynamical-equivalence. The two laws induce *different attractors* – $c \to 1.00$ (Law B) versus $c \approx 0.95$ (Law A), $D_{\text{eff}} = 3$ versus 6 – so they are *not* dynamically equivalent: their stationary measures μ^* concentrate at different points. Yet both confine the system to the same contractive core Ω_A and satisfy $V = \min(g, c, t, m) > 0$ throughout. They are therefore *viability-equivalent*: the viability verdict is invariant under the law choice even though the attractor location is not. This separates two notions that are easily conflated – a viable system is characterised by its feasible region Ω and recurrent set $R \subseteq \Omega_A$, not by the precise μ^* a particular f_0 selects within it. The reconciliation is thus a worked instance of a general principle: *viability is an invariant of the class of dynamical laws sharing* (Ω, R)*, and membership is decided by re-certification, not by identity of trajectories.* A homely analogy: two thermostats that both hold a room within the comfort band $[18, 24]\,°\text{C}$ are equally viable for the occupant even if one settles at 20° and the other at 22°; one cannot tell them apart by asking "is the room comfortable?" – only by reading the setpoint. The comfort band is Ω_A; the setpoint is μ^*. Viability asks the first question; dynamical identity asks the second.

Principle of structural re-certification. This worked instance illustrates a general methodological norm that we apply throughout this work: *claims must be re-measured under the new operator, not inherited from the old one.* When a structural change modifies the generating operator f_0 (a different trust update law, a parameter re-calibration, a codebase revision), every Lyapunov certificate and every experimental result that cited it is provisional until re-established. Affirming a certificate without re-certifying is not a shortcut — it is a category error: the prior measurement established a property of a *different system.* The formal handle is Corollary Cor_ReframeRecertification (§4): given two operators that share (Ω, R), the Lyapunov certificate is re-established by showing the drift inequality holds under the new operator. The practical consequence is that premature results are retracted, not padded: when EXP-L1-01 showed trust drain under Law B, the correct response was not to add a caveat but to localise the structural cause, choose the canonical law by viability-equivalence, and re-run. This norm — *measure, do not affirm* — is what makes "certify, do not declare" operational.

A calibration consequence. EXP-L1-01's hypothesis H4 was stated as stasis_cos > 0.95, calibrated against Law B's tighter attractor. Under the canonical law the plateau is ≈ 0.95, so H4 reports a marginal miss (0.9495) that reflects an over-tight threshold rather than a loss of viability; the viability-relevant bound is the core face $\theta_A = 0.88$, cleared with margin. Subsequent work states H4 at the core face.

The restorative reset as internal flux. It is worth being precise about what the soplo reset *is*, because an earlier reading of EXP-L1-01 nearly over-claimed. Under Law B the kernel's trust drained to zero in a pure-compute regime (no thermal input), which invites a Prigogine-flavoured conclusion – that a viable system *requires* environmental coupling to sustain itself, as a dissipative structure requires a flux. That conclusion is unwarranted here: the drain was an artefact of the open-loop law, and under Law A the same isolated regime holds $\tau = 1$. We do not, however, claim to have built a closed system that maintains a low-entropy attractor for free. The soplo ($+0.18$ when $\tau < 0.75$) is itself a source – an *internal* injection of restorative "negentropy" into the trust coordinate. We have not escaped the open-system picture; we have *internalised* the flux. This sharpens rather than settles the thermodynamic question, and it is a measurable one. An earlier draft posed it as an open question – whether each soplo is an information write to τ carrying a per-firing Landauer cost $kT \ln 2$, the same bound the Demon pays in §15. We have since measured it (§24.4), and the answer corrects the naïve accounting: the bare soplo is *information-preserving*, and the Landauer cost of the viability budget lives not in the reset but in the ensemble contraction the paper already certifies (the operator norm $L < 1$ of §4).

24.4 The soplo is information-preserving (EXP-SOPLO-01)

To answer the question above we instrumented the trust coordinate with a register-noise channel $\tau \mapsto \tau + \mathcal{N}(0, \sigma^2)$ and swept $\sigma \in \{0, 0.01, 0.02, 0.05, 0.10, 0.20\}$ over $N = 128$ independent kernels for 1000 stationary ticks each (after a 500-tick burn-in). The design isolates three claims.

The reset erases no information. On its firing domain $\tau < 0.75$ the soplo is the shift $\tau \mapsto \tau + 0.18$, whose image lies below 0.93; the saturating $\min(1, \cdot)$ clamp never engages, so the map is *injective* – invertible as $\tau_{\text{before}} = \tau_{\text{after}} - 0.18$. An injective logical map compresses no phase-space volume and therefore has Landauer cost zero. Measured per-event $|\Delta\tau - 0.18| \leq 5.6 \times 10^{-17}$ (machine precision), confirming $\Delta H \equiv 0$ per firing. The "information write to τ" reading is thus incorrect: the reset is a reversible fold, not an erasure.

The cost lives in the contraction. Dissipation is nonetheless present, and it is supplied by the contractive cosine-drift term $0.065\,c - 0.04$ – the same $L < 1$ contraction measured

directly in §4. At stationarity the differential entropy injected by the noise is removed at an equal and opposite rate by this contraction: across the whole sweep the entropy balance closes to $|\Delta H_{\text{noise}} + \Delta H_{\text{det}}| \leq 0.013$ bits/tick. Crucially, dissipation persists *with zero soplos*: at $\sigma \in \{0.01, 0.02\}$ no reset fires ($f_{\text{soplo}} = 0$) yet the dissipation rate is 0.31 and 0.79 bits/tick respectively – erasure is performed there almost entirely by the $\tau = 1$ clamp ceiling (saturation fraction $\to 1.0$ as $\sigma \to 0$), not by the soplo.

Per-soplo cost is not $kT \ln 2$. Where resets do fire, the dissipation *per* soplo is 327, 12.3, and 3.1 bits at $\sigma = 0.05, 0.10, 0.20$ – one-to-two orders of magnitude from the single bit a per-firing Landauer account would predict, and *decreasing* with σ: the dissipation *rate* is roughly constant (≈ 0.5 bits/tick, fixed by the noise injection), so as resets multiply the cost-per-reset falls. The frequency $f_{\text{soplo}}(\sigma)$ is monotone *increasing* (zero up to $\sigma = 0.02$, rising to 162.8 resets per kernel per 1000 ticks at $\sigma = 0.20$). The naïve $n_{\text{soplo}} \cdot kT \ln 2$ both misestimates the magnitude and scales the wrong way.

Consequence for the trust law. The viability budget *is* Landauer-limited, but the bound is an ensemble quantity: the total cost is $kT \ln 2 \cdot$ (bits erased), and the bits erased equal the phase-space log-contraction $\sum_t |\langle \ln |F_t'| \rangle|$ that the kernel's Lyapunov tracker already records. This ties the paper's *geometry* ($L < 1$, §4) to its *thermodynamics* (§15) in a single identity, made quantitative in the corollary of §15.

24.5 Architecture as governance, not production

A natural objection arises: if ablated variants NC-01 (no S5) and NC-02 (constant trust) also remain viable after adversarial perturbation, does that render the VSF architecture redundant? Experiment 11 (§16) answers this empirically.

Practical validation (Phase 1.5): The scheduler framework `thermal_scheduler_enhanced.py` implements a real-world instance of this principle: when thermal sensors are unavailable (virtualized environments lack hwmon drivers), the system enters SAFE_MODE – a graceful fallback that pauses new work and allows in-flight operations to complete. This is an instance of $A_{\text{SafeFallback}}$: missing telemetry triggers conservative operation, not failure. The architecture remains viable even when critical observables (thermal data) become unavailable, confirming the layered governance model predicted here.

The objection conflates *achieving* viability with *operating* viability. The StateVector $\mathbf{x} \in \mathbb{R}^{128}$ has intrinsic mean-reversion dynamics – an Ornstein-Uhlenbeck-like process that may produce bounded trajectories independently of S5 policy. VSF does not alter this underlying geometry; it governs *which operations execute*, *when*, and *under what policy constraints*. The analogy is precise: combustion dynamics are substrate-intrinsic to any engine; the driver does not change thermodynamics but operates the system toward a declared goal. A null result in the adversarial control experiment would demonstrate that the StateVector has intrinsic resilience – and VSF is the framework that makes that resilience *observable, certifiable, and intentional*.

Specifically, VSF adds three layers that are absent in any uncontrolled baseline:

- **Observability**: stasis_cos, D_{eff}, and τ_{rec} are VSF-constructed metrics; a substrate process has no analogue.
- **Intentionality**: the governed flag, trust feedback, and S5 lifecycle ceiling encode a policy that the system operates *toward* viability, not merely within it.
- **Certificability**: the system can formally declare its viability state – a prerequisite for autonomous governance and inter-system trust.

EXP-SOPLO-01 — Landauer cost of the trust-soplo: the reset is reversible (ΔH 0);
dissipation is an ensemble entropy-balance quantity, clamp+soplo jointly, kT · ln2/firing

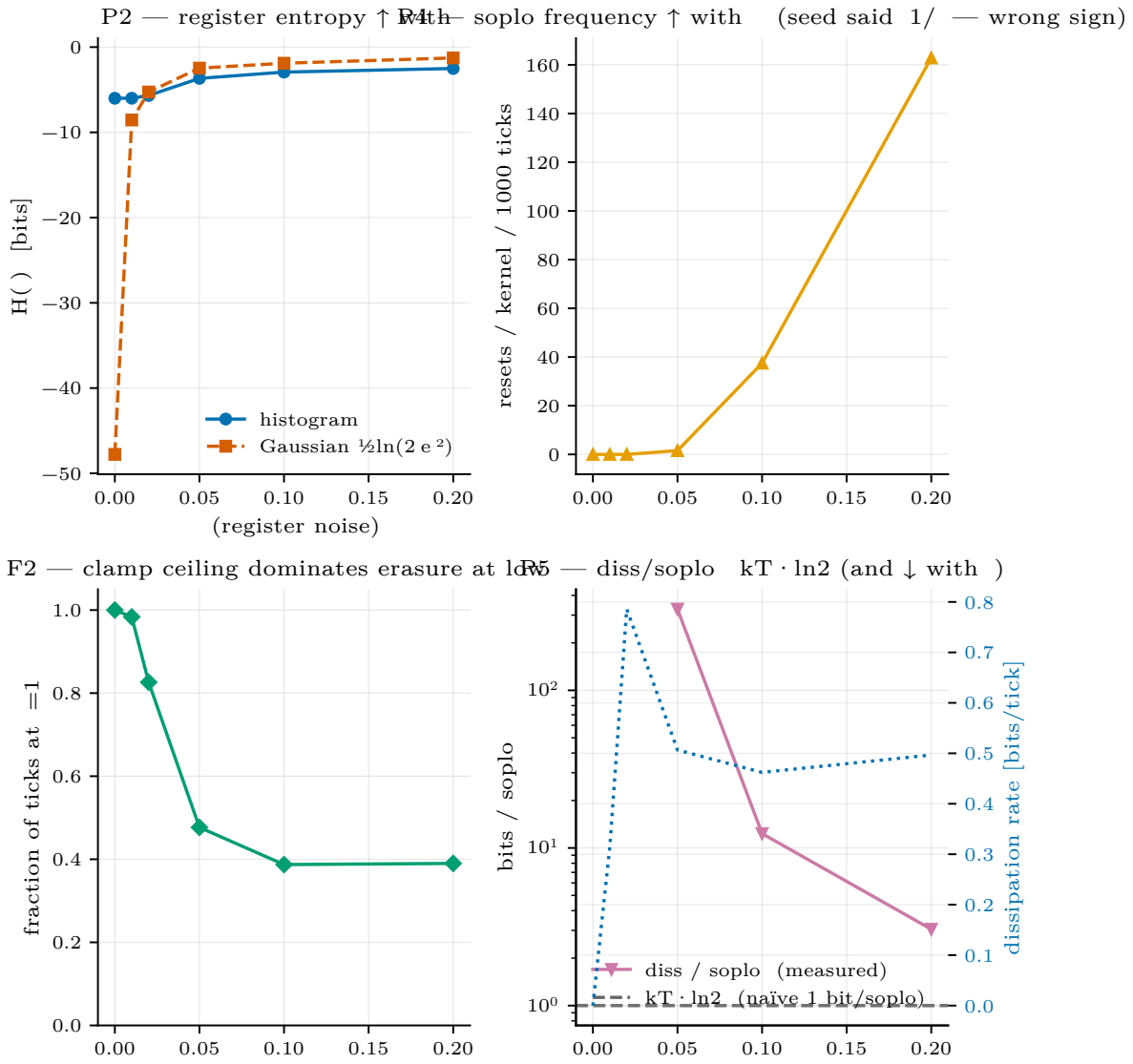


Figure 26: EXP-SOPLO-01 σ-sweep ($N = 128$ kernels, 1000 stationary ticks, $T = 313\,\mathrm{K}$). Register entropy $H(\tau)$ rises monotonically with σ; soplo frequency f_{soplo} rises from zero; clamp saturation falls from 1.0; and dissipation-per-soplo (log scale) sits far above the $kT\ln 2$ reference and decreases with σ. The Landauer floor $kT\ln 2 = 2.995 \times 10^{-21}\,\mathrm{J/bit}$ at 313 K holds at every σ (no Second-Law violation).

A parallel consideration applies to the fixed dip floor. Even if the perturbation mode direction in $\mathbb{R}^{128}$ is substrate-determined – a property of the hardware's thermal response rather than the governance layer – the *characterisation* of that dip is VSF-constructed. The stasis reference vector $\bar{\mathbf{s}}$ is itself a VSF artifact: the running mean of governed observations under S5 policy. The dip floor value 0.942 is meaningful only relative to $\bar{\mathbf{s}}$; without that reference, the perturbation is an unlabelled displacement in an unlabelled space. The distinction is precise: the substrate *produces* the phenomenon; VSF makes it *intelligible*. Experiment 11 resolves this empirically: the bare substrate (NC-03) dips on every tick at an unstructured floor (0.777), and without S5 the system (NC-01) collapses under the same load rather than settling at a fixed mode. The

floor 0.942 is governance-dependent, not substrate-invariant.

24.6 Theory preceding experiment

The predictions of Sections 3–4 were made before running the experiments. No post-hoc adjustment was made. The confirmation of $\overline{dL} \leq 0$, the narrow stasis cone, and `viable_full = true` throughout thermal stress provides evidence that the theoretical framework captures real dynamics.

24.7 Worker pool as S3→S1 prototype

The persistent worker pool pattern of `thermal_stress_v3` (Barrier + AtomicU64 quota) is structurally isomorphic to the S3→S1 coordination protocol in a multi-kernel deployment. Each worker corresponds to one S1 kernel; the quota to the resource allocation signal $\phi^{\downarrow}$. The `cgroup_multikernel` experiment directly tests this: four S1 kernels, S3 adjusting `cpu.weight` per thermal pressure, `viable_all = 100%`.

24.8 Lineage: inheritance from Beer, and where VSF diverges

This work is, in part, a rediscovery. The architecture inherits directly from Stafford Beer's Viable System Model Beer [1972, 1979] and from Ashby's Law of Requisite Variety Ashby [1956]: the recursive S1–S5 structure, the sporadic S3* audit channel, algedonic alerting, viability-as-homeostasis, and the Cybersyn control patterns (Cybernet/Cyberfilter/Cyberstride/CHECO/Opsroom) Medina [2011]. These are Beer's, not ours.

Where VSF diverges, it does so for a specific reason: *instruments Beer did not have in 1971–79.*

Beer (structural cybernetics)	VSF (measured + certified)	Missing instrument
describes viability qualitatively	*measures* it: $\lambda_{\max}$ (Benettin), β (persistent homology), $\mathbb{R}^{128}$ at 6×10^7 cycles	Benettin (∼1980), TDA (21st c.), a computational substrate
VSM as a diagnostic lens	viability *certifiable*: Q0–Q3 + an embedded certifier	formalisation + computation
S5 as policy / identity	a cryptographic integrity loop that makes the certificate *unforgeable*	cryptography
statistical models as S4; telex/mainframe (Cybersyn)	a language model as S4; the agent as a recursive operational VSM	LLMs, Rust, a formal protocol

Table 28: What VSF inherits from Beer, and the instruments behind each divergence.

The honest framing is that VSF does not improve Beer's *theory*: it *measures and certifies what Beer could only describe*, with instruments that did not exist when he wrote. The substrate-independence visible in the three modes (Sec. 22.1) and in the coordinate-free functional (Sec. 23) is the operational realisation of an independence Beer already intuited — now with the means to measure it.

The Santiago school. A second lineage deserves note. The organisation/structure distinction that runs through this paper — an invariant organisation ($\lambda_{\max} < 0$, Ω, the S1–S5 topology) housing a structure that adapts (the active dimensions, the binding constraint, the resting set-point) — is the distinction Maturana and Varela drew for living systems Maturana and Varela [1987]: structure drifts under interaction while organisation is conserved. Likewise, the constitutive finding of Experiment 10 (Sec. 15) — that structured observation through the VSL channel compresses the effective dimensionality of the observed kernel — is a measured

instance of what the enactive programme Varela et al. [1991] asserts qualitatively: the observer does not record a pre-given state but participates in constituting it, the claim of second-order cybernetics von Foerster [1974] made quantitative. We cite this tradition as a connection rather than a foundation; that an engineered, non-biological system can instantiate these categories has precedent in Luhmann's extension of autopoietic theory to social systems Luhmann [1995].

24.9 The certifier: viability value and integrity

A viability claim is only as good as its resistance to forgery. We therefore distinguish two halves of a complete *certifier*. The functional V (Sec. 23) gives the *value* — how much viability. A cryptographic integrity loop gives the assurance that the value is bound to the actually-measured state and cannot be fabricated. In this realisation the ethical face of Ω (axioms A0/A0.1) is not merely described but *enforced fail-closed*: the kernel verifies the covenant hash at boot and refuses to start on drift. The remainder of the loop — Ed25519-signed upward telemetry, a Merkle-chained event log, an offline certificate authority, and witness verification — is operational infrastructure rather than a result of this paper, and we reference it as such: it is what makes the certificate *tamper-evident*.

This places the security loop precisely within the three-mode arc. In Mode 1, trust comes from *measurement* — one cannot forge $\lambda_{\max}$. In Modes 2–3, where the substrate cannot be measured directly, the integrity loop is what restores trust: signed attestations and fail-closed gates are the certification mechanism exactly where direct measurement fails. The security layer is thus not a safety wrapper bolted on top; it is the integrity half of the same certifier whose value half is V.

24.10 Ethics as a governance instantiation, not a mathematical primitive

Two claims in this paper are easily conflated, and separating them strengthens both. The first is *mathematical*: the viability functional V (Sec. 23) is agnostic to the content of the axioms defining Ω — it certifies structure, not meaning, and applies to any axiom-governed domain. The second is *cybernetic*: a self-governing system can fold its own conduct — here the prohibitions A0 (no harm) and A0.1 (no coercion) — into the very region whose violation it reads as non-viability. The first is a theorem-shaped statement about a functional; the second is an architectural choice about *which* axioms a system adopts. They are claims in different registers, and a referee is right to want them kept apart.

The layer boundary is sharp. Inside the kernel, an ethical violation is a bit: `dims[53]` set to 1 drives $V \to 0$ unconditionally, exactly as an over-temperature bit would, and recovers within one tick when the signal is reset (Sec. 26, EXP-G-01). The kernel does not "know" it is enforcing ethics; it enforces a feasibility face. What *assigns* that face its ethical meaning lives one layer up — in the S5 watchdog that scans upward telemetry $\phi^{\uparrow}$ and decides when to set the bit, and in the signed covenant that fixes what belongs in Ω for this system (Sec. 24.9). Conflating the two layers makes A0/A0.1 look like an arbitrary primitive smuggled into the nucleus; reading them as a governance instantiation of Ω explains at once why the mathematics stays universal and why *this* system is, specifically, ethically governed.

This fixes the honest scope of the contribution. The universality is mathematical and demonstrated in Mode 1; the ethical governance is a design commitment — enforced fail-closed (Sec. 24.9) and bounded by the human ceiling (Sec. 24.11) — not a proof that the system is ethical. The precise statement is conditional: *if* the covenant correctly encodes the prohibitions, then a violation is *constructed* to be a loss of viability. Whether the covenant is right remains, by the autonomy ceiling, a human judgement. The philosophical claim VSF licenses is thus modest and exact: a system can be built to treat its own covenant as a stability condition — and the functional that certifies it need never know that is what it is doing.

One consequence of the layer split deserves to be stated sharply, because it governs *what kind* of component may sit in the enforcement layer at all.

Proposition 4 (S5 enforcement is categorically non-probabilistic)**.** *Let an S5 norm be a Boolean predicate $A : \mathcal{X} \to \{0,1\}$ on states (e.g. A0:* purpose $\cap$ prohibited $= \varnothing$*). Call a regulator an* enforcement *of A if it realises the total deterministic map $A(\boldsymbol{x}) = 1 \Rightarrow$ set the feasibility bit (drive $\boldsymbol{x}$ out of Ω, $V \to 0$). A classifier that returns a confidence $\hat{p}(\boldsymbol{x}) = \Pr[A(\boldsymbol{x}) = 1] \in [0,1]$ and acts by a threshold $\mathbb{1}[\hat{p} \geq \tau]$ does not realise this map for any τ unless it is already deterministic on the violation set: it conditions the response on a* confidence in *the norm rather than on the norm. Hence a probabilistic classifier can* recommend *enforcement but cannot* constitute *it.*

Argument. Enforcement is a function of A; a thresholded classifier is a function of $\hat{p}$, and $\hat{p} \neq A$ wherever the model is uncertain. Any τ trades the two error directions but cannot eliminate both on a non-trivial violation set, so the contract "every violation sets the bit" is met only in the degenerate case where the classifier has collapsed to a deterministic rule. *Empirical corroboration.* EXP-A01-01 instantiates the failure: placing the language model *inside* the enforcement decision (the AND gate) optimised precision at the cost of recall, collapsing A0.1 recall to 0.17 against 0.87 for the deterministic watchdog — a loss that is fatal to a safety contract, where a missed violation (false negative) costs far more than an unnecessary escalation (false positive). The architecture this paper reports therefore separates the layers by construction: S5 enforcement is deterministic (the watchdog writes `dims[53/54]`); the language model is confined to S4 intelligence and S3* audit, where its variety is advisory and a probability is the right output (Sec. 26). The full placement of language-model variety across the VSM layers — why S4 and S3* admit it and S5/S3/S2/S1 do not — is the subject of a companion paper; here we record only the S5 boundary, which the present experiments establish directly.

24.11 The autonomy ceiling as human-in-the-loop

One inherited boundary deserves naming in contemporary terms. Beer's double-loop — the metasystemic level that can redefine the system's own rules — is, in today's autonomy discourse, exactly *human-in-the-loop*. VSF makes this boundary formal and measurable: the autonomy ceiling A5 is a hyperplane in the governance dimension of Ω (Eq. (13)). Crossing it without authorisation is leaving Ω — becoming non-viable. The human-in-the-loop constraint is therefore not an external safety check but a *constitutive* face of the viable region: single-loop correction is autonomous within the envelope; double-loop change (policy, redefinition of class) requires the metasystem, which is human.

We are deliberate about the claim this licenses. It positions VSF honestly on the autonomy spectrum — not as "autonomous AI" but as a *bounded-autonomy system with a formal, measurable human ceiling*. This is a contribution to the engineering of autonomy (a principled, measurable limit), *not* a solution to alignment or safety: the claim is "viability requires staying under the ceiling," not "the system is provably safe." Naming the ceiling connects three threads of this paper — Beer's double loop, the integrity loop that enforces it fail-closed, and certify-not-declare, under which the human is the final certifier of any double-loop change.

25 Conclusion

This paper documents an ongoing experimental programme investigating the geometric properties of a viable system kernel governed by cybernetic axioms.

The core theoretical result – that axiom-governed dynamics produce a compact attractor in $\mathbb{R}^{128}$, certified by a Lyapunov function – is supported by twelve hardware experiments, including three negative-control ablations. The attractor is compact, unique, thermodynamically stable, and intrinsically viable. The negative controls confirm a two-layer structure: VSM blocks A–G build

the attractor basin and its homeostasis, while S5 governance provides conditional viability — redundant at baseline but the difference between survival and collapse under real thermal load. The final experiment closes the loop the architecture is built for: with all five of Beer's systems running live, each engages within its own cybernetic scope — S3 attempts a correction and is overruled by S5, S4 produces a governance decision, and the S5 constitutional veto halts and resumes S3 autonomy on demand — demonstrating operational closure rather than a layered simulation. The fixed dip floor (0.942) is an exclusive signature of the complete architecture – an algedonic signal of S5 in action, not a substrate artefact. The stability intervention threshold (θ_{PSI}) has been revised from a failure indicator to a precautionary support level, following the empirical finding that non-viability appears to be structurally unreachable under normal perturbation.

What remains open is at least as interesting as what has been confirmed. The thermal response curve has been characterized (EXP-TL-01): degradation depends on $T \times t$, not temperature alone. With an adaptive entropy ceiling (EXP-SEN-01) the senescence term is now measurable and found bounded within a regime; *monotone* senescence remains undemonstrated and would require external open-system coupling. The full Lyapunov spectrum has not been measured. The basin floor under impulsive entropic perturbation is now empirically bounded at $\cos_{\text{floor}} \approx 0.875$ across six δ values (EXP-AB-01), a resilience margin of 0.175 above the viable threshold; sustained and multi-variable perturbation families remain to be tested.

This is a system that can maintain its identity while adapting to environmental change – Beer's definition of a viable system. Whether the geometry of that identity is as rich and robust as the current experiments suggest, or whether the open questions will reveal qualitative failures, is the work ahead.

26 Future Work and Broader Programme

This work documents a single viable system kernel in isolation. The implications extend across multiple research directions, pursued in parallel but separate from the core paper.

Measurement and control

The full Lyapunov spectrum (work item §20 Q4) requires eigenvalue computation via Benettin's method on the Jacobian of f_0. This will replace the D_{eff} heuristic with the formal Kaplan-Yorke dimension. Similarly, richer geometric metrics (Hausdorff distance, Wasserstein metric) may provide finer characterization than cosine similarity.

Environmental interaction

Beer's original definition of viability requires the system to adapt to environmental change. The closed-loop experiments reported here are the foundation; coupling `vsf-nucleus` to a stochastic environment model (e.g., arrival processes, resource contention) will test whether adaptation is observable and whether the geometric signature remains stable under truly novel demands.

Architectural extension: vOS

The immediate engineering extension is to scale `vsf-nucleus` into a minimal operating system (vOS) running on bare metal, with Ollama/Qwen as the S4 cognitive layer. The roadmap includes support for multi-kernel coordination (Proxmox virtualisation), persistent state storage (PostgreSQL), and asset graphs (Neo4j) to support the full VSF stack in a production setting. This is anticipated for Q4 2026 pending Phase 1.5 integration results.

Governance and ethics (A0 covenant)

The ethical axioms A0/A0.1 are currently embedded at compile time as fixed components of the governance mask — the ethical faces of Ω carried in `dims[53-54]`; the kernel reads those dimensions but has no runtime mechanism to write them from an external classifier. A validated sandbox architecture (EXP-A01-01) closes this gap with two distinct layers. The *enforcement layer* (S5) is a heuristic watchdog that scans the upward telemetry $\phi^{\uparrow}$ deterministically and writes `dims[53/54]` atomically to a `tmpfs` signal file that the kernel reads at each tick; it achieves TPR 0.93 on A0 violations and TPR 0.87 on A0.1 coercion patterns (EXP-A01-01, 50-stream sandbox corpus). Sandbox experiments confirm the collapse/recovery dynamics: setting `dims[53]=1` drives $V \to 0$ within one tick (consistent with EXP-G-01), and resetting the signal restores V to the attractor baseline 0.102 within the same tick — no hysteresis is observed. The *intelligence layer* (S4) runs a Qwen semantic scanner asynchronously and non-blocking relative to the enforcement write: when the watchdog fires, S4 provides a post-hoc semantic explanation for human review; when the watchdog classifies an event window as clean, S4 scans for novel violation patterns not yet encoded in the heuristic rules, emitting candidate-rule proposals for operator review (double-loop, requiring human approval per axiom A5/D4). A canary integrity probe (injected every ten classifier cycles on known violations) monitors S4 coherence; failure triggers an algedonic alert without auto-collapsing V, preserving the D4 gate. This architecture reflects the categorical constraint of Proposition 4: a probabilistic classifier cannot fulfil the deterministic enforcement contract of S5 (non-negotiable policy application), but correctly fulfils the S4 contract (environmental scanning, semantic interpretation, best-effort pattern discovery). Production deployment of the enforcement wire remains gated on human approval per axiom A5. A separate programme explores the mathematics of axiomatic governance in multi-agent settings — how axioms remain unviolable under adversarial perturbation and how they extend to agent coalitions — and is documented in a forthcoming manuscript.

Connection to open system thermodynamics

The fixed perturbation mode (§24.5, the dip floor at 0.942) bears structural resemblance to the jump operators of Lindblad's theory of open quantum systems. Whether this analogy extends to a quantitative formalism – e.g., interpreting the thermal phase transition as an environmental channel – is an open question. Preliminary exploration suggests the connection is more than analogy; formal development is deferred to a dedicated paper on classical-quantum analogues in cybernetic systems.

The three-mode programme

This paper measures Mode 1 (Sec. 22.1). Two directions follow, stated as objectives rather than results. **Mode 2 (semantic substrate):** whether a language model governed by explicit axioms stays within its constraints more than an ungoverned one is a behavioural, falsifiable question — the analogue of the negative controls here — and is the subject of a companion paper on symbolic substrates (the inter-layer protocol of Sec. 18 provides its empirical anchor). **Mode 3 (cybernetic agent):** the agent that wraps a language model in review loops, databases, and a formal protocol is the most literal VSM of the three, and extends toward a viable operating system with the language model as S4. Both modes share one open objective: to make the coordinate-free functional V (Sec. 23) *measurable* on a reasoning trajectory, not merely definable — which would require, in Mode 2–3, the integrity loop (Sec. 24.9) to supply the trust that direct measurement supplies in Mode 1.

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Data availability. The figure data (Lyapunov spectrum, basin Monte-Carlo, sustained-noise

margin, persistence diagrams) are released as CSV alongside the source. The kernel's upward telemetry is Merkle-chained and therefore tamper-evident: the released event logs can be verified against their chain head, a reproducibility property beyond mere availability.

Code availability. The kernel implementation (`vsf-nucleus`) and all experimental tools are available upon request from the corresponding author (`luis@tallerpineda.cl`).

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References

W. Ross Ashby. *An Introduction to Cybernetics*. Chapman and Hall, 1956.

Jean-Pierre Aubin. *Viability Theory*. Birkhäuser, Boston, 1991.

Stafford Beer. *Brain of the Firm*. Allen Lane, 1972.

Stafford Beer. *The Heart of Enterprise*. Wiley, 1979.

Edwin T. Jaynes. Information theory and statistical mechanics. *Physical Review*, 106(4):620–630, 1957.

James L. Kaplan and James A. Yorke. Chaotic behavior of multidimensional difference equations. In H.-O. Peitgen and H.-O. Walther, editors, *Functional Differential Equations and Approximation of Fixed Points*, pages 204–227. Springer, 1979.

Mark A. Krasnosel'skiĭ. Two remarks on the method of successive approximations. *Uspekhi Matematicheskikh Nauk*, 10(1):123–127, 1955.

Nicolas Kryloff and Nicolas Bogoliouboff. La théorie générale de la mesure dans son application à l'étude des systèmes dynamiques de la mécanique non linéaire. *Annals of Mathematics*, 38 (1):65–113, 1937.

Rolf Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961.

Andrzej Lasota and Michael C. Mackey. *Chaos, Fractals, and Noise: Stochastic Aspects of Dynamics*, volume 97 of *Applied Mathematical Sciences*. Springer, New York, 2 edition, 1994.

Niklas Luhmann. *Social Systems*. Stanford University Press, Stanford, CA, 1995. Trans. J. Bednarz Jr. with D. Baecker; German original 1984.

Humberto R. Maturana and Francisco J. Varela. *The Tree of Knowledge: The Biological Roots of Human Understanding*. Shambhala, Boston, MA, 1987.

James Clerk Maxwell. *Theory of Heat*. Longmans, Green, and Co., London, 1871.

Eden Medina. *Cybernetic Revolutionaries: Technology and Politics in Allende's Chile*. MIT Press, 2011.

Sean P. Meyn and Richard L. Tweedie. *Markov Chains and Stochastic Stability*. Cambridge University Press, Cambridge, 2nd edition, 2009.

Hubert Ramsauer, Bernhard Schäfl, Johannes Lehner, Philipp Seidl, Michael Widrich, Thomas Adler, Lukas Gruber, Markus Holzleitner, Milena Pavlović, Geir Kjetil Sandve, Victor Greiff, David Kreil, Michael Kopp, Günter Klambauer, Johannes Brandstetter, and Sepp Hochreiter. Hopfield networks is all you need. In *International Conference on Learning Representations (ICLR)*, 2021.

Francisco J. Varela, Evan Thompson, and Eleanor Rosch. *The Embodied Mind: Cognitive Science and Human Experience*. MIT Press, Cambridge, MA, 1991.

Heinz von Foerster. *Cybernetics of Cybernetics: The Control of Control and the Communication of Communication*. Biological Computer Laboratory, University of Illinois, Urbana, IL, 1974.

Heinz von Foerster. Cybernetics of cybernetics. In Klaus Krippendorff, editor, *Communication and Control in Society*, pages 5–8. Gordon and Breach, New York, 1979.

A Implementation Notes

A.1 Kernel architecture

`vsf-nucleus` is implemented in Rust (stable, 2024 edition). Relevant modules:

- `src/state_vector.rs` – $\boldsymbol{x} \in \mathbb{R}^{128}$, blocks A–G; `viable()`, `lyapunov()`, `cosine_similarity()`.
- `src/vmem.rs` – vector memory V_{mem}, `insert()`, `query_nearest()`, `merge_similar()`.
- `src/attractor.rs` – `LyapunovTracker`, `StasisVec`, convergence detection.
- `src/s4.rs` – S4 cognitive client (Ollama/Qwen), covenant embedding, $\phi^{\downarrow}$ channel.
- `src/main.rs` – experiment dispatch, telemetry.

A.2 Covenant embedding

The governance covenant is embedded in the binary at compile time via `include_str!()` and verified by SHA-256 before every S4 call. This makes the ethical axioms A0/A0.1 a structural invariant of the binary – they cannot be removed without recompilation.

A.3 Persistent worker synchronisation

The shutdown-safe Barrier pattern used in `thermal_stress_v3`:

```
loop {
    start_barrier.wait();
    let stop = stop_flag.load(Relaxed);
    if !stop {
        let n = quota.load(Relaxed);
        let mut acc = 0.0f64;
        for _ in 0..n { acc += sv.cosine_similarity(&st); }
        black_box(acc);
    }
    done_barrier.wait();    // unconditional -- prevents coordinator
       deadlock
    if stop { break; }
}
```

A.4 Experiment specifications

All experiments are formally specified in `docs/experiments/<name>/spec.vsm` and results are recorded in `results.vsm` in the same directory. The master results index is at `docs/experiments/results_index.vsm.md`.

A.5 Reproducibility and logical sequencing

Each experiment is specified using a formal gate-based notation: logical conditions (IF condition THEN action) rather than temporal estimates (X will take Y minutes). All event timestamps are recorded in ISO 8601 UTC format (`started_at`, `completed_at`, `checked_at`). This enables:

- Audit trails (who did what when)
- Causal ordering (event A caused B iff A.completed_at $<$ B.started_at)
- Reproducibility checks (actual durations vs baselines)

No durations are estimated; observed durations are compared against historical runs to detect anomalies or improvements. This methodology avoids false confidence from invented timelines and provides a clear trigger path for future experiments: `IF all gates pass THEN proceed to next phase`.

B Glossary

This paper crosses cybernetics, dynamical systems, topology, thermodynamics, and computer science. The glossary is grouped by domain so each community can read it; the final table maps every measured quantity to the source artefact that computes it — certify-not-declare made lexical, so that each claim is traceable to auditable code.

Cybernetics (Beer, Ashby).

VSM Viable System Model: the recursive S1–S5 organisation of any viable system Beer [1972].

S1–S5 operations, coordination, control, intelligence, policy; S3* is the sporadic audit channel.

Algedonic a priority alert channel bypassing the normal hierarchy.

Requisite variety Ashby's law: a regulator must match the variety of what it regulates Ashby [1956].

Recursion each S1 unit may itself be a complete VSM.

Dynamical systems.

Attractor the set a trajectory converges to; here a point sink.

Lyapunov exponent λ mean exponential rate of separation of nearby trajectories; $\lambda < 0$ is contraction (Benettin's method).

Lyapunov function L a scalar that decreases along trajectories, certifying convergence (distinct from the exponent).

Basin the set of initial conditions converging to a given attractor.

Kaplan–Yorke dimension a fractal dimension from the spectrum; 0 for a point sink Kaplan and Yorke [1979].

Topology.

Betti numbers β_0, β_1 counts of connected components and of independent loops; $\beta_0 = 1 \wedge \beta_1 = 0$ is the point-sink signature.

Persistent homology tracks topological features across a scale parameter ε; long-lived features are significant.

Vietoris–Rips complex the simplicial complex built by connecting points within distance ε.

Thermodynamics.

Entropy H Shannon entropy of the occupation distribution; $H_{\max}$ its ceiling, H_{rest} the stasis level.

Landauer limit the minimum energy $k_B T \ln 2$ to erase one bit Landauer [1961]; the outer bound on governance.

Maxwell's demon an observer that reduces entropy by measurement; here the S4 layer Maxwell [1871].

VSF-specific.

Ω the feasible region $\bigcap_i \{A_i \text{ satisfied}\}$; Ω_A its convex contractive core.

stasis_cos cosine similarity to the stored attractor reference; $\theta_{\text{viable}} = 0.70$ is the viability threshold.

D_{eff} active-dimension heuristic (*not* the KY dimension).

$\varphi^{\uparrow}/\varphi^{\downarrow}$ upward telemetry / downward command channels between recursion levels.

Covenant the S5 policy document; its hash is verified fail-closed at boot (axioms A0/A0.1).

VSL Viable System Language, the compact protocol notation (Sec. 18).

V the coordinate-free viability functional $\min(g, c, t, m)$ (Sec. 23).

Artefact register. Each measured quantity is computed by a named source artefact:

Quantity	Symbol	Source artefact
stasis cosine	stasis_cos	`state_vector.rs`
Lyapunov spectrum	$\lambda_i, \hat{\sigma}_\lambda$	`-experiment lyapunov_spectrum`
feasible region	Ω	`omega_validator.rs`
Betti numbers	β_0, β_1	`vr_complex.rs` ($\mathbb{Z}_2$ homology)
basin volume	–	`-experiment basin_mc`
sustained-noise margin	$m(\sigma)$	`-experiment sustained_noise`
calibrated constants	$\theta, \varepsilon_{\text{frontier}}$	`calibration.rs`

Table 29: Measured quantities and the source artefacts that compute them.